

Problems in general relativity

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Problems marked with an asterisk (*) have incomplete solutions.

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Wald, General Relativity

Problem 2.1

- (a) Show that the overlap functions $f_i^\pm \circ (f_j^\pm)^{-1}$ are C^∞ , thus completing the demonstration given in section 2.1 that S^2 is a manifold.
- (b) Show by explicit construction that two coordinate systems (as opposed to the six used in the text) suffice to cover S^2 . (It is impossible to cover S^2 with a single chart, as follows from the fact that S^2 is compact, but every open subset of $\mathbb{R}^2$ is noncompact; see appendix A.)

Solution. (a) We have

$$f_1^\pm(x^1, x^2, x^3) = (x^2, x^3),$$

$$f_2^\pm(x^1, x^2, x^3) = (x^1, x^3),$$

$$f_3^\pm(x^1, x^2, x^3) = (x^1, x^2),$$

and it is readily seen that

$$(f_1^\pm)^{-1}(u^1, u^2) = \left(\pm\sqrt{1 - (u^1)^2 - (u^2)^2}, u^1, u^2 \right),$$

$$(f_2^\pm)^{-1}(u^1, u^2) = \left(u^1, \pm\sqrt{1 - (u^1)^2 - (u^2)^2}, u^2 \right),$$

$$(f_3^\pm)^{-1}(u^1, u^2) = \left(u^1, u^2, \pm\sqrt{1 - (u^1)^2 - (u^2)^2} \right),$$

which are defined on the open disk $D = \{(u^1, u^2) \in \mathbb{R}^2 \mid (u^1)^2 + (u^2)^2 < 1\}$. For $i = j$, we have

$$f_i^\pm \circ (f_i^\pm)^{-1}(u^1, u^2) = (u^1, u^2),$$

and this is clearly C^∞ . [There's no need to check $f_i^\pm \circ (f_i^\mp)^{-1}$, since $O_i^+ \cap O_i^- = \emptyset$.] Next,

$$f_1^\pm \circ (f_2^\pm)^{-1} = \left(\pm\sqrt{1 - (u^1)^2 - (u^2)^2}, u^2 \right),$$

$$f_1^\pm \circ (f_3^\pm)^{-1} = \left(u^2, \pm\sqrt{1 - (u^1)^2 - (u^2)^2} \right),$$

$$f_2^\pm \circ (f_3^\pm)^{-1} = \left(u^1, \pm\sqrt{1 - (u^1)^2 - (u^2)^2} \right),$$

with similar expressions for $f_2^\pm \circ (f_1^\pm)^{-1}$, etc., which are the inverses of the functions above. Since these are defined on the open disk D , we conclude that all the overlap functions are C^∞ .

It's not much harder to use a similar construction to show that S^n is a manifold.

(b) We can cover S^2 with the charts (O_N, ψ_N) and (O_S, ψ_S) , which correspond to stereographic projection from the north and south poles, respectively, onto a plane bisecting the sphere:

$$\begin{aligned} O_N &= S^2 \setminus (0, 0, 1), \\ \psi_N(x^1, x^2, x^3) &= \left(\frac{x^1}{1 - x^3}, \frac{x^2}{1 - x^3} \right), \\ O_S &= S^2 \setminus (0, 0, -1), \\ \psi_S(x^1, x^2, x^3) &= \left(\frac{x^1}{1 + x^3}, \frac{x^2}{1 + x^3} \right). \end{aligned}$$

To see that this works, note that every point in S^2 is contained in either O_N or O_S , and that the maps ψ_N and ψ_S are bijective. Indeed, these have the inverses

$$\begin{aligned} \psi_N^{-1}(u^1, u^2) &= \left(\frac{2u^1}{1 + (u^1)^2 + (u^2)^2}, \frac{2u^2}{1 + (u^1)^2 + (u^2)^2}, \frac{-1 + (u^1)^2 + (u^2)^2}{1 + (u^1)^2 + (u^2)^2} \right), \\ \psi_S^{-1}(u^1, u^2) &= \left(\frac{2u^1}{1 + (u^1)^2 + (u^2)^2}, \frac{2u^2}{1 + (u^1)^2 + (u^2)^2}, \frac{1 - (u^1)^2 - (u^2)^2}{1 + (u^1)^2 + (u^2)^2} \right). \end{aligned}$$

Finally, a direct calculation shows that the overlap functions $\psi_N \circ (\psi_S)^{-1}$ and $\psi_S \circ (\psi_N)^{-1}$ are the same C^∞ mapping,

$$(u^1, u^2) \mapsto \left(\frac{u^1}{(u^1)^2 + (u^2)^2}, \frac{u^2}{(u^1)^2 + (u^2)^2} \right).$$

This completes the proof. □

Problem 2.2

Prove that any smooth function $F : \mathbb{R}^n \rightarrow \mathbb{R}$ can be written in the form of equation (2.2.2).

Proof. Fix $x, a \in \mathbb{R}^n$ and let $u(t) = F[t(x - a) + a]$, so that by the chain rule

$$u'(t) = \sum_{\mu=1}^n (x^\mu - a^\mu) \frac{\partial F}{\partial x^\mu} [t(x - a) + a].$$

Then apply the Fundamental Theorem of Calculus to obtain

$$u(1) - u(0) = \int_0^1 u'(t) dt,$$

which, by the definition of $u(t)$, is equivalent to

$$F(x) = F(a) + \sum_{\mu=1}^n (x^\mu - a^\mu) \int_0^1 \frac{\partial F}{\partial x^\mu} [t(x - a) + a] dt.$$

To see that this is in the form of equation (2.2.2), we just need to identify

$$H_\mu(x) \equiv \int_0^1 \frac{\partial F}{\partial x^\mu} [t(x - a) + a] dt,$$

since each H_μ is a C^∞ function with the property $H_\mu(a) = \partial F / \partial x^\mu|_{x=a}$. □

Problem 2.3

- (a) Verify that the commutator, defined by equation (2.2.14), satisfies the linearity and Leibnitz properties, and hence defines a vector field.
- (b) Let X, Y, Z be smooth vector fields on a manifold M . Verify that their commutator satisfies the Jacobi identity:

$$[[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y] = 0.$$

- (c) Let $Y_1, \dots, Y_n$ be smooth vector fields on an n -dimensional manifold M such that at each $p \in M$ they form a basis of the tangent space V_p . Then, at each point, we may expand each commutator $[Y_\alpha, Y_\beta]$ in this basis, thereby defining the functions $C^\gamma_{\alpha\beta} = -C^\gamma_{\beta\alpha}$ by

$$[Y_\alpha, Y_\beta] = \sum_{\gamma} C^\gamma_{\alpha\beta} Y_\gamma.$$

Use the Jacobi identity to derive an equation satisfied by $C^\gamma_{\alpha\beta}$. (This equation is a useful algebraic relation if the $C^\gamma_{\alpha\beta}$ are constants, as will be the case if $Y_1, \dots, Y_n$ are left [or right] invariant vector fields on a Lie group [see section 7.2].)

Solution. (a) Let $f, g \in \mathcal{F}$ and $a, b \in \mathbb{R}$, and let v, w be smooth vector fields, so that $v(f)$, etc., are also smooth functions. Then

$$\begin{aligned} [v, w](af + bg) &= v[w(af + bg)] - w[v(af + bg)] \\ &= v[aw(f) + bw(g)] - w[av(f) + bv(g)] \\ &= av[w(f)] + bw[w(g)] - aw[v(f)] - bw[v(g)] \\ &= a\{v[w(f)] - w[v(f)]\} + b\{v[w(g)] - w[v(g)]\} \\ &= a[v, w](f) + b[v, w](g), \end{aligned}$$

which proves linearity. To show that $[v, w]$ obeys the Leibniz rule, let $p \in M$. Then at p ,

$$\begin{aligned} [v, w](fg) &= v[w(fg)] - w[v(fg)] \\ &= v[f(p)w(g) + g(p)w(f)] - w[f(p)v(g) + g(p)v(f)] \\ &= f(p)v[w(g)] + g(p)v[w(f)] - f(p)w[v(g)] - g(p)w[v(f)] \\ &= f(p)\{v[w(g)] - w[v(g)]\} + g(p)\{v[w(f)] - w[v(f)]\} \\ &= f(p)[v, w](g) + g(p)[v, w](f). \end{aligned}$$

Therefore, the commutator of two vector fields defines another vector field.

(b) Let $f \in \mathcal{F}$. To ease notation, we'll use the shorthand $XYZ(f) \equiv X(Y(Z(f)))$, and so on. Then we have

$$\begin{aligned} [[X, Y], Z](f) &= [X, Y]Z(f) - Z[X, Y](f) \\ &= XYZ(f) - YXZ(f) - ZXY(f) + ZYX(f). \end{aligned}$$

Cycling through X, Y, Z gives

$$\begin{aligned} [[Y, Z], X](f) &= YZX(f) - ZYX(f) - XYZ(f) + XZY(f), \\ [[Z, X], Y](f) &= ZXY(f) - XZY(f) - YZX(f) + YXZ(f), \end{aligned}$$

and using these, it is easy to verify that

$$\left([[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y] \right)(f) = 0.$$

Since f was arbitrary, the sum of commutators in the parentheses must vanish, which establishes the Jacobi identity.

(c) Let $Y_1, \dots, Y_n$ be smooth vector fields as described in the problem statement, let $f \in \mathcal{F}$, and pick some point $p \in M$. Then at p ,

$$\begin{aligned} [[Y_\alpha, Y_\beta], Y_\gamma](f) &= [C^\mu_{\alpha\beta} Y_\mu, Y_\gamma](f) \\ &= (C^\mu_{\alpha\beta} Y_\mu)(Y_\gamma(f)) - Y_\gamma(C^\mu_{\alpha\beta} Y_\mu(f)) \\ &= C^\mu_{\alpha\beta}(p) Y_\mu(Y_\gamma(f)) - C^\mu_{\alpha\beta}(p) Y_\gamma(Y_\mu(f)) - Y_\gamma(C^\mu_{\alpha\beta}) Y_\mu(f) \\ &= C^\mu_{\alpha\beta}(p) [Y_\mu, Y_\gamma](f) - Y_\gamma(C^\mu_{\alpha\beta}) Y_\mu(f) \\ &= \left[C^\mu_{\alpha\beta}(p) C^\nu_{\mu\gamma}(p) - Y_\gamma(C^\nu_{\alpha\beta}) \right] Y_\nu(f), \end{aligned}$$

where in the last line, we also relabel a bound index on the second term. Cycling through the indices α, β, γ then gives

$$\begin{aligned} [[Y_\beta, Y_\gamma], Y_\alpha](f) &= \left[C^\mu_{\beta\gamma}(p) C^\nu_{\mu\alpha}(p) - Y_\alpha(C^\nu_{\beta\gamma}) \right] Y_\nu(f), \\ [[Y_\gamma, Y_\alpha], Y_\beta](f) &= \left[C^\mu_{\gamma\alpha}(p) C^\nu_{\mu\beta}(p) - Y_\beta(C^\nu_{\gamma\alpha}) \right] Y_\nu(f). \end{aligned}$$

Adding these three expressions, applying the Jacobi identity, then using the fact that f is arbitrary, we get

$$\begin{aligned} \left[C^\mu_{\alpha\beta}(p) C^\nu_{\mu\gamma}(p) - Y_\gamma(C^\nu_{\alpha\beta}) + C^\mu_{\beta\gamma}(p) C^\nu_{\mu\alpha}(p) - Y_\alpha(C^\nu_{\beta\gamma}) \right. \\ \left. + C^\mu_{\gamma\alpha}(p) C^\nu_{\mu\beta}(p) - Y_\beta(C^\nu_{\gamma\alpha}) \right] Y_\nu = 0. \end{aligned}$$

But by assumption $Y_1, \dots, Y_n$ are a basis at p , hence they are linearly independent, so the

expression in brackets must vanish. Finally, since $C^\mu_{\alpha\beta} = -C^\mu_{\beta\alpha}$, this can be neatly written in terms of antisymmetrization brackets:

$$C^\nu_{\mu[\alpha} C^\mu_{\beta\gamma]} - Y_{[\gamma} (C^\nu_{\alpha\beta]}) = 0.$$

If the functions $C^\mu_{\alpha\beta}$ are constant, then only the first term survives. □

Problem 2.4

- (a) Show that in any coordinate basis, the components of the commutator of two vector fields v and w are given by

$$[v, w]^\mu = \sum_\nu \left(v^\nu \frac{\partial w^\mu}{\partial x^\nu} - w^\nu \frac{\partial v^\mu}{\partial x^\nu} \right).$$

- (b) Let $Y_1, \dots, Y_n$ be as in problem 3(c). Let $Y^{1*}, \dots, Y^{n*}$ be the dual basis. Show that the components $(Y^{\gamma*})_\mu$ of $Y^{\gamma*}$ in any coordinate basis satisfy

$$\frac{\partial(Y^{\gamma*})_\mu}{\partial x^\nu} - \frac{\partial(Y^{\gamma*})_\nu}{\partial x^\mu} = \sum_{\alpha, \beta} C^\gamma_{\alpha\beta} (Y^{\alpha*})_\mu (Y^{\beta*})_\nu.$$

Solution. (a) Let ψ be a chart, and let $\{X_\mu\}$ be the associated coordinate basis. Then for $f \in \mathcal{F}$, we have

$$\begin{aligned} [v, w](f) &= v[w(f)] - w[v(f)] \\ &= v[(w^\mu X_\mu)(f)] - w[(v^\mu X_\mu)(f)] \\ &= (v^\nu X_\nu)(w^\mu X_\mu(f)) - (w^\nu X_\nu)(v^\mu X_\mu(f)) \\ &= v^\nu \frac{\partial w^\mu}{\partial x^\nu} X_\mu(f) + v^\nu w^\mu \frac{\partial^2}{\partial x^\nu \partial x^\mu} (f \circ \psi^{-1}) - w^\nu \frac{\partial v^\mu}{\partial x^\nu} X_\mu(f) - w^\nu v^\mu \frac{\partial^2}{\partial x^\nu \partial x^\mu} (f \circ \psi^{-1}) \\ &= \left(v^\nu \frac{\partial w^\mu}{\partial x^\nu} - w^\nu \frac{\partial v^\mu}{\partial x^\nu} \right) X_\mu(f), \end{aligned}$$

where in the fourth line we use the Leibniz property of tangent vectors, and in the last line we use the commutativity of mixed partial derivatives. This shows that

$$[v, w]^\mu = v^\nu \frac{\partial w^\mu}{\partial x^\nu} - w^\nu \frac{\partial v^\mu}{\partial x^\nu}.$$

- (b) Feed $Y^{\gamma*}$ to the equation defining the functions $C^\mu_{\sigma\rho}$ to get

$$[Y_\sigma, Y_\rho](Y^{\gamma*}) = C^\mu_{\sigma\rho} \underbrace{Y_\mu(Y^{\gamma*})}_{=\delta_\mu^\gamma},$$

and with a coordinate basis, this becomes

$$[Y_\sigma, Y_\rho]^\mu (Y^{\gamma*})_\mu = C^\gamma_{\sigma\rho}. \quad (1)$$

Note also that

$$\begin{aligned} 0 &= \frac{\partial}{\partial x^\nu} \delta_\beta^\alpha = \frac{\partial((Y^{\alpha*})_\mu (Y_\beta)^\mu)}{\partial x^\nu} \\ &= (Y^{\alpha*})_\mu \frac{\partial(Y_\beta)^\mu}{\partial x^\nu} + \frac{\partial(Y^{\alpha*})_\mu}{\partial x^\nu} (Y_\beta)^\mu. \end{aligned} \quad (2)$$

Expanding out the left-hand side of Eq. (1) using the result of part (a), we get

$$\begin{aligned}
[Y_\sigma, Y_\rho]^\mu (Y^{\gamma*})_\mu &= (Y_\sigma)^\nu (Y^{\gamma*})_\mu \frac{\partial (Y_\rho)^\mu}{\partial x^\nu} - (Y_\rho)^\nu (Y^{\gamma*})_\mu \frac{\partial (Y_\sigma)^\mu}{\partial x^\nu} \\
&= -(Y_\sigma)^\nu (Y_\rho)^\mu \frac{\partial (Y^{\gamma*})_\mu}{\partial x^\nu} + (Y_\rho)^\nu (Y_\sigma)^\mu \frac{\partial (Y^{\gamma*})_\mu}{\partial x^\nu} \\
&= \left(\frac{\partial (Y^{\gamma*})_\mu}{\partial x^\nu} - \frac{\partial (Y^{\gamma*})_\nu}{\partial x^\mu} \right) (Y_\sigma)^\mu (Y_\rho)^\nu,
\end{aligned}$$

where in the second line we use Eq. (2). Meanwhile, the right-hand side of Eq. (1) is

$$\begin{aligned}
C^\gamma_{\sigma\rho} &= C^\gamma_{\alpha\beta} \delta^\alpha_\sigma \delta^\beta_\rho \\
&= C^\gamma_{\alpha\beta} (Y^{\alpha*})_\mu (Y_\sigma)^\mu (Y^{\beta*})_\nu (Y_\rho)^\nu,
\end{aligned}$$

so we obtain

$$\left[\left(\frac{\partial (Y^{\gamma*})_\mu}{\partial x^\nu} - \frac{\partial (Y^{\gamma*})_\nu}{\partial x^\mu} \right) - C^\gamma_{\alpha\beta} (Y^{\alpha*})_\mu (Y^{\beta*})_\nu \right] (Y_\sigma)^\mu (Y_\rho)^\nu = 0.$$

Since Y_σ and Y_ρ are arbitrary basis vectors, the expression in brackets must vanish, which establishes the identity. (this solution is kind of awkward – it would be clearer to start by feeding arbitrary basis vectors to this tensor, instead of doing a bunch of work and then invoking that argument at the end.) $\square$

Problem 2.5 (*)

Let $Y_1, \dots, Y_n$ be smooth vector fields on an n -dimensional manifold M which form a basis of V_p at each $p \in M$. Suppose $[Y_\alpha, Y_\beta] = 0$ for all α, β . Prove that in a neighborhood of each $p \in M$ there exist coordinates $y_1, \dots, y_n$ such that $Y_1, \dots, Y_n$ are the coordinate vector fields, $Y_\mu = \partial/\partial y^\mu$.

Solution. (unfinished)

□

Problem 2.6

- (a) Verify that the dual vectors $\{v^{\mu*}\}$ defined by equation (2.3.1) constitute a basis of V^* .
- (b) Let $v_1, \dots, v_n$ be a basis of the vector space V and let $v^{1*}, \dots, v^{n*}$ be its dual basis. Let $w \in V$ and let $\omega \in V^*$. Show that

$$w = \sum_{\alpha} v^{\alpha*}(w) v_{\alpha},$$

$$\omega = \sum_{\alpha} \omega(v_{\alpha}) v^{\alpha*}.$$

- (c) Prove that the operation of contraction, equation (2.3.2), is independent of the choice of basis.

Solution. (a) Equation (2.3.1) reads

$$v^{\mu*}(v_{\nu}) = \delta_{\nu}^{\mu}.$$

The dual vectors defined this way are linearly independent, for if

$$\alpha_{\mu} v^{\mu*} = 0,$$

then acting v_{ν} on this equation gives $\alpha_{\nu} = 0$. Going through this process for $\nu = 1, \dots, n$ shows that all the coefficients are zero. These also span V^* , for if $\omega \in V^*$, then we can write

$$\omega = \omega(v_{\mu}) v^{\mu*}.$$

To see this, simply feed basis vectors v_{ν} to both sides. The left-hand side becomes $\omega(v_{\nu})$, while the right-hand side becomes

$$\omega(v_{\mu}) v^{\mu*}(v_{\nu}) = \omega(v_{\mu}) \delta_{\nu}^{\mu} = \omega(v_{\nu}).$$

Since we have two linear maps that agree on a basis, these must be the same linear map, which proves our claim. Hence $\{v^{\mu*}\}$ is a basis for V^* , which also shows that $\dim V^* = n$.

- (b) We showed that every $\omega \in V^*$ can be decomposed in this way, so consider $w \in V$. Recalling that we can associate to every element of V a corresponding element of V^{**} , we have

$$(v^{\alpha*}(w) v_{\alpha})(v^{\beta*}) = v^{\alpha*}(w) v_{\alpha}(v^{\beta*}) = v^{\alpha*}(w) \delta_{\alpha}^{\beta} = v^{\beta*}(w)$$

and

$$w(v^{\beta*}) = v^{\beta*}(w).$$

We have two linear maps that agree on a basis, which shows that these are equal.

(c) Let $\{v_\sigma\}$ be a basis of V , and let $\{v^{\sigma*}\}$ be the corresponding dual basis. Given some other basis $\{w_\alpha\}$, we may use the result of part (b) to write

$$\begin{aligned} v_\sigma &= w^{\alpha*}(v_\sigma)w_\alpha, \\ v^{\sigma*} &= v^{\sigma*}(w_\alpha)w^{\alpha*}. \end{aligned}$$

Therefore, if $T \in \mathcal{T}(k, l)$, then its contraction in terms of the $\{v_\sigma\}$ basis is

$$\begin{aligned} CT &= T(\dots, v^{\sigma*}, \dots; \dots, v_\sigma, \dots) \\ &= T(\dots, v^{\sigma*}(w_\alpha)w^{\alpha*}, \dots; \dots, w^{\beta*}(v_\sigma)w_\beta, \dots) \\ &= v^{\sigma*}(w_\alpha)w^{\beta*}(v_\sigma)T(\dots, w^{\alpha*}, \dots; \dots, w_\beta, \dots) \\ &= w^{\beta*}(w_\alpha)T(\dots, w^{\alpha*}, \dots; \dots, w_\beta, \dots) \\ &= \delta_\alpha^\beta T(\dots, w^{\alpha*}, \dots; \dots, w_\beta, \dots) \\ &= T(\dots, w^{\alpha*}, \dots; \dots, w_\alpha, \dots), \end{aligned}$$

where the fourth line follows from the fact that $w^{\beta*} = w^{\beta*}(v_\sigma)v^{\sigma*}$. The last expression is what we'd write for CT in terms of the $\{w_\alpha\}$ basis, which shows that the operation of contraction is basis-independent. $\square$

Problem 2.7

Let V be an n -dimensional vector space and let g be a metric on V .

- (a) Show that one can always find an orthonormal basis $v_1, \dots, v_n$ of V , i.e. a basis such that $g(v_\alpha, v_\beta) = \pm \delta_{\alpha\beta}$.
- (b) Show that the signature of g is independent of the choice of orthonormal basis.

Lemma (Polarization identity).

For a symmetric bilinear form $f : V \times V \rightarrow \mathbb{R}$ and $u, v \in V$, we have

$$f(u, v) = \frac{1}{4}q(u+v) - \frac{1}{4}q(u-v),$$

where q is the quadratic form associated with f , i.e., $q(u) = f(u, u)$.

Proof. A direct computation. □

Solution. (a) This follows Section 10.2 of Hoffman and Kunze. The proof is by induction on n . For $n = 1$, pick any $v \neq 0$ such that $g(v, v) \neq 0$, and then normalize it to get an orthonormal basis. Such a vector v exists, for if $g(v, v) = 0$ for all $v \in V$, then by the polarization identity, g must send every pair of vectors to zero. But this contradicts nondegeneracy.

Next assume that the hypothesis holds for some n . As earlier, pick a vector $v \in V$ such that $v \neq 0$ and $g(v, v) \neq 0$; without loss of generality, we may assume $g(v, v) = \pm 1$. Now let W be the one-dimensional subspace of V spanned by v , and let $W^\perp$ be the set of all $u \in V$ such that $g(u, v) = 0$. We claim that $W^\perp$ is a subspace of V , and that

$$V = W \oplus W^\perp.$$

The first claim is easily verified, so we focus on the second. To see that the subspaces are independent, pick some element of W , which can in general be written cv for $c \in \mathbb{R}$. Then $g(cv, v) = cg(v, v) \neq 0$ unless $c = 0$, which shows that $W \cap W^\perp = \{0\}$. To see that any element of V can be written as a sum of elements of W and $W^\perp$, take $x \in V$ and write

$$x = \frac{g(x, v)}{g(v, v)} v + \left(x - \frac{g(x, v)}{g(v, v)} v \right).$$

It's easy to check that the vector in parentheses is orthogonal to v , thus $V = W \oplus W^\perp$. Finally, by our inductive hypothesis, there exists an orthonormal basis $v_1, \dots, v_n$ of $W^\perp$. If we let $v_{n+1} \equiv v$, then $v_1, \dots, v_{n+1}$ is an orthonormal basis of V , which completes the proof.

(b) This also follows Hoffman and Kunze. Let $v_1, \dots, v_n$ be an orthonormal basis of V , and let p be the number of basis vectors such that $g(v_\mu, v_\mu) = +1$. Define V^+ to be the vector subspace of V spanned by the basis vectors for which $g(v_\mu, v_\mu) = +1$, and define V^- similarly by the condition $g(v_\mu, v_\mu) = -1$. Then by construction, $V = V^+ \oplus V^-$.

It follows that g is positive definite on V^+ , for if $x \in V^+$ is nonzero, then

$$\begin{aligned} g(x, x) &= g(x^\alpha v_\alpha, x^\beta v_\beta) = x^\alpha x^\beta g(v_\alpha, v_\beta) \\ &= x^\alpha x^\beta \delta_{\alpha\beta} \\ &= x^\alpha x_\alpha > 0. \end{aligned}$$

Similarly, one can show that g is negative definite on V^- . Now let W be any subspace of V on which g is positive definite. We claim that $W \cap V^- = \{0\}$. To see this, take $x \in W$, $y \in V^-$ such that $x + y = 0$. Then

$$\begin{aligned} 0 &= g(x, x + y) = g(x, x) + g(x, y), \\ 0 &= g(y, x + y) = g(y, x) + g(y, y), \end{aligned}$$

and due to the symmetry of g , this implies $g(x, x) = g(y, y)$. But by the positive and negative-definiteness of g on W and V^- , respectively, we must have $x = y = 0$. This implies that $W \cap V^- = \{0\}$, and since $V = V^+ \oplus V^-$, we have $\dim W \leq \dim V^+$.

Finally, let $\mathcal{B}_1$ be any orthonormal basis of V . Repeating the construction from earlier, we obtain the corresponding subspaces V_1^+ and V_1^- . Since g is positive definite on V_1^+ , we have $\dim V_1^+ \leq \dim V^+$. But we can apply the same argument starting from $\mathcal{B}_1$ to conclude that $\dim V^+ \leq \dim V_1^+$, hence $\dim V_1^+ = \dim V^+ = p$. That is, any orthonormal basis of V has the same signature. $\square$

Problem 2.8

- (a) The metric of flat, three-dimensional Euclidean space is

$$ds^2 = dx^2 + dy^2 + dz^2.$$

Show that the metric components $g_{\mu\nu}$ in spherical coordinates r, θ, ϕ defined by

$$r = (x^2 + y^2 + z^2)^{1/2},$$

$$\cos \theta = z/r,$$

$$\tan \phi = y/x,$$

are given by

$$ds^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2.$$

- (b) The spacetime metric of special relativity is

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2.$$

Find the components, $g_{\mu\nu}$ and $g^{\mu\nu}$, of the metric and inverse metric in “rotating coordinates”, defined by

$$t' = t,$$

$$x' = (x^2 + y^2)^{1/2} \cos(\phi - \omega t),$$

$$y' = (x^2 + y^2)^{1/2} \sin(\phi - \omega t),$$

$$z' = z,$$

where $\tan \phi = y/x$.

Solution. Both of these are straightforward applications of the tensor transformation law. For part (b), we note that the definitions can be rewritten in the form

$$\left\{ \begin{array}{l} t' = t \\ x' = x \cos \omega t + y \sin \omega t \\ y' = -x \sin \omega t + y \cos \omega t \\ z' = z \end{array} \right. \longleftrightarrow \left\{ \begin{array}{l} t = t' \\ x = x' \cos \omega t' - y' \sin \omega t' \\ y = x' \sin \omega t' + y' \cos \omega t' \\ z = z' \end{array} \right.$$

The second set of expressions is what we use to compute the relevant Jacobian matrix. We then

find that

$$\begin{aligned} ds^2 = & [\omega^2 x'^2 + \omega^2 y'^2 - 1](dt')^2 - \omega y'(dt' dx' + dx' dt') + \omega x'(dt' dy' + dy' dt') \\ & + (dx')^2 + (dy')^2 + (dz')^2, \end{aligned}$$

where we omit the tensor product symbols. Note that if $\omega = 0$, then we recover the flat metric, as we expect. Also note that something weird seems to happen when $\omega^2 x'^2 + \omega^2 y'^2 = 1$; what this indicates is not immediately obvious. $\square$

Problem 2.8

(a)

```
In[1]:= xs = {r,  $\theta$ ,  $\phi$ };
transition = {r Sin[ $\theta$ ] Cos[ $\phi$ ], r Sin[ $\theta$ ] Sin[ $\phi$ ], r Cos[ $\theta$ ]};
(* cartesian to spherical *)

In[3]:= jacobian = Table[D[transition[[ $\alpha$ ]], xs[[ $\beta$ ]]], { $\alpha$ , 1, 3}, { $\beta$ , 1, 3}] // Simplify;
jacobian // MatrixForm

Out[4]//MatrixForm=

$$\begin{pmatrix} \text{Cos}[\phi] \text{Sin}[\theta] & r \text{Cos}[\theta] \text{Cos}[\phi] & -r \text{Sin}[\theta] \text{Sin}[\phi] \\ \text{Sin}[\theta] \text{Sin}[\phi] & r \text{Cos}[\theta] \text{Sin}[\phi] & r \text{Cos}[\phi] \text{Sin}[\theta] \\ \text{Cos}[\theta] & -r \text{Sin}[\theta] & 0 \end{pmatrix}$$


In[5]:= oldmetric = {{1, 0, 0}, {0, 1, 0}, {0, 0, 1}}; (* old components *)

In[6]:= Table[Sum[jacobian[[ $\mu$ ]] [[ $\alpha$ ]]  $\times$  jacobian[[ $\nu$ ]] [[ $\beta$ ]]  $\times$  oldmetric[[ $\mu$ ]] [[ $\nu$ ]], { $\mu$ , 1, 3}, { $\nu$ , 1, 3}],
{ $\alpha$ , 1, 3}, { $\beta$ , 1, 3}] // Simplify //
MatrixForm (* apply tensor transformation law *)

Out[6]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \text{Sin}[\theta]^2 \end{pmatrix}$$

```

(b)

```
In[7]:= xs = {tp, xp, yp, zp};
transition = {tp, xp Cos[ $\omega$  tp] - yp Sin[ $\omega$  tp], xp Sin[ $\omega$  tp] + yp Cos[ $\omega$  tp], zp};
(* unprimed to primed (rotating) coordinates *)

In[9]:= jacobian = Table[D[transition[[ $\alpha$ ]], xs[[ $\beta$ ]]], { $\alpha$ , 1, 4}, { $\beta$ , 1, 4}] // Simplify;
jacobian // MatrixForm

Out[10]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ -\omega (yp \text{Cos}[\omega tp] + xp \text{Sin}[\omega tp]) & \text{Cos}[\omega tp] & -\text{Sin}[\omega tp] & 0 \\ \omega (xp \text{Cos}[\omega tp] - yp \text{Sin}[\omega tp]) & \text{Sin}[\omega tp] & \text{Cos}[\omega tp] & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$


In[11]:= oldmetric = {{-1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, 1, 0}, {0, 0, 0, 1}};
(* old components *)
```

```
In[12]:= newmetric = Table[
  Sum[jacobian[[μ]][[α]] × jacobian[[ν]][[β]] × oldmetric[[μ]][[ν]], {μ, 1, 4}, {ν, 1, 4}],
  {α, 1, 4}, {β, 1, 4}] // Simplify; (* tensor transformation law *)
newmetric // MatrixForm
```

Out[13]//MatrixForm=

$$\begin{pmatrix} -1 + xp^2 \omega^2 + yp^2 \omega^2 & -yp \omega & xp \omega & 0 \\ -yp \omega & 1 & 0 & 0 \\ xp \omega & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

```
In[14]:= Inverse[newmetric] // Simplify // MatrixForm (* Inverse[newmetric] [[a,b]] ≡ gab *)
```

Out[14]//MatrixForm=

$$\begin{pmatrix} -1 & -yp \omega & xp \omega & 0 \\ -yp \omega & 1 - yp^2 \omega^2 & xp yp \omega^2 & 0 \\ xp \omega & xp yp \omega^2 & 1 - xp^2 \omega^2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Problem 3.1

Let property (5) (the “torsion free” condition) be dropped from the definition of derivative operator ∇_α in section 3.1.

- (a) Show that there exists a tensor T_{ab}^c (called the *torsion tensor*) such that for all smooth functions, f , we have $\nabla_a \nabla_b f - \nabla_b \nabla_a f = -T_{ab}^c \nabla_c f$.
- (b) Show that for any smooth vector fields X^a, Y^a we have

$$T_{ab}^c X^a Y^b = X^a \nabla_a Y^c - Y^a \nabla_a X^c - [X, Y]^c.$$

- (c) Given a metric, g_{ab} , show that there exists a unique derivative operator ∇_a with torsion T_{ab}^c such that $\nabla_c g_{ab} = 0$. Derive the analog of equation (3.1.29), expressing this derivative operator in terms of an ordinary derivative ∂_a and T_{ab}^c .

Solution. (a) Let ∇_a and $\tilde{\nabla}_a$ be derivative operators, and let f be a smooth function. Then

$$\begin{aligned} \nabla_a \nabla_b f - \nabla_b \nabla_a f &= \tilde{\nabla}_a \nabla_b f - C_{ab}^c \nabla_c f - \tilde{\nabla}_b \nabla_a f + C_{ba}^c \nabla_c f \\ &= \tilde{\nabla}_a \tilde{\nabla}_b f - \tilde{\nabla}_b \tilde{\nabla}_a f - (C_{ab}^c - C_{ba}^c) \nabla_c f, \end{aligned}$$

where we’ve used the fact that $\nabla_a f = \tilde{\nabla}_a f$. The choice $\tilde{\nabla}_a = \partial_a$ causes the first two terms to vanish (since partial derivatives commute), which leaves us with

$$\nabla_a \nabla_b f - \nabla_b \nabla_a f = -(\Gamma_{ab}^c - \Gamma_{ba}^c) \nabla_c f.$$

Then the desired identity follows if we let

$$T_{ab}^c = \Gamma_{ab}^c - \Gamma_{ba}^c,$$

since the two terms on the right are $(1, 2)$ tensor fields by definition.

- (b) First we derive the analogue of equation (3.1.2) for the case where ∇_a is not torsion-free. Let v^a and w^a be smooth vector fields, and let f be a smooth function. Then

$$\begin{aligned} [v, w](f) &= v\{w(f)\} - w\{v(f)\} \\ &= v^a \nabla_a (w^b \nabla_b f) - w^a \nabla_a (v^b \nabla_b f) \\ &= (v^a \nabla_a w^b - w^a \nabla_a v^b) \nabla_b f - v^a w^b (\nabla_a \nabla_b f - \nabla_b \nabla_a f), \end{aligned}$$

where we’ve used properties (2) and (4) of a derivative operator. Using the result of part (a), relabeling bound indices, and recalling that f was arbitrary, we get

$$[v, w]^b = v^a \nabla_a w^b - w^a \nabla_a v^b - T_{ac}^b v^a w^c.$$

Therefore, if X^a, Y^a are smooth vector fields, then

$$\begin{aligned} X^a \nabla Y^c - Y^a \nabla_a X^c - [X, Y]^c &= \cancel{X^a \nabla_a Y^c} - \cancel{Y^a \nabla_a X^c} - \cancel{X^a \nabla_a Y^c} + \cancel{Y^a \nabla_a X^c} + T^c_{ab} X^a Y^b \\ &= T^c_{ab} X^a Y^b, \end{aligned}$$

as we wanted to show.

(c) Let g_{ab} be a metric, and let ∇_a be a derivative operator satisfying $\nabla_a g_{bc} = 0$. Then

$$0 = \nabla_a g_{bc} = \partial_a g_{bc} - \Gamma^d_{ab} g_{dc} - \Gamma^d_{ac} g_{bd},$$

and by cycling through indices, we can get two more equations:

$$\Gamma_{cab} + \Gamma_{bac} = \partial_a g_{bc},$$

$$\Gamma_{abc} + \Gamma_{cba} = \partial_b g_{ca},$$

$$\Gamma_{bca} + \Gamma_{acb} = \partial_c g_{ab}.$$

Subtracting the first and second equations from the third, then adding and subtracting by $2\Gamma_{cab}$, we obtain

$$T_{bca} + T_{acb} + T_{cab} - 2\Gamma_{cab} = \partial_c g_{ab} - \partial_b g_{ca} - \partial_a g_{bc},$$

and by rearranging terms and raising an index, we get

$$\Gamma^c_{ab} = \frac{1}{2} g^{cd} (T_{dab} + T_{adb} + T_{bda}) + \frac{1}{2} g^{cd} (\partial_a g_{bd} + \partial_b g_{da} - \partial_d g_{ab}).$$

As a check, this reduces to equation (3.1.29) when ∇_a is torsion-free, as it should. $\square$

Problem 3.2

Let M be a manifold with metric g_{ab} and associated derivative operator ∇_a . A solution of the equation $\nabla_a \nabla^a \alpha = 0$ is called a *harmonic function*. In the case where M is a two-dimensional manifold, let α be harmonic and let ϵ_{ab} be an antisymmetric tensor field satisfying $\epsilon_{ab} \epsilon^{ab} = 2(-1)^s$, where s is the number of minuses occurring in the signature of the metric. Consider the equation $\nabla_a \beta = \epsilon_{ab} \nabla^b \alpha$.

- (a) Show that the integrability conditions (see problem 5 of chapter 2 or appendix B) for this equation are satisfied, and thus, locally, there exists a solution, β . Show that β also is harmonic, $\nabla_a \nabla^a \beta = 0$. (β is called the harmonic function *conjugate* to α .)
- (b) By choosing α and β as coordinates, show that the metric takes the form

$$ds^2 = \pm \Omega^2(\alpha, \beta) [d\alpha^2 + (-1)^s d\beta^2].$$

Solution.

□

Problem 3.3

- (a) Show that $R_{abcd} = R_{cdab}$.
- (b) In n dimensions, the Riemann tensor has n^4 components. However, on account of the symmetries (3.2.13), (3.2.14), and (3.2.15), not all of these components are independent. Show that the number of independent components is $n^2(n^2 - 1)/12$.

Solution. (a) First we enumerate the algebraic properties of the Riemann tensor:

1. $R_{abc}{}^d = -R_{bac}{}^d$
2. $R_{[abc]d} = 0$
3. $R_{abcd} = -R_{abdc}$

Using property (1), it is straightforward to show that (2) is equivalent to $R_{abcd} + R_{cabd} + R_{bcad} = 0$, which is what we'll use below. Now

$$\begin{aligned}
 R_{abcd} &\stackrel{(2)}{=} -R_{cabd} - R_{bcad} \\
 &\stackrel{(3)}{=} -R_{cabd} + R_{bcda} \\
 &\stackrel{(2)}{=} -R_{cabd} - R_{dbca} - R_{cdba} \\
 &\stackrel{(1)}{=} -R_{cabd} - R_{dbca} + R_{cdab},
 \end{aligned}$$

and also

$$\begin{aligned}
 R_{abcd} &\stackrel{(1),(3)}{=} R_{badc} \\
 &\stackrel{(2)}{=} -R_{dbac} - R_{adbc} \\
 &\stackrel{(1)}{=} -R_{dbca} - R_{adbc}.
 \end{aligned}$$

Adding these expressions gives

$$\begin{aligned}
 2R_{abcd} &= -R_{cabd} - R_{adbc} + R_{cdab} \\
 &\stackrel{(1),(3)}{=} -R_{acdb} - R_{dacb} + R_{cdab} \\
 &\stackrel{(2)}{=} 2R_{cdab},
 \end{aligned}$$

which is what we wanted to show.

(b) Using the property we just derived, the Riemann tensor is symmetric under interchanges of the first and second pair of indices. Due to properties (1) and (3), each pair has $n(n-1)/2$ independent combinations. This gives

$$\frac{1}{2} \left[\frac{n(n-1)}{2} \right] \left[\frac{n(n-1)}{2} + 1 \right] = \frac{1}{8} n(n-1)(n^2 - n + 2).$$

Furthermore, by properties (1) and (3), the cyclic sum $R_{abcd} + R_{cabd} + R_{bcad}$ is totally antisymmetric, so that property (2) imposes an additional number of constraints equal to

$$\binom{n}{4} = \frac{n!}{4!(n-4)!} = \frac{1}{24} n(n-1)(n-2)(n-3).$$

Subtracting this from the previous expression yields

$$\frac{1}{12} n^2 (n^2 - 1)$$

for the number of independent components of the Riemann tensor, as we wanted to show. **For this to be more convincing, you want to show that these properties yield independent constraints, so that you aren't overcounting.** □

Problem 3.4

- (a) Show that in two dimensions, the Riemann tensor takes the form $R_{abcd} = Rg_{a[c}g_{d]b}$.
- (b) By similar arguments, show that in three dimensions the Weyl tensor vanishes identically, i.e. for $n = 3$, equation (3.2.28) holds with $C_{abcd} = 0$.

Solution. (a) By part (b) of problem 3.3, in $n = 2$ dimensions, if X_{abcd} is a tensor sharing the symmetries of the Riemann tensor, namely

$$X_{abcd} = -X_{abdc}, \quad (1)$$

$$X_{abcd} = -X_{bacd}, \quad (2)$$

$$X_{a[bcd]} = 0, \quad (3)$$

then X_{abcd} has only one independent component. In particular, then, all such tensors differ only by an overall multiplicative constant. Now consider the tensor $g_{a[c}g_{d]b}$. We have

$$g_{a[c}g_{d]b} = \frac{1}{2}(g_{ac}g_{db} - g_{ad}g_{cb}),$$

so that if we interchange $a \leftrightarrow b$, we obtain

$$\frac{1}{2}(g_{bc}g_{da} - g_{bd}g_{ca}) = -g_{a[c}g_{d]b}$$

by the symmetry of g_{ab} . Similarly, the interchange $c \leftrightarrow d$ leads to

$$\frac{1}{2}(g_{ad}g_{cb} - g_{ac}g_{db}) = -g_{a[c}g_{d]b},$$

which shows that $g_{a[c}g_{d]b}$ possesses symmetries (1) and (2). Next, we antisymmetrize $g_{a[c}g_{d]b}$ over b, c, d to find

$$\frac{1}{2}(g_{a[c}g_{d]b} - g_{a[d}g_{c]b}) = 0,$$

because g_{ab} is symmetric. This establishes symmetry (3). Finally, it's straightforward to verify that symmetries (1)–(3) imply the only nonzero component of this tensor is

$$g_{1[2}g_{1]2} = \frac{1}{2}[(g_{12})^2 - g_{11}g_{22}] = \frac{1}{2}\det(g).$$

Since g is nondegenerate, its determinant is nonzero, and we conclude that

$$R_{abcd} = \kappa g_{a[c}g_{d]b},$$

where κ is some constant. To fix the value of κ , we contract over all four indices to get

$$\begin{aligned} R &= \frac{1}{2} \kappa g^{ac} g^{bd} (g_{ac} g_{db} - g_{ad} g_{cb}) \\ &= \frac{1}{2} \kappa ((2)(2) - \delta_d^c \delta_c^d) \\ &= \frac{1}{2} \kappa (4 - 2) = \kappa, \end{aligned}$$

where we've used the fact that $\delta_a^a = 2$. Thus $R_{abcd} = R g_{a[c} g_{d]b}$, as we wanted, and in particular the Ricci scalar is given explicitly by

$$R = \frac{2}{\det(g)} R_{1212}.$$

(b) One can show from the definition that the Weyl tensor shares the algebraic symmetries (1)–(3) of the Riemann tensor, which implies $n^2(n^2 - 1)/12$ independent components. But the Weyl tensor also has the property that $C^a_{abc} = 0$, and by using symmetries (1)–(3), one can show that all six possible traces are equal. Thus the Weyl tensor is trace-free on all pairs of indices, and this statement,

$$C^\mu_{\nu\mu\sigma} = 0,$$

represents $\binom{n+2-1}{2}$ independent constraints (this is the number of ways to choose two values, with replacement, from n possible values). Thus, the Weyl tensor has

$$\begin{aligned} \frac{n^2(n^2 - 1)}{12} - \binom{n+2-1}{2} &= \frac{n^2(n^2 - 1)}{12} - \frac{n(n+1)}{2} \\ &= \frac{1}{12} n(n+1)(n+2)(n-3) \end{aligned}$$

independent components. This vanishes when $n = 3$, as we wanted to show. $\square$

Problem 3.5

- (a) Show that any curve whose tangent satisfies equation (3.3.2) can be reparametrized so that equation (3.3.1) is satisfied.
- (b) Let t be an affine parameter of a geodesic γ . Show that all other affine parameters of γ take the form $at + b$, where a and b are constants.

Solution. (a) Suppose we have a curve whose tangent T^a satisfies equation (3.3.2),

$$T^a \nabla_a T^b = \alpha T^b,$$

where $\alpha = \alpha(t)$ is an arbitrary function along the curve. Reparametrizing by $\tilde{t}$, and noting that

$$\begin{aligned} \frac{d}{dt} &= \frac{d\tilde{t}}{dt} \frac{d}{d\tilde{t}}, \\ \frac{d^2}{dt^2} &= \frac{d^2\tilde{t}}{dt^2} \frac{d}{d\tilde{t}} + \left(\frac{d\tilde{t}}{dt}\right)^2 \frac{d^2}{d\tilde{t}^2}, \end{aligned}$$

we obtain (working in a coordinate basis)

$$\begin{aligned} T^\alpha \nabla_\alpha T^\mu &= \frac{d^2 x^\mu}{dt^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{dt} \frac{dx^\beta}{dt} \\ &= \frac{d^2 \tilde{t}}{dt^2} \frac{dx^\mu}{d\tilde{t}} + \left(\frac{d\tilde{t}}{dt}\right)^2 \frac{d^2 x^\mu}{d\tilde{t}^2} + \Gamma^\mu_{\alpha\beta} \left(\frac{d\tilde{t}}{dt}\right)^2 \frac{dx^\alpha}{d\tilde{t}} \frac{dx^\beta}{d\tilde{t}} \\ &= \frac{d^2 \tilde{t}}{dt^2} \frac{dx^\mu}{d\tilde{t}} + \left(\frac{d\tilde{t}}{dt}\right)^2 \left[\frac{d^2 x^\mu}{d\tilde{t}^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\tilde{t}} \frac{dx^\beta}{d\tilde{t}} \right] \end{aligned}$$

and

$$\alpha T^\mu = \alpha \frac{dx^\mu}{dt} = \alpha \frac{d\tilde{t}}{dt} \frac{dx^\mu}{d\tilde{t}}.$$

These two expressions are equal by assumption, which gives

$$\left[\frac{d^2 \tilde{t}}{dt^2} - \alpha(t) \frac{d\tilde{t}}{dt} \right] \frac{dx^\mu}{d\tilde{t}} + \left(\frac{d\tilde{t}}{dt}\right)^2 \left[\frac{d^2 x^\mu}{d\tilde{t}^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\tilde{t}} \frac{dx^\beta}{d\tilde{t}} \right] = 0.$$

We see that $\tilde{t}$ will be an affine parameter only if

$$\frac{d^2 \tilde{t}}{dt^2} - \alpha(t) \frac{d\tilde{t}}{dt} = 0, \quad \frac{d\tilde{t}}{dt} \neq 0,$$

and it is readily seen that we can get this by choosing $\tilde{t}$ such that

$$\frac{d\tilde{t}}{dt} = \exp \left[\int^t \alpha(t') dt' \right].$$

Thus it is always possible to reparametrize in order for equation (3.3.1) to be satisfied.

(b) If t is an affine parameter, then

$$T^\alpha \nabla_\alpha T^\mu = \frac{d^2 \tilde{t}}{dt^2} \frac{dx^\mu}{d\tilde{t}} + \left(\frac{d\tilde{t}}{dt} \right)^2 \left[\frac{d^2 x^\mu}{d\tilde{t}^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\tilde{t}} \frac{dx^\beta}{d\tilde{t}} \right] = 0.$$

If we take $\tilde{t}$ to be an affine parameter as well, then the expression in brackets vanishes, and

$$\frac{d^2 \tilde{t}}{dt^2} \frac{dx^\mu}{d\tilde{t}} = 0.$$

Then since $x^\mu(\tilde{t})$ is arbitrary, we obtain

$$\frac{d^2 \tilde{t}}{dt^2} = 0 \implies \tilde{t} = at + b,$$

where a and b are integration constants. □

Problem 3.6

The metric of Euclidean $\mathbb{R}^3$ in spherical coordinates is $ds^2 = dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2)$.

- (a) Calculate the Christoffel components $\Gamma^\sigma_{\mu\nu}$ in this coordinate system.
- (b) Write down the components of the geodesic equation in this coordinate system and verify that the solutions correspond to straight lines in Cartesian coordinates.

Solution. (a) A straightforward calculation shows that the nonzero Christoffel symbols are

$$\begin{aligned}\Gamma^r_{\theta\theta} &= -r, & \Gamma^\theta_{\phi\phi} &= -\sin\theta \cos\theta, \\ \Gamma^r_{\phi\phi} &= -r \sin^2\theta, & \Gamma^\phi_{r\phi} &= \frac{1}{r}, \\ \Gamma^\theta_{r\theta} &= \frac{1}{r}, & \Gamma^\phi_{\theta\phi} &= \cot\theta.\end{aligned}$$

- (b) The components of the geodesic equation are then

$$\begin{aligned}\ddot{r} - r\dot{\theta}^2 - r\dot{\phi}^2 \sin^2\theta &= 0, \\ \ddot{\theta} + \frac{2\dot{r}\dot{\theta}}{r} - \dot{\phi}^2 \sin\theta \cos\theta &= 0, \\ \ddot{\phi} + \frac{2\dot{r}\dot{\phi}}{r} + 2\dot{\theta}\dot{\phi} \cot\theta &= 0,\end{aligned}$$

where dots denote differentiation with respect to t . I'm unsure whether it is possible to solve these equations as they stand, but here's one approach. Suppose $x^\alpha(t) = (r(t), \theta(t), \phi(t))$ is a geodesic, so that the equations above are satisfied. Then we can transform back to Cartesian coordinates via

$$\begin{aligned}r &= \sqrt{x^2 + y^2 + z^2}, \\ \cos\theta &= \frac{z}{\sqrt{x^2 + y^2 + z^2}}, \\ \tan\phi &= y/x,\end{aligned}$$

and in this way obtain a system of ordinary differential equations in x, y, z (see the Mathematica notebook that follows). Any solution to *these* is a straight line, as we wanted. **This approach is dumb. It is easier to solve the geodesic equation in Cartesian coordinates, and then transform the solution to spherical coordinates. Alternatively, write down a parametrization of a straight line in spherical coordinates and check that it satisfies the geodesic equation.** $\square$

Problem 3.6

(a)

```
In[1]:= xx = {r,  $\theta$ ,  $\phi$ }; (* coordinates *)
      nD = Length[xx]; (* number of dimensions *)

In[3]:= g = {{1, 0, 0}, {0, r2, 0}, {0, 0, r2 Sin[ $\theta$ ]2}}; (* g[a,b]  $\equiv$  gab *)
      gin = Inverse[g]; (* gin[a,b]  $\equiv$  gab *)
      g // MatrixForm
      gin // MatrixForm
```

Out[5]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2[\theta] \end{pmatrix}$$

Out[6]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & \frac{\csc^2[\theta]}{r^2} \end{pmatrix}$$

Return nonzero components of the given tensor

```
In[7]:= nonzero[tensor_] := Module[{
  inds = SparseArray[tensor] ["NonzeroPositions"],
  vals = SparseArray[tensor] ["NonzeroValues"]
},
Module[{
  (* replace indices with symbols, e.g. {1,1,2}  $\rightarrow$  {t,t,r} *)
  syms = Table[
    AssociationThread[Table[i, {i, nD}]  $\rightarrow$  xx] /@ inds[[j]], {j, Length[inds]}]
  },
  Transpose[{syms, vals}]
]
(* Transpose[{inds,vals}] *)
]
```

```
In[8]:= chr = Table[
  Sum[ $\frac{1}{2}$  gin[[c, d]]  $\times$ 
    (D[g[[a, d]], xx[[b]]] + D[g[[b, d]], xx[[a]]] - D[g[[a, b]], xx[[d]]], {d, nD}],
  {c, nD}, {a, nD}, {b, nD}]; (* chr[[c,a,b]]  $\equiv$   $\Gamma^c_{ab}$  *)
```

In[9]:= nonzero[chr]

Out[9]= $\left\{ \left\{ \{r, \theta, \theta\}, -r \right\}, \left\{ \{r, \phi, \phi\}, -r \sin[\theta]^2 \right\}, \left\{ \{\theta, r, \theta\}, \frac{1}{r} \right\}, \right.$
 $\left. \left\{ \{\theta, \theta, r\}, \frac{1}{r} \right\}, \left\{ \{\theta, \phi, \phi\}, -\cos[\theta] \sin[\theta] \right\}, \left\{ \{\phi, r, \phi\}, \frac{1}{r} \right\}, \right.$
 $\left. \left\{ \{\phi, \theta, \phi\}, \cot[\theta] \right\}, \left\{ \{\phi, \phi, r\}, \frac{1}{r} \right\}, \left\{ \{\phi, \phi, \theta\}, \cot[\theta] \right\} \right\}$

(b)

In[10]:= $r[t_]:= \sqrt{x[t]^2 + y[t]^2 + z[t]^2};$
 $\theta[t_]:= \text{ArcCos}\left[\frac{z[t]}{\sqrt{x[t]^2 + y[t]^2 + z[t]^2}}\right];$
 $\phi[t_]:= \text{ArcTan}\left[\frac{y[t]}{x[t]}\right];$

In what follows, we'll assume $x(t)^2 + y(t)^2 + z(t)^2 > 0$ and also $x(t)^2 + y(t)^2 > 0$; spherical coordinates are singular along the z axis, anyway

In[13]:= Assuming[$x[t]^2 + y[t]^2 + z[t]^2 > 0$,
 $r'[t] - r[t] \theta'[t]^2 - r[t] \phi'[t]^2 \sin[\theta[t]]^2 == 0 // \text{Simplify}$]

Assuming[$x[t]^2 + y[t]^2 + z[t] > 0 \&\& x[t]^2 + y[t]^2 > 0$,
 $\theta''[t] + \frac{2 r'[t] \times \theta'[t]}{r[t]} - \phi'[t]^2 \sin[\theta[t]] \cos[\theta[t]] == 0 // \text{Simplify}$]

Assuming[$x[t]^2 + y[t]^2 + z[t]^2 > 0 \&\& x[t]^2 + y[t]^2 > 0$,
 $\phi''[t] + \frac{2 r'[t] \times \phi'[t]}{r[t]} + 2 \theta'[t] \times \phi'[t] \cot[\theta[t]] == 0 // \text{Simplify}$]

Out[13]= $x[t] x''[t] + y[t] y''[t] + z[t] z''[t] == 0$

Out[14]= $(x[t]^2 + y[t]^2) z''[t] == z[t] \times (x[t] x''[t] + y[t] y''[t])$

Out[15]= $y[t] x''[t] == x[t] y''[t]$

In[16]:= DSolve[{
 $x[t] x''[t] + y[t] y''[t] + z[t] z''[t] == 0,$
 $(x[t]^2 + y[t]^2) z''[t] == z[t] \times (x[t] x''[t] + y[t] y''[t]),$
 $y[t] x''[t] == x[t] y''[t]$
 $\}, \{x[t], y[t], z[t]\}, t]$

Out[16]= $\{\{x[t] \rightarrow c_1 + t c_2, y[t] \rightarrow c_3 + t c_4, z[t] \rightarrow c_5 + t c_6\}\}$

Problem 3.7

As shown in problem 2, an arbitrary Lorentz metric on a two-dimensional manifold locally can always be put in the form $ds^2 = \Omega^2(x, t)[-dt^2 + dx^2]$. Calculate the Riemann curvature tensor of this metric (a) by the coordinate basis methods of section 3.4a and (b) by the tetrad methods of section 3.4b.

Solution.

□

Problem 3.8

Using the antisymmetry of $\omega_{\lambda\mu\nu}$ in μ and ν , equation (3.4.15), show that

$$\omega_{\lambda\mu\nu} = 3\omega_{[\lambda\mu\nu]} - 2\omega_{[\mu\nu]\lambda}.$$

Use this formula together with equation (3.4.23) to solve for $\omega_{\lambda\mu\nu}$ in terms of commutators (or antisymmetrized derivatives) of the orthonormal basis vectors.

Solution.

□

Verifying some assertions in Chapter 4

Show that in the Newtonian limit, the equation of motion of a perfect fluid subject to no external forces reduces to the conservation of mass and Euler's equation.

Solution. Rewrite Eq. (4.2.10) in the form

$$T_{ab} = (P + \rho)u_a u_b + P\eta_{ab}.$$

Then the equation of motion of the fluid can be written

$$0 = T^{ab}{}_{,a} = (P + \rho)_{,a} u^a u^b + (P + \rho) u^a{}_{,a} u^b + (P + \rho) u^a u^b{}_{,a} + P_{,a} \eta^{ab}. \quad (1)$$

Project this equation parallel to u^b to obtain

$$\begin{aligned} 0 &= -(\cancel{P} + \rho)_{,a} u^a - (P + \rho) u^a{}_{,a} + (P + \rho) \cancel{u^a u^b{}_{,a} u_b} + \cancel{P_{,a} \eta^{ab}} \\ &= -\rho_{,a} u^a - (P + \rho) u^a{}_{,a} \end{aligned} \quad (2)$$

where we've used the fact that $u_b u^b{}_{,a} = \frac{1}{2}(u^b u_b)_{,a} = 0$. This reproduces Eq. (4.2.12) in Wald. Next, project the equation of motion perpendicular to u^b . Since the first two terms on the right-hand side of Eq. (1) are parallel to u^b , we obtain

$$\begin{aligned} 0 &= (\eta_{bc} + u_b u_c)(P + \rho) u^a u^b{}_{,a} + (\eta_{bc} + u_b u_c) P_{,a} \eta^{ab} \\ &= (P + \rho) u^a u_{c,a} + (P + \rho) u^a u_c \cancel{u_b u^b{}_{,a}} + P_{,c} + u^a u_c P_{,a} \\ &= (P + \rho) u^a u_{c,a} + (\delta_c^a + u_c u^a) P_{,a} \end{aligned} \quad (3)$$

which is equivalent to Eq. (4.2.13) in Wald. We now take the Newtonian limit of these equations. We have $P \ll \rho$ and $u^\mu = (1, v^i)$, where the three-velocity v^i is assumed to be small. Thus Eq. (2) reduces to

$$\begin{aligned} 0 &= \rho_{,\mu} u^\mu + \rho u^\mu{}_{,\mu} \\ &= (\rho_{,0} + v^i \rho_{,i}) + \rho v^i{}_{,i} \\ &= \rho_{,0} + (\rho v^i)_{,i} \\ &\equiv \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}), \end{aligned}$$

which expresses the conservation of mass in differential form. Now we examine Eq. (3) in the same limit, which gives

$$\begin{aligned} 0 &= \rho u^\nu u_{\mu,\nu} + P_{,\mu} \\ &= \rho \{u^\nu v_{i,\nu}\} + P_{,\mu}. \end{aligned}$$

The spatial part of this equation is equivalent to

$$0 = \rho \left\{ \frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \vec{\nabla}) \vec{v} \right\} + \vec{\nabla} P,$$

which is the Euler equation. Meanwhile, the zeroth component yields $\partial P / \partial t = 0$. $\square$

Suppose you have a scalar field ϕ satisfying the Klein-Gordon equation,

$$\partial^a \partial_a \phi - m^2 \phi = 0,$$

with stress-energy tensor given by

$$T_{ab} = \partial_a \phi \partial_b \phi - \frac{1}{2} \eta_{ab} (\partial^c \phi \partial_c \phi + m^2 \phi^2).$$

Check that the stress-energy tensor T_{ab} satisfies the weak energy condition, i.e. $T_{ab} v^a v^b \geq 0$ for all timelike vectors v^a , and verify that it is conserved.

Solution. It is trivial to show that $\partial^a T_{ab} = 0$ using the field equation, so let us consider the weak energy condition. Suppose v^a is timelike. Then there exists a frame such that $v^\mu = (1, 0, 0, 0)$, and using this frame we find

$$\begin{aligned} T_{\mu\nu} v^\mu v^\nu &= T_{00} = (\partial_0 \phi)^2 - \frac{1}{2} \eta_{00} (\partial^c \phi \partial_c \phi + m^2 \phi^2) \\ &= (\partial_0 \phi)^2 + \partial^c \phi \partial_c \phi \\ &= \cancel{(\partial_0 \phi)^2} + [-(\cancel{\partial_0 \phi})^2 + (\partial_i \phi)^2] \\ &\equiv |\vec{\nabla} \phi|^2 \geq 0, \end{aligned}$$

as we wanted to show.

There is a nice demonstration in *A Relativist's Toolkit* of energy conditions for a perfect fluid. Suppose we have an orthonormal basis $\{\hat{e}_\mu^\alpha\}$ such that the stress-energy tensor admits the decomposition

$$T^{\alpha\beta} = \rho \hat{e}_0^\alpha \hat{e}_0^\beta + p_1 \hat{e}_1^\alpha \hat{e}_1^\beta + p_2 \hat{e}_2^\alpha \hat{e}_2^\beta + p_3 \hat{e}_3^\alpha \hat{e}_3^\beta.$$

Then let v^a be arbitrary, so that $v^\alpha = \gamma(\hat{e}_0^\alpha + a \hat{e}_1^\alpha + b \hat{e}_2^\alpha + c \hat{e}_3^\alpha)$ and $\gamma = (1 - a^2 - b^2 - c^2)^{-1/2}$, where a , b , and c are arbitrary functions of the coordinates subject to the constraint $a^2 + b^2 + c^2 < 1$. A direct calculation then shows that the weak energy condition

$$T_{\alpha\beta} v^\alpha v^\beta \geq 0$$

is equivalent to

$$\rho + a^2 p_1 + b^2 p_2 + c^2 p_3 \geq 0.$$

In particular, if we choose $a = b = c = 0$, which amounts to the choice of an observer comoving with the fluid, then we get $\rho \geq 0$. A similar argument shows that $\rho + p_i > 0$ for each i . However, this time the weak energy condition becomes something to impose on ρ , rather than something that follows automatically, as in the case of the scalar field satisfying the Klein-Gordon equation. It's also worth noting that ρ is a quantity defined using a preferred frame, i.e., the one comoving with the fluid. □

Problem 4.1

Show that Maxwell's equation (4.3.12) implies strict charge conservation, $\nabla_a j^a = 0$.

Solution.

□

Problem 4.2

- (a) Let α be a p -form on an n -dimensional oriented manifold with metric g_{ab} , i.e., $\alpha_{a_1 \dots a_p}$ is a totally antisymmetric tensor field (see appendix B). We define the *dual*, ${}^*\alpha$, of α by

$${}^*\alpha_{b_1 \dots b_{n-p}} = \frac{1}{p!} \alpha^{a_1 \dots a_p} \epsilon_{a_1 \dots a_p b_1 \dots b_{n-p}},$$

where $\epsilon_{a_1 \dots a_n}$ is the natural volume element on M , i.e., the totally antisymmetric tensor field determined up to sign by equation (B.2.9). Show that ${}^{**}\alpha = (-1)^{s+p(n-p)}\alpha$, where s is the number of minuses occurring in the signature of g_{ab} .

- (b) Show that in differential forms notation (see appendix B), Maxwell's equations (4.3.12) and (4.3.13) can be written as

$$d^*\mathbf{F} = 4\pi^*j$$

$$d\mathbf{F} = 0.$$

Note that if we apply Stokes's theorem (see appendix B) to the first equation, we obtain $\int_{\Sigma} {}^*j = (1/4\pi) \int_S {}^*F$, where Σ is a three-dimensional hypersurface with two-dimensional boundary S . But $-\int_{\Sigma} {}^*j = -\int_{\Sigma} j^a t_a d\Sigma$ is just the total electric charge e in the volume Σ , where t^a is the unit normal to Σ , and $-\int_S {}^*F = \int_S E^a n_a dA$ is just the integral of the normal component of $E_a = F_{ab} t^b$ on S . Thus, Gauss's law of electromagnetism continues to hold in curved spacetime.

- (c) Define for each $\beta \in [0, 2\pi]$ the tensor field $\tilde{F}_{ab} = F_{ab} \cos \beta + {}^*F_{ab} \sin \beta$. We call $\tilde{F}_{ab}$ a *duality rotation* of F_{ab} by “angle” β . It follows immediately from part (b) that if F_{ab} satisfies the source-free Maxwell's equations ($j^a = 0$), then so does $\tilde{F}_{ab}$. Show that the stress-energy, T_{ab} , of the solution is the same as that of F_{ab} .

Solution.

□

Problem 4.3

- (a) Derive equation (4.4.24).
- (b) Show that the “gravitational electric and magnetic fields” $\vec{E}$ and $\vec{B}$ inside a spherical shell of mass M and radius R (with $M \ll R$) slowly rotating with angular velocity $\vec{\omega}$ are

$$\vec{E} = 0, \quad \vec{B} = \frac{2}{3} \frac{M}{R} \vec{\omega}.$$

- (c) An observer at rest at the center of the shell of part (b) parallelly propagates along his (geodesic) world line a vector S^a with $S^a u_a = 0$, where u^a is the tangent to his world line. Show that the inertial components, $\vec{S}$, precess according to $d\vec{S}/dt = \vec{\Omega} \times \vec{S}$, where $\vec{\Omega} = 2\vec{B} = \frac{4}{3}(M/R)\vec{\omega}$. This effect, first analyzed by Thirring and Lense (1918) and discussed further by Brill and Cohen (1966), may be interpreted as a “dragging of inertial frames” cause by the rotating shell. At the center of the shell the local standard of “nonrotating”, defined by parallel propagation along a geodesic, is changed from what it would be without the shell, in a manner in accord with Mach’s principle.

Solution.

□

Problem 4.4

Starting with equation (3.4.5) for R_{ab} , derive the formula, equation (4.4.51), for $R_{ab}^{(2)}$ by substituting $\eta_{ab} + \gamma_{ab}$ for g_{ab} and keeping precisely the terms quadratic in γ_{ab} .

Solution.

□

Problem 4.5

Let T_{ab} be a symmetric, conserved tensor field (i.e., $T_{ab} = T_{ba}$, $\partial^a T_{ab} = 0$), in Minkowski spacetime. Show that there exists a tensor field U_{acbd} with the symmetries $U_{acbd} = U_{[acbd]} = U_{ac[bd]} = U_{bdac}$ such that $T_{ab} = \partial^c \partial^d U_{acbd}$.

Solution.

□

Problem 4.6

As discussed in the text, in general relativity no meaningful expression is known for the local stress-energy of the gravitational field. However, a four-index tensor T_{abcd} can be constructed out of the curvature in a manner closely analogous to the way in which the stress tensor of the electromagnetic field is constructed out of F_{ab} (eq. [4.2.27]). We define the *Bel-Robinson tensor* in terms of the Weyl tensor by

$$\begin{aligned} T_{abcd} &= C_{aecf} C_b{}^e{}_d{}^f + \frac{1}{4} \epsilon_{ae}{}^{hi} \epsilon_b{}^{ej}{}_{ik} \\ &= C_{aecf} C_b{}^e{}_d{}^f - \frac{3}{2} g_{a[b} C_{jk]cf} C^{jk}{}_d{}^f, \end{aligned}$$

where ϵ_{abcd} is defined in appendix B and equation (B.2.13) was used. It follows that $T_{abcd} = T_{(abcd)}$. (This is established most easily from the spinor decomposition of the Weyl tensor given in chapter 13.)

- (a) Show that $T^a{}_{acd} = 0$.
- (b) Using the Bianchi identity (3.2.16), show that in vacuum, $R_{ab} = 0$, we have $\nabla^a T_{abcd} = 0$.

Solution.

□

Problem 4.7

- (a) Show that the total energy E , equation (4.4.55), is time independent, i.e., the value of E is unchanged if the integral is performed over a time translate, Σ' , of Σ .
- (b) Let ξ_a be a gauge transformation which vanishes outside a bounded region of space. Show that $E[\gamma_{ab}] = E[\gamma_{ab} + 2\partial_{(a}\xi_{b)}]$ by comparing them with $E[\gamma_{ab} + 2\partial_{(a}\xi'_{b)}]$ where ξ'_a is a new gauge transformation which agrees with ξ_a in a neighborhood of the hyperplane Σ but vanishes in a neighborhood of another hyperplane Σ' .

Solution.

□

Problem 4.8

Two point masses of mass M are attached to the ends of a spring of spring constant K . The spring is set into oscillation. In the quadrupole approximation, equation (4.4.58), what fraction of the energy of oscillation of the spring is radiated away during one cycle of oscillation?

Solution.

□

Problem 4.9

A binary star system consists of two stars of mass M and of negligible size in a nearly Newtonian circular orbit of radius R around each other. Assuming the validity of equation (4.4.58) for this system, calculate the rate of increase of the orbital frequency due to emission of gravitational radiation.

Solution.

□

Problem 6.1

Let M be a three-dimensional manifold possessing a spherically symmetric Riemannian metric with $\nabla_a r \neq 0$, where r is defined by equation (6.1.3).

- (a) Show that a new “isotropic” radial coordinate $\tilde{r}$ can be introduced so that the metric takes the form $ds^2 = H(\tilde{r})[d\tilde{r}^2 + \tilde{r}^2 d\Omega^2]$. (This shows that every spherically symmetric three-dimensional space is conformally flat.)
- (b) Show that in isotropic coordinates the Schwarzschild metric is

$$ds^2 = -\frac{(1 - M/2\tilde{r})^2}{(1 + M/2\tilde{r})^2} dt^2 \left(1 + \frac{M}{2\tilde{r}}\right)^4 [d\tilde{r}^2 + \tilde{r}^2 d\Omega^2].$$

Solution.

□

Problem 6.2

Calculate the Ricci tensor, R_{ab} , for a static, spherically symmetric spacetime, equation (6.1.5), using the coordinate component method of section 3.4a. Compare the amount of labor involved with that of the tetrad approach given in the text.

Solution.

□

Problem 6.3

Consider the source-free ($j^a = 0$) Maxwell's equations (4.3.12) and (4.3.13) in a static, spherically symmetric spacetime, equation (6.1.5).

- (a) Argue that the general form of a Maxwell tensor which shares the static and spherical symmetries of the spacetime is $F_{ab} = 2A(r)(e_0)_{[a}(e_1)_{b]} + 2B(r)(e_2)_{[a}(e_3)_{b]}$, where the $(e_\mu)_a$ are defined by equation (6.1.6).
- (b) Show that if $B(r) = 0$, the general solution of Maxwell's equations with the form of part (a) is $A(r) = -q/r^2$, where q may be interpreted as the total charge. [The solution obtained with $B(r) \neq 0$ is a “duality rotation” of this solution, representing the field of a magnetic monopole.]
- (c) Write down and solve Einstein's equation, $G_{ab} = 8\pi T_{ab}$, with the electromagnetic stress-energy tensor corresponding to the solution of part (b). Show that the general solution is the *Reissner-Nordstrom metric*,

$$ds^2 = -\left(1 - \frac{2M}{r} + \frac{q^2}{r^2}\right)dt^2 + \left(1 - \frac{2M}{r} + \frac{q^2}{r^2}\right)^{-1}dr^2 + r^2 d\Omega^2.$$

Solution.

□

Problem 6.4

Let (M, g_{ab}) be a stationary spacetime with timelike Killing field ξ^a . Let $V^2 = -\xi^a \xi_a$.

- (a) Show that the acceleration $a^b = u^a \nabla_a u^b$ of a stationary observer is given by $a^b = \nabla^b \ln V$.
- (b) Suppose in addition that (M, g_{ab}) is asymptotically flat, i.e., that there exist coordinates t, x, y, z [with $\xi^a = (\partial/\partial t)^a$] such that the components of g_{ab} approach $\text{diag}(-1, 1, 1, 1)$ as $r \rightarrow \infty$, where $r = (x^2 + y^2 + z^2)^{1/2}$. (See chapter 11 for further discussion of asymptotic flatness.) As in the case of the Schwarzschild metric, the “energy as measured at infinity” of a particle of mass m is held stationary by a (massless) string, with the other end of the string held by a stationary observer at large r . Let F denote the force exerted by the string on the particle. According to part (a) we have $F = mV^{-1}[\nabla_a V \nabla^a V]^{1/2}$. Use conservation-of-energy arguments to show that the force exerted by the observer at infinity on the other end of the string is $F_\infty = VF$. Thus, the magnitude of the force exerted at infinity differs from the force exerted locally by the redshift factor.

Solution.

□

Problem 6.5

Derive the formula, equation (6.3.45), for the general relativistic time delay.

Solution.

□

Problem 6.6

Show that any particle (not necessarily in geodesic motion) in region II ($r < 2M$) of the extended Schwarzschild spacetime, Figure 6.9, must decrease its radial coordinate at a rate given by $|dr/d\tau| \geq [2M/r - 1]^{1/2}$. Hence, show that the maximum lifetime of any observer in region II is $\tau = \pi M [\sim 10^{-5}(M/M_\odot) \text{ s}]$, i.e., any observer in region II will be pulled into the singularity at $r = 0$ within this proper time. Show that this maximum time is approached by freely falling (i.e., geodesic) motion from $r = 2M$ with $E \rightarrow 0$.

Solution.

□

Carroll, Spacetime and Geometry

Proof of Eq. (3.33)

From the definition, we have

$$\Gamma^\mu_{\mu\nu} = \frac{1}{2} g^{\mu\rho} (g_{\nu\rho,\mu} + g_{\rho\mu,\nu} - g_{\mu\nu,\rho}).$$

But since

$$g^{\mu\rho} g_{\nu\rho,\mu} = g^{\mu\rho} g_{\mu\nu,\rho}$$

by the symmetry of $g_{\mu\nu}$ and a relabeling of dummy indices, we have

$$\Gamma^\mu_{\mu\nu} = \frac{1}{2} g^{\mu\rho} g_{\rho\mu,\nu}.$$

The contracted Christoffel symbol can be written in terms of the metric determinant $G := \det g_{ij}$ as follows. First note that for any choice of μ , we can write G as an expansion in minors:

$$G = \sum_{\nu} (-1)^{\mu+\nu} g_{\mu\nu} M_{\mu\nu},$$

where $M_{\mu\nu}$ is the (μ, ν) -minor of the metric. (To avoid confusion in what follows, we will explicitly indicate which indices are summed over.) We also have

$$g^{\mu\nu} = \frac{1}{G} M_{\mu\nu} (-1)^{\mu+\nu},$$

with no sum on the indices. Regarding G as a function of $g_{\mu\nu}$, we have

$$\begin{aligned} \frac{\partial G}{\partial x^\lambda} &= \sum_{\mu,\nu} \frac{\partial g_{\mu\nu}}{\partial x^\lambda} \frac{\partial G}{\partial g_{\mu\nu}} \\ &= \sum_{\mu,\nu} g_{\mu\nu,\lambda} \frac{\partial}{\partial g_{\mu\nu}} \left(\sum_{\gamma} (-1)^{\mu+\gamma} g_{\mu\gamma} M_{\mu\gamma} \right) \\ &= \sum_{\mu,\nu} g_{\mu\nu,\lambda} \sum_{\gamma} (-1)^{\mu+\gamma} \delta_{\gamma\nu} M_{\mu\gamma} \\ &= \sum_{\mu,\nu} g_{\mu\nu,\lambda} (-1)^{\mu+\nu} M_{\mu\nu} \\ &= \sum_{\mu,\nu} g_{\mu\nu,\lambda} G g^{\mu\nu} \end{aligned}$$

where in the second line, we choose the same index μ in the calculation of G as is used in the first sum. (A fairly subtle point is that $\partial M_{\mu\gamma} / \partial g_{\mu\nu} = 0$ since, regarding μ as some fixed index, $M_{\mu\gamma}$ does not depend on any of the components $g_{\mu\nu}$ in the same row by definition.) Returning to the Einstein summation convention, we conclude that $g^{\mu\nu} g_{\mu\nu,\lambda} = G^{-1} G_{,\lambda}$, and finally that

$$\boxed{\Gamma^\mu_{\mu\nu} = (\ln G^{1/2})_{,\lambda}}$$

Exercise 3.1

Verify these consequences of metric compatibility ($\nabla_\sigma g_{\mu\nu} = 0$):

$$\nabla_\sigma g^{\mu\nu} = 0$$

$$\nabla_\lambda \epsilon_{\mu\nu\rho\sigma} = 0.$$

Solution. For the first claim, we note that

$$\begin{aligned} 0 &= \nabla_\sigma \delta_\beta^\alpha \\ &= \nabla_\sigma (g^{\alpha\mu} g_{\mu\beta}) \\ &= (\nabla_\sigma g^{\alpha\mu}) g_{\mu\beta} + g^{\alpha\mu} (\nabla_\sigma g_{\mu\beta}). \end{aligned}$$

Multiplying by $g^{\beta\gamma}$ and relabeling indices gives

$$\nabla_\sigma g^{\mu\nu} = 0,$$

as we wanted to show. This means that we can freely raise and lower indices on tensors being acted on by a covariant (metric-compatible) derivative operator.

For the second claim, we use the definitions

$$\begin{aligned} \epsilon_{\mu_1\mu_2\cdots\mu_n} &= \sqrt{|g|} \tilde{\epsilon}_{\mu_1\mu_2\cdots\mu_n} \\ \epsilon^{\mu_1\mu_2\cdots\mu_n} &= \frac{1}{\sqrt{|g|}} \tilde{\epsilon}^{\mu_1\mu_2\cdots\mu_n} \end{aligned}$$

and the result (2.71) for the contraction of Levi-Civita tensors using a metric of signature $s = 1$:

$$\epsilon^{\mu_1\mu_2\cdots\mu_p\alpha_1\cdots\alpha_{n-p}} \epsilon_{\mu_1\mu_2\cdots\mu_p\beta_1\cdots\beta_{n-p}} = (-1)^p p! (n-p)! \delta_{\beta_1}^{[\alpha_1} \cdots \delta_{\beta_{n-p}}^{\alpha_{n-p}]}$$

Then

$$\begin{aligned} 0 &= \nabla_\lambda (\epsilon_{\mu\nu\rho\sigma} \epsilon^{\mu'\nu'\rho'\sigma'}) \\ &= (\nabla_\lambda \epsilon_{\mu\nu\rho\sigma}) \epsilon^{\mu'\nu'\rho'\sigma'} + \epsilon_{\mu\nu\rho\sigma} (\nabla_\lambda \epsilon^{\mu'\nu'\rho'\sigma'}), \end{aligned}$$

where the first line follows since the expression in parentheses is a sum of Kronecker deltas. Multiplying through by $\epsilon_{\mu'\nu'\rho'\sigma'}$ gives (since $n = 4$, and we are contracting over $p = 4$ indices)

$$0 = -4! (\nabla_\lambda \epsilon_{\mu\nu\rho\sigma}) + \epsilon_{\mu'\nu'\rho'\sigma'} \epsilon_{\mu\nu\rho\sigma} (\nabla_\lambda \epsilon^{\mu'\nu'\rho'\sigma'}).$$

For the second term, use metric compatibility to simultaneously raise and lower indices, then

apply (2.71):

$$\begin{aligned}
\epsilon_{\mu'\nu'\rho'\sigma'}\epsilon_{\mu\nu\rho\sigma}(\nabla_\lambda\epsilon^{\mu'\nu'\rho'\sigma'}) &= \epsilon_{\mu'\nu'\rho'\sigma'}\epsilon^{\mu\nu\rho\sigma}(\nabla_\lambda\epsilon_{\mu'\nu'\rho'\sigma'}) \\
&= -4!\delta_\mu^{[\mu'}\delta_\nu^{\nu'}\delta_\rho^{\rho'}\delta_\sigma^{\sigma']}\nabla_\lambda\epsilon_{\mu'\nu'\rho'\sigma'} \\
&= -4!\delta_\mu^{\mu'}\delta_\nu^{\nu'}\delta_\rho^{\rho'}\delta_\sigma^{\sigma'}\nabla_\lambda\epsilon_{\mu'\nu'\rho'\sigma'} \\
&= -4!\nabla_\lambda\epsilon_{\mu\nu\rho\sigma},
\end{aligned}$$

where the third line follows because the contraction involves only the part of $\nabla_\lambda\epsilon_{\mu'\nu'\rho'\sigma'}$ that is antisymmetric in the primed indices, and the Levi-Civita tensor is completely antisymmetric by definition. Substituting into our previous expression and clearing out an overall factor, we get

$$\nabla_\lambda\epsilon_{\mu\nu\rho\sigma} = 0,$$

as we wanted to show. Note that the same reasoning applies to the covariant derivative of a Levi-Civita tensor with an arbitrary number of indices. $\square$

Exercise 3.2

You are familiar with the operations of gradient $(\nabla\phi)$, divergence $(\nabla \cdot \mathbf{V})$, and curl $(\nabla \times \mathbf{V})$ in ordinary vector analysis in three-dimensional Euclidean space. Using covariant derivatives, derive formulae for these operations in spherical polar coordinates $\{r, \theta, \phi\}$ defined by

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

Compare your results to those in Jackson (1999) or an equivalent text. Are they identical? Should they be?

Solution. We define

$$\text{grad } \phi = (\nabla^i \phi) \partial_i$$

$$\text{div } \mathbf{X} = \nabla_i X^i$$

$$\text{curl } \mathbf{X} = \epsilon^{ijk} (\nabla_i X_j) \partial_k$$

where $\{\partial_i\}$ are the coordinate basis vectors. In Cartesian coordinates, the covariant derivative reduces to an ordinary (partial) derivative and $\epsilon^{ijk} = \tilde{\epsilon}^{ijk}$, so this definition is equivalent to the standard one. Before proceeding, we will derive a few expressions that are easier to work with, since they turn the covariant derivatives into ordinary ones. We begin with the observation that

$$\begin{aligned} \text{div } \mathbf{X} &= X^i_{;i} \\ &= X^i_{,i} + \Gamma^i_{ik} X^k \end{aligned}$$

and using the expression (3.33) for the contracted Christoffel symbol, we get

$$\begin{aligned} \text{div } \mathbf{X} &= X^i_{,i} + \Gamma^i_{ik} X^k \\ &= X^i_{,i} + G^{-1/2} (G^{1/2})_{,k} X^k. \end{aligned}$$

Now since

$$(G^{1/2} X^i)_{,i} = (G^{1/2})_{,i} X^i + G^{1/2} X^i_{,i},$$

we have

$$\boxed{\text{div } \mathbf{X} = G^{-1/2} (G^{1/2} X^i)_{,i}}$$

which was what we wanted to derive. Then since $\nabla^i f = g^{ij} f_{;j} = g^{ij} f_{,j}$, we get the following expression for the Laplacian of a scalar function:

$$\boxed{\nabla^2 f = G^{-1/2} (G^{1/2} g^{ij} f_{,j})_{,i}}$$

Now, the Euclidean metric in spherical polar coordinates is

$$(g_{ij}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}.$$

To make our expressions agree with those in the standard references, we consider the orthonormal frame field $\{e_{\hat{r}}, e_{\hat{\theta}}, e_{\hat{\phi}}\}$ defined by

$$e_{\hat{r}} = \frac{\partial}{\partial r}, \quad e_{\hat{\theta}} = \frac{1}{r} \frac{\partial}{\partial \theta}, \quad e_{\hat{\phi}} = \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}.$$

That this frame is orthonormal can be verified by inspection. With this we can write

$$\begin{aligned} \mathbf{X} &= X^{\hat{r}} e_{\hat{r}} + X^{\hat{\theta}} e_{\hat{\theta}} + X^{\hat{\phi}} e_{\hat{\phi}} \\ &= X^r \partial_r + X^\theta \partial_\theta + X^\phi \partial_\phi \\ &= X^r e^{\hat{r}} + (r X^\theta) e^{\hat{\theta}} + (r \sin \theta X^\phi) e^{\hat{\phi}}, \end{aligned}$$

which shows that

$$X^{\hat{r}} = X^r, \quad X^{\hat{\theta}} = r X^\theta, \quad X^{\hat{\phi}} = r \sin \theta X^\phi.$$

By inspection of the metric, we have $G = r^4 \sin^2 \theta$, so that $G^{-1/2} = 1/(r^2 \sin \theta)$. The gradient is easiest to compute:

$$\begin{aligned} \text{grad } f &= (\nabla^i f) \partial_i \\ &= (g^{ij} \nabla_i f) \partial_j \\ &= (g^{ij} \partial_i f) \partial_j \\ &= \left(g^{rr} \frac{\partial f}{\partial r} \right) \partial_r + \left(g^{\theta\theta} \frac{\partial f}{\partial \theta} \right) \partial_\theta + \left(g^{\phi\phi} \frac{\partial f}{\partial \phi} \right) \partial_\phi \\ &= \frac{\partial f}{\partial r} \partial_r + \frac{1}{r^2} \frac{\partial f}{\partial \theta} \partial_\theta + \frac{1}{r^2 \sin^2 \theta} \frac{\partial f}{\partial \phi} \partial_\phi. \end{aligned}$$

In terms of the orthonormal frame, we have

$$\boxed{\text{grad } f = \frac{\partial f}{\partial r} \partial_r + \frac{1}{r} \frac{\partial f}{\partial \theta} e_{\hat{\theta}} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} e_{\hat{\phi}}}$$

Next we compute the divergence of a vector field:

$$\begin{aligned}
\operatorname{div} \mathbf{X} &= G^{-1/2} (G^{1/2} X^i)_{,i} \\
&= \frac{1}{r^2 \sin \theta} \left[\frac{\partial}{\partial r} (r^2 \sin \theta X^r) + \frac{\partial}{\partial \theta} (r^2 \sin \theta X^\theta) + \frac{\partial}{\partial \phi} (r^2 \sin \theta X^\phi) \right] \\
&= \frac{1}{r^2 \sin \theta} \left[\frac{\partial}{\partial r} (r^2 \sin \theta X^{\hat{r}}) + \frac{\partial}{\partial \theta} (r \sin \theta X^{\hat{\theta}}) + \frac{\partial}{\partial \phi} (r X^{\hat{\phi}}) \right],
\end{aligned}$$

which upon simplifying gives

$$\boxed{\operatorname{div} \mathbf{X} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 X^{\hat{r}}) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta X^{\hat{\theta}}) + \frac{1}{r \sin \theta} \frac{\partial X^{\hat{\phi}}}{\partial \phi}},$$

as we wanted to show. Next we compute the Laplacian of a scalar function:

$$\begin{aligned}
\nabla^2 f &= G^{-1/2} (G^{1/2} g^{ij} f_{,j})_{,i} \\
&= G^{-1/2} \left[\frac{\partial}{\partial r} \left(G^{1/2} \frac{\partial f}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(\frac{G^{1/2}}{r^2} \frac{\partial f}{\partial \theta} \right) + \frac{\partial}{\partial \phi} \left(\frac{G^{1/2}}{r^2 \sin^2 \theta} \frac{\partial f}{\partial \phi} \right) \right] \\
&= \frac{1}{r^2 \sin \theta} \left[\frac{\partial}{\partial r} \left(r^2 \sin \theta \frac{\partial f}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{\partial}{\partial \phi} \left(\frac{1}{\sin \theta} \frac{\partial f}{\partial \phi} \right) \right],
\end{aligned}$$

and upon simplification we get

$$\boxed{\nabla^2 f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2}}.$$

To compute the curl of a vector field, we first note that by the result of Exercise 3.3 (and a bit of algebra, not shown here), the nonzero Christoffel symbols are

$$\begin{aligned}
\Gamma_{\theta\theta}^r &= -r & \Gamma_{\phi r}^\phi &= \frac{1}{r} \\
\Gamma_{\phi\phi}^r &= -r \sin^2 \theta & \Gamma_{\phi\theta}^\phi &= \cot \theta \\
\Gamma_{\phi\phi}^\theta &= -\sin \theta \cos \theta & \Gamma_{\theta r}^\theta &= \frac{1}{r}
\end{aligned}$$

We also note that

$$\epsilon^{ijk} = G^{-1/2} \tilde{\epsilon}^{ijk} = \frac{1}{r^2 \sin \theta} \tilde{\epsilon}^{ijk}$$

Now we are ready to compute

$$\operatorname{curl} \mathbf{X} = \epsilon^{ijk} (\nabla_i X_j) \partial_k.$$

We do this component-by-component in the coordinate basis, then translate to the orthonormal

frame field at the end. The r component of $\text{curl } \mathbf{X}$ is

$$\begin{aligned}
\epsilon^{ijr}(\nabla_i X_j) &= \frac{1}{r^2 \sin \theta} \tilde{\epsilon}^{ijr}(\nabla_i X_j) \\
&= \frac{1}{r^2 \sin \theta} (\nabla_\theta X_\phi - \nabla_\phi X_\theta) \\
&= \frac{1}{r^2 \sin \theta} (g_{\phi\mu} \nabla_\theta X^\mu - g_{\theta\mu} \nabla_\phi X^\mu) \\
&= \frac{1}{r^2 \sin \theta} (r^2 \sin^2 \theta \nabla_\theta X^\phi - r^2 \nabla_\phi X^\theta) \\
&= \sin \theta \nabla_\theta X^\phi - \frac{1}{\sin \theta} \nabla_\phi X^\theta,
\end{aligned}$$

where we used metric compatibility in the third line. Then

$$\begin{aligned}
\nabla_\theta X^\phi &= \partial_\theta X^\phi + \Gamma_{\theta\mu}^\phi X^\mu \\
&= \partial_\theta X^\phi + \Gamma_{\theta\phi}^\phi X^\phi \\
&= \frac{\partial}{\partial \theta} \left(\frac{X^\phi}{r \sin \theta} \right) + \Gamma_{\theta\phi}^\phi \left(\frac{X^\phi}{r \sin \theta} \right) \\
&= \frac{1}{r} \frac{\partial}{\partial \theta} \left(\frac{X^\phi}{\sin \theta} \right) + \cot \theta \left(\frac{X^\phi}{r \sin \theta} \right) \\
&= \frac{1}{r \sin \theta} \frac{\partial X^\phi}{\partial \theta},
\end{aligned}$$

where the last line follows after applying the product rule to the first term. Next

$$\begin{aligned}
\nabla_\phi X^\theta &= \partial_\phi X^\theta + \Gamma_{\phi\mu}^\theta X^\mu \\
&= \partial_\phi X^\theta + \Gamma_{\phi\phi}^\theta X^\phi \\
&= \partial_\phi \left(\frac{X^\theta}{r} \right) + (-\sin \theta \cos \theta) \left(\frac{X^\phi}{r \sin \theta} \right) \\
&= \frac{1}{r} \frac{\partial X^\theta}{\partial \phi} - \frac{X^\phi \cos \theta}{r}.
\end{aligned}$$

Substituting these back into the expression for the r component of the curl, we get

$$\begin{aligned}
\epsilon^{ijr}(\nabla_i X_j) &= \sin \theta \left[\frac{1}{r \sin \theta} \frac{\partial X^\phi}{\partial \theta} \right] - \frac{1}{\sin \theta} \left[\frac{1}{r} \frac{\partial X^\theta}{\partial \phi} - \frac{X^\phi \cos \theta}{r} \right] \\
&= \frac{1}{r} \frac{\partial X^\phi}{\partial \theta} - \frac{1}{r \sin \theta} \frac{\partial X^\theta}{\partial \phi} + \frac{1}{r \sin \theta} X^\phi \cos \theta \\
&= \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (X^\phi \sin \theta) - \frac{\partial X^\theta}{\partial \phi} \right],
\end{aligned}$$

which is what we wanted. Next we compute the θ component:

$$\begin{aligned}
\epsilon^{ij\theta}(\nabla_i X_j) &= \frac{1}{r^2 \sin \theta} \tilde{\epsilon}^{ij\theta}(\nabla_i X_j) \\
&= \frac{1}{r^2 \sin \theta} (\nabla_\phi X_r - \nabla_r X_\phi) \\
&= \frac{1}{r^2 \sin \theta} (g_{\mu r} \nabla_\phi X^\mu - g_{\mu \phi} \nabla_r X^\mu) \\
&= \frac{1}{r^2 \sin \theta} (\nabla_\phi X^r - r^2 \sin^2 \theta \nabla_r X^\phi) \\
&= \frac{1}{r^2 \sin \theta} \nabla_\phi X^r - \sin \theta \nabla_r X^\phi.
\end{aligned}$$

Now

$$\begin{aligned}
\nabla_\phi X^r &= \partial_\phi X^r + \Gamma_{\phi\mu}^r X^\mu \\
&= \partial_\phi X^r + \Gamma_{\phi\phi}^r X^\phi \\
&= \partial_\phi X^r - r \sin^2 \theta X^\phi \\
&= \frac{\partial X^{\hat{r}}}{\partial \phi} - \sin \theta X^{\hat{\phi}},
\end{aligned}$$

and

$$\begin{aligned}
\nabla_r X^\phi &= \partial_r X^\phi + \Gamma_{r\mu}^\phi X^\mu \\
&= \partial_r X^\phi + \Gamma_{r\phi}^\phi X^\phi \\
&= \frac{\partial}{\partial r} \left(\frac{X^{\hat{\phi}}}{r \sin \theta} \right) + \frac{1}{r} \frac{X^{\hat{\phi}}}{r \sin \theta} \\
&= \frac{1}{r \sin \theta} \frac{\partial X^{\hat{\phi}}}{\partial r}.
\end{aligned}$$

Substituting back, we get

$$\begin{aligned}
\epsilon^{ij\theta}(\nabla_i X_j) &= \frac{1}{r^2 \sin \theta} \left(\frac{\partial X^{\hat{r}}}{\partial \phi} - \sin \theta X^{\hat{\phi}} \right) - \sin \theta \left(\frac{1}{r \sin \theta} \frac{\partial X^{\hat{\phi}}}{\partial r} \right) \\
&= \frac{1}{r^2 \sin \theta} \frac{\partial X^{\hat{r}}}{\partial \phi} - \frac{1}{r^2} X^{\hat{\phi}} - \frac{1}{r} \frac{\partial X^{\hat{\phi}}}{\partial r} \\
&= \frac{1}{r^2 \sin \theta} \frac{\partial X^{\hat{r}}}{\partial \phi} - \frac{1}{r^2} \left(X^{\hat{\phi}} + r \frac{\partial X^{\hat{\phi}}}{\partial r} \right) \\
&= \frac{1}{r^2} \left[\frac{1}{\sin \theta} \frac{\partial X^{\hat{r}}}{\partial \phi} - \frac{\partial}{\partial r} (r X^{\hat{\phi}}) \right].
\end{aligned}$$

Finally, we compute the ϕ component:

$$\begin{aligned}
\epsilon^{ij\phi}(\nabla_i X_j) &= \frac{1}{r^2 \sin \theta} \tilde{\epsilon}^{ij\phi}(\nabla_i X_j) \\
&= \frac{1}{r^2 \sin \theta} (\nabla_r X_\theta - \nabla_\theta X_r) \\
&= \frac{1}{r^2 \sin \theta} (g_{\mu\theta} \nabla_r X^\mu - g_{\mu r} \nabla_\theta X^\mu) \\
&= \frac{1}{r^2 \sin \theta} (r^2 \nabla_r X^\theta - \nabla_\theta X^r) \\
&= \frac{1}{\sin \theta} \nabla_r X^\theta - \frac{1}{r^2 \sin \theta} \nabla_\theta X^r.
\end{aligned}$$

It is straightforward to show, in a similar way as before, that

$$\begin{aligned}
\nabla_r X^\theta &= \frac{\partial}{\partial r} \left(\frac{X^\theta}{r} \right) + \frac{1}{r^2} X^\theta \\
&= \frac{1}{r} \frac{\partial X^\theta}{\partial r}, \\
\nabla_\theta X^r &= \frac{\partial X^{\hat{r}}}{\partial \theta} - X^\theta.
\end{aligned}$$

This then gives

$$\begin{aligned}
\epsilon^{ij\phi}(\nabla_i X_j) &= \frac{1}{\sin \theta} \left[\frac{1}{r} \frac{\partial X^\theta}{\partial r} \right] - \frac{1}{r^2 \sin \theta} \left[\frac{\partial X^{\hat{r}}}{\partial \theta} - X^\theta \right] \\
&= \frac{1}{r^2 \sin \theta} \left[r \frac{\partial X^\theta}{\partial r} + X^\theta - \frac{\partial X^{\hat{r}}}{\partial \theta} \right] \\
&= \frac{1}{r^2 \sin \theta} \left[\frac{\partial}{\partial r} (r X^\theta) - \frac{\partial X^{\hat{r}}}{\partial \theta} \right].
\end{aligned}$$

Putting everything together, this shows that

$$\begin{aligned}
\text{curl } \mathbf{X} &= \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (X^{\hat{\phi}} \sin \theta) - \frac{\partial X^{\hat{\theta}}}{\partial \phi} \right] \partial_r + \frac{1}{r^2} \left[\frac{1}{\sin \theta} \frac{\partial X^{\hat{r}}}{\partial \phi} - \frac{\partial}{\partial r} (r X^{\hat{\phi}}) \right] \partial_\theta \\
&\quad + \frac{1}{r^2 \sin \theta} \left[\frac{\partial}{\partial r} (r X^\theta) - \frac{\partial X^{\hat{r}}}{\partial \theta} \right] \partial_\phi.
\end{aligned}$$

In terms of the orthonormal frame field, we get

$$\begin{aligned}
\text{curl } \mathbf{X} &= \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (X^{\hat{\phi}} \sin \theta) - \frac{\partial X^{\hat{\theta}}}{\partial \phi} \right] e_{\hat{r}} + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial X^{\hat{r}}}{\partial \phi} - \frac{\partial}{\partial r} (r X^{\hat{\phi}}) \right] e_{\hat{\theta}} \\
&\quad + \frac{1}{r} \left[\frac{\partial}{\partial r} (r X^\theta) - \frac{\partial X^{\hat{r}}}{\partial \theta} \right] e_{\hat{\phi}}
\end{aligned}$$

This is the same expression as you'd see in, e.g., Jackson. In the same way, you can obtain expressions for gradient, divergence and curl in arbitrary coordinate systems, e.g. paraboloidal. This is done in Exercise 3.4. But there is probably a more efficient way to compute the curl without having to look at Christoffel symbols; look up how scale factors work. $\square$

Exercise 3.3

Imagine we have a *diagonal* metric $g_{\mu\nu}$. Show that the Christoffel symbols are given by

$$\begin{aligned}\Gamma^\lambda_{\mu\nu} &= 0 \\ \Gamma^\lambda_{\mu\mu} &= -\frac{1}{2}(g_{\lambda\lambda})^{-1}\partial_\lambda g_{\mu\mu} \\ \Gamma^\lambda_{\mu\lambda} &= \partial_\mu \left(\ln \sqrt{|g_{\lambda\lambda}|} \right) \\ \Gamma^\lambda_{\lambda\lambda} &= \partial_\lambda \left(\ln \sqrt{|g_{\lambda\lambda}|} \right)\end{aligned}$$

In these expressions, $\mu \neq \nu \neq \lambda$, and repeated indices are not summed over.

Solution. See attached scan. $\square$

Exercise 3.4

In Euclidean three-space, we can define paraboloidal coordinates (u, v, ϕ) via

$$x = uv \cos \phi \quad y = uv \sin \phi \quad z = \frac{1}{2}(u^2 - v^2)$$

- Find the coordinate transformation matrix between paraboloidal and Cartesian coordinates $\partial x^\alpha / \partial x^{\beta'}$ and the inverse transformation. Are there any singular points in the map?
- Find the basis vectors and basis one-forms in terms of Cartesian basis vectors and forms.
- Find the metric and inverse metric in paraboloidal coordinates.
- Calculate the Christoffel symbols.
- Calculate the divergence $\nabla_\mu V^\mu$ and Laplacian $\nabla_\mu \nabla^\mu f$.

Solution. See attached scan. The main results are that

$$\nabla_\mu X^\mu = \frac{(u+v)^2}{u^2+v^2} \left(\frac{1}{v} X^u + \frac{1}{u} X^v \right) + \partial_\mu X^\mu$$

and

$$\nabla_\mu \nabla^\mu f = \frac{1}{u^2+v^2} \left[\frac{1}{v} \frac{\partial f}{\partial u} + \frac{1}{u} \frac{\partial f}{\partial v} + \frac{\partial^2 f}{\partial u^2} + \frac{\partial^2 f}{\partial v^2} \right] + \frac{1}{u^2 v^2} \frac{\partial^2 f}{\partial \phi^2}$$

Best to check these against another method, e.g., using scale factors. $\square$

Exercise 3.5 (*)

Consider a 2-sphere with coordinates (θ, ϕ) and metric

$$ds^2 = d\theta^2 + \sin^2 \theta d\phi^2. \quad (3.217)$$

- (a) Show that lines of constant longitude ($\phi = \text{constant}$) are geodesics, and that the only line of constant latitude ($\theta = \text{constant}$) that is a geodesic is the equator ($\theta = \pi/2$).
- (b) Take a vector with components $V^\mu = (1, 0)$ and parallel-transport it once around a circle of constant latitude. What are the components of the resulting vector, as a function of θ ?

Solution. See attached scan. The main result is that a vector with components $V^\mu = (1, 0)$, when parallel-transported around a circle of constant latitude θ , ends up with components

$$\left(\cos(2\pi \cos \theta), -\csc \theta \sin(2\pi \cos \theta) \right).$$

It is probably more enlightening to have a solution for the parallel-transported vector as a function of arc length along the path. Also, compare the angles of the initial and final vectors. In what sense does the parallel-transported vector rotate? □

carroll 3.3

diagonal metric $g_{\mu\nu}$. we know that

$$\Gamma^\sigma_{\mu\nu} = \frac{1}{2} g^{\sigma\rho} (\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}).$$

1. $\Gamma^\lambda_{\mu\nu} = 0$

Butter equations than words,
Now since $g_{\mu\nu}$ is diagonal, we have $g_{\mu\nu} = 0$

for $\mu \neq \nu$, hence the last term is zero. And

$g^{\sigma\rho}$ is also diagonal, so we get a $g^{\nu\sigma}$

and $g_{\sigma\mu}$ in the first two, which are zero. $\square$

2. $\Gamma^\lambda_{\mu\mu} = -\frac{1}{2} (g_{\lambda\lambda})^{-1} \partial_\lambda g_{\mu\mu}$

Now

$$\begin{aligned} \Gamma^\lambda_{\mu\mu} &= \frac{1}{2} g^{\lambda\rho} (\partial_\mu g_{\mu\rho} + \partial_\mu g_{\rho\mu} - \partial_\rho g_{\mu\mu}) \\ &= \frac{1}{2} g^{\lambda\lambda} (\partial_\mu g_{\mu\lambda}^{\rightarrow 0} + \partial_\mu g_{\lambda\mu}^{\rightarrow 0} - \partial_\lambda g_{\mu\mu}) \\ &= -\frac{1}{2} (g_{\lambda\lambda})^{-1} \partial_\lambda g_{\mu\mu}. \end{aligned} \quad \square$$

3. $\Gamma^\lambda_{\mu\lambda} = \frac{1}{2} g^{\lambda\rho} (\partial_\mu g_{\lambda\rho} + \partial_\lambda g_{\rho\mu} - \partial_\rho g_{\mu\lambda})$

$$= \frac{1}{2} g^{\lambda\lambda} (\partial_\mu g_{\lambda\lambda} + \partial_\lambda g_{\lambda\mu}^{\rightarrow 0})$$

$$= \frac{1}{2 g_{\lambda\lambda}} \partial_\mu g_{\lambda\lambda}$$

$$= \partial_\mu (\ln \sqrt{|g_{\lambda\lambda}|}) \cdot \frac{1}{\sqrt{|g_{\lambda\lambda}|}} \cdot \frac{1}{2} \frac{1}{\sqrt{|g_{\lambda\lambda}|}} \partial_\mu g_{\lambda\lambda} \quad \textcircled{1}$$

lastly

$$\begin{aligned}
 4. \Gamma^{\lambda}_{\lambda\lambda} &= \frac{1}{2} g^{\lambda\rho} \left(\cancel{\partial_{\lambda} g_{\lambda\rho}} + \partial_{\lambda} g_{\rho\lambda} - \cancel{\partial_{\rho} g_{\lambda\lambda}} \right) \\
 &= \frac{1}{2} g^{\lambda\lambda} \left(\partial_{\lambda} g_{\lambda\lambda} + \cancel{\partial_{\lambda} g_{\lambda\lambda}} - \cancel{\partial_{\lambda} g_{\lambda\lambda}} \right) \\
 &= \frac{1}{2 g_{\lambda\lambda}} \partial_{\lambda} g_{\lambda\lambda} \\
 &= \partial_{\lambda} \left(\ln \sqrt{|g_{\lambda\lambda}|} \right). \quad \square.
 \end{aligned}$$

3, 4

paraboloidal coordinates (u, v, ϕ)

$$x = uv \cos \phi, \quad y = uv \sin \phi,$$

$$z = \frac{1}{2} (u^2 - v^2)$$

$$(a) \quad \left(\frac{\partial x^{\alpha}}{\partial x^{\beta'}} \right) = \begin{pmatrix} v \cos \phi & u \cos \phi & -uv \sin \phi \\ v \sin \phi & u \sin \phi & uv \cos \phi \\ u & -v & 0 \end{pmatrix}$$

can invert this normally, or use

$$x^2 = u^2 v^2 \cos^2 \phi$$

$$y^2 = u^2 v^2 \sin^2 \phi$$

$$x^2 + y^2 = u^2 v^2$$

$$\rightarrow v^2 = \frac{x^2 + y^2}{u^2}$$

$$z = \frac{1}{2} \left(u^2 - \frac{x^2 + y^2}{u^2} \right)$$

quadratic eq. for u^2

(2)

$$2z = u^2 - \frac{x^2 + y^2}{u^2}$$

$$\Rightarrow 2zu^2 = u^4 - (x^2 + y^2)$$

$$\Rightarrow u^4 - 2zu^2 - (x^2 + y^2) = 0$$

$$\Rightarrow u^2 = \frac{2z \pm \sqrt{4z^2 - 4(1)(x^2 + y^2)}}{2(1)}$$

$$= z \pm \sqrt{z^2 - (x^2 + y^2)}$$

hmm

$$\frac{y}{x} = \frac{u \sin \phi}{u \cos \phi} = \tan \phi$$

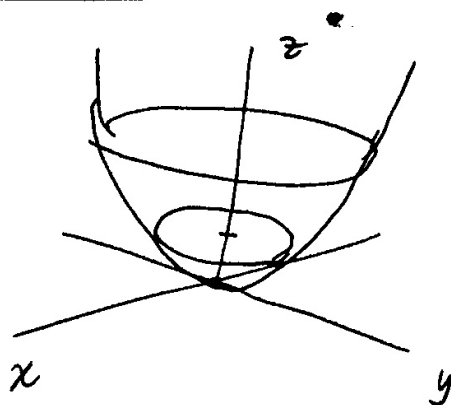
$$u = \frac{x}{v \cos \phi}$$

$$z = \frac{1}{2} \left(\frac{x^2}{v^2 \cos^2 \phi} - v^2 \right)$$

$$\begin{aligned} x^2 + y^2 + z^2 &= u^2 v^2 + \frac{1}{4} (u^2 - v^2)^2 \\ &= u^2 v^2 + \frac{1}{4} (u^4 - 2u^2 v^2 + v^4) \\ &= \frac{1}{4} u^4 + \frac{1}{2} u^2 v^2 + \frac{1}{4} v^4 \\ &= \frac{1}{4} (u^2 + v^2)^2 \end{aligned}$$

now since $u^2 - v^2 = 2z \Rightarrow u^2 = 2z + v^2$

$$= \frac{1}{4} (2z + 2v^2)^2 = (z + v^2)^2 \quad (3)$$



$$z + v^2 = \sqrt{x^2 + y^2 + z^2}$$

$$v^2 = \sqrt{x^2 + y^2 + z^2} - z$$

similar argument;

$$u^2 = \sqrt{x^2 + y^2 + z^2} + z.$$

Thus

$$u = \sqrt{\sqrt{x^2 + y^2 + z^2} + z}$$

$$v = \sqrt{\sqrt{x^2 + y^2 + z^2} - z}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

Then

$$\frac{\partial u}{\partial x} =$$

much! instead

$$\frac{\partial x^{\alpha'}}{\partial x^{\beta}} = \begin{pmatrix} \frac{v \cos \phi}{u^2 + v^2} & \frac{v \sin \phi}{u^2 + v^2} & \frac{u}{u^2 + v^2} \\ \frac{u \cos \phi}{u^2 + v^2} & \frac{u \sin \phi}{u^2 + v^2} & -\frac{v}{u^2 + v^2} \\ -\frac{\sin \phi}{uv} & \frac{\cos \phi}{uv} & 0 \end{pmatrix}$$

star straight forward to compute Jacobian:

$$\left| \frac{\partial x^{\alpha'}}{\partial x^{\beta'}} \right| = uv(u^2 + v^2),$$

singular where $u = 0$ or $v = 0$.

(b) Basis vectors and basis one-forms.

$$\frac{\partial}{\partial x^{\alpha'}} = \frac{\partial x^{\mu}}{\partial x^{\alpha'}} \frac{\partial}{\partial x^{\mu}}$$

Thus

$$\partial_u = \frac{\partial x^1}{\partial x^u} \partial_1 + \frac{\partial x^2}{\partial x^u} \partial_2 + \frac{\partial x^3}{\partial x^u} \partial_3$$

$$= v \cos \phi \partial_x + v \sin \phi \partial_y + u \partial_z$$

$$\partial_v = u \cos \phi \partial_x + u \sin \phi \partial_y - v \partial_z$$

$$\partial_\phi = -uv \sin \phi \partial_x + uv \cos \phi \partial_y$$

$$\frac{dx^\mu}{dx^\alpha} =$$

$$dx^{\alpha'} = \frac{\partial x^{\alpha'}}{\partial x^\mu} dx^\mu$$

$$dx^u = \frac{v \cos \phi}{u^2 + v^2} dx^x + \frac{u \cos \phi}{u^2 + v^2} dx^y - \frac{\sin \phi}{uv} dx^z$$

$$\text{check: } \frac{v \cos \phi}{u^2 + v^2} dx^1 + \frac{v \sin \phi}{u^2 + v^2} dx^2 + \frac{u}{u^2 + v^2} dx^3$$

$$dx^u(\partial_u) = \frac{v^2 \cos^2 \phi + uv \sin \phi \cos \phi}{u^2 + v^2} - \frac{u \sin \phi}{uv}$$

$$= \frac{uv^3 \cos^2 \phi + u^2 v^2 \sin \phi \cos \phi - u \sin \phi (u^2 + v^2)}{(u^2 + v^2) uv}$$

$$= \frac{uv^3 \cos^2 \phi + u^2 v^2 \sin \phi \cos \phi - u^3 \sin \phi - uv^2 \sin \phi}{() ()}$$

we can just write off everything from the matrices:

$$\partial_u = v \cos \phi \partial_1 + v \sin \phi \partial_2 + u \partial_3$$

$$\partial_v = u \cos \phi \partial_1 + u \sin \phi \partial_2 - v \partial_3$$

$$\partial_\phi = -uv \sin \phi \partial_1 + uv \cos \phi \partial_2$$

and

$$dx^u = \frac{v \cos \phi}{u^2 + v^2} dx^1 + \frac{v \sin \phi}{u^2 + v^2} dx^2 + \frac{u}{u^2 + v^2} dx^3$$

$$dx^v = \frac{u \cos \phi}{u^2 + v^2} dx^1 + \frac{u \sin \phi}{u^2 + v^2} dx^2 - \frac{v}{u^2 + v^2} dx^3$$

$$dx^\phi = -\frac{\sin \phi}{uv} dx^1 + \frac{\cos \phi}{uv} dx^2$$

(c) metric components:

$$g = g_{ij} dx^i \otimes dx^j$$

$$g(\partial_u, \partial_u) = g_{ij} dx^i(\partial_u) \otimes dx^j(\partial_u)$$

$$\stackrel{\text{now}}{=} \delta_{ij} dx^i(\partial_u) \otimes dx^j(\partial_u)$$

$$dx^i(\partial_u) \rightarrow \text{depends!}$$

$$= dx^i_u(\partial_u) dx^i(\partial_u)$$

$$= v^2 \cos^2 \phi + v^2 \sin^2 \phi + u^2$$

$$= u^2 + v^2.$$

$$\begin{aligned}
 g(\partial_u, \partial_v) &= \delta_{ij} dx^i(\partial_u) dx^j(\partial_v) \\
 &= dx_i(\partial_u) dx^i(\partial_v) \\
 &= uv \cos^2 \phi + uv \sin^2 \phi - uv \\
 &= 0.
 \end{aligned}$$

$$\begin{aligned}
 g(\partial_u, \partial_\phi) &= dx_i(\partial_u) dx^i(\partial_\phi) \\
 &= -uv^2 \sin \phi \cos \phi + uv^2 \sin \phi \cos \phi \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 g(\partial_v, \partial_v) &= dx_i(\partial_v) dx^i(\partial_v) \\
 &= u^2 \cos^2 \phi + u^2 \sin^2 \phi + v^2 \\
 &= u^2 + v^2
 \end{aligned}$$

$$\begin{aligned}
 g(\partial_v, \partial_\phi) &= -uv^2 \sin \phi \cos \phi + uv^2 \sin \phi \cos \phi \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 g(\partial_\phi, \partial_\phi) &= u^2 v^2 \sin^2 \phi + u^2 v^2 \cos^2 \phi \\
 &= u^2 v^2.
 \end{aligned}$$

Hence, in the paraboloidal coords.

$$(g_{ij}) = \begin{pmatrix} u^2 + v^2 & 0 & 0 \\ 0 & u^2 + v^2 & 0 \\ 0 & 0 & u^2 v^2 \end{pmatrix}$$

$$ds^2 = (u^2 + v^2)(du^2 + dv^2) + u^2 v^2 d\phi^2.$$

inverse metric is

$$(g^{ij}) = \begin{pmatrix} \frac{1}{u^2+v^2} & & \\ & \frac{1}{u^2+v^2} & \\ & & \frac{1}{u^2v^2} \end{pmatrix}$$

(a) christoffels via 3.3,

$$\Gamma^u_{v\phi} = \Gamma^u_{\phi v} = 0$$

$$\Gamma^u_{vv} = -\frac{1}{2} (g_{uu})^{-1} \partial_u g_{vv}$$

$$= -\frac{1}{2(u^2+v^2)} 2u$$

$$= -\frac{u}{u^2+v^2}$$

$$\Gamma^u_{\phi\phi} = -\frac{1}{2} (g_{uu})^{-1} \partial_u g_{\phi\phi}$$

$$= -\frac{1}{2(u^2+v^2)} 2uv^2$$

$$= -\frac{uv^2}{u^2+v^2}$$

$$\Gamma^u_{uv} = \partial_v (\ln \sqrt{|g_{uu}|})$$

$$= \partial_v (\ln \sqrt{u^2+v^2})$$

$$= \frac{1}{\sqrt{u^2+v^2}} \frac{1}{2} \frac{1}{\sqrt{u^2+v^2}} 2v$$

$$= \frac{v}{u^2+v^2}$$

$$\Gamma^u_{u\phi} = \partial_\phi (\ln \sqrt{|g_{uu}|})$$

$$= 0$$

$$\Gamma^u_{uu} = \partial_u (\ln \sqrt{|g_{uu}|})$$

$$= \frac{u}{u^2+v^2}$$

$$\Gamma^{\nu}_{uu} = -\frac{1}{2} (g_{\nu\nu})^{-1} \partial_{\nu} g_{uu}$$

$$= -\frac{1}{2(u^2+v^2)} \partial_{\nu}$$

$$= -\frac{v}{u^2+v^2}$$

$$\Gamma^{\nu}_{\nu\nu} = \frac{v}{u^2+v^2}$$

$$\Gamma^{\nu}_{\phi\phi} = -\frac{1}{2} (g_{\nu\nu})^{-1} \partial_{\nu} g_{\phi\phi}$$

$$= -\frac{1}{2(u^2+v^2)} \partial_{\nu} u^2$$

$$= -\frac{u^2 v}{u^2+v^2}$$

$$\Gamma^{\nu}_{\nu u} = \partial_u (\ln \sqrt{|g_{\nu\nu}|})$$

$$= \partial_u (\ln \sqrt{u^2+v^2})$$

$$= \frac{u}{u^2+v^2}$$

$$\Gamma^{\nu}_{\nu\phi} = \partial_{\phi} ()$$

$$= 0$$

Lastly

$$\Gamma^{\phi}_{uu} = 0$$

$$\Gamma^{\phi}_{\nu\nu} = 0$$

$$\Gamma^{\phi}_{\phi u} = \partial_u \ln \sqrt{g_{\phi\phi}}$$

$$= \partial_u \ln(uv)$$

$$= \frac{1}{uv} v = \frac{1}{u}$$

$$\Gamma^{\phi}_{\phi\nu} = \frac{1}{v}$$

$$\Gamma^{\phi}_{\phi\phi} = 0$$

Thus the nonzero christoffels are

$$\Gamma^u_{\sqrt{v}} = -\frac{u}{u^2+v^2}$$

$$\Gamma^u_{\cancel{\phi\phi}} = -\frac{uv^2}{u^2+v^2}$$

$$\Gamma^u_{\cancel{uv}} = \Gamma^u_{vu} = \frac{v}{u^2+v^2}$$

~~$$\Gamma^u_{u\phi} = \Gamma^u_{\phi u} = 0$$~~

$$\Gamma^u_{\cancel{uu}} = \frac{u}{u^2+v^2}$$

$$\Gamma^v_{\cancel{uu}} = -\frac{v}{u^2+v^2}$$

$$\Gamma^v_{\cancel{\phi\phi}} = -\frac{u^2v}{u^2+v^2}$$

$$\Gamma^v_{\cancel{vu}} = \Gamma^v_{\cancel{uv}} = \frac{u}{u^2+v^2}$$

$$\Gamma^v_{\cancel{vv}} = \frac{v}{u^2+v^2}$$

$$\Gamma^\phi_{\cancel{\phi u}} = \Gamma^\phi_{\cancel{u\phi}} = \frac{1}{u}$$

$$\Gamma^\phi_{\cancel{\phi v}} = \Gamma^\phi_{\cancel{v\phi}} = \frac{1}{v}$$

when you call the "nonzero" function you should count 14 symbols.

(2) Divergence and Laplacian.

First note that

$$G = \det(g_{ij}) \\ = (u^2 + v^2)^2 u^2 v^2,$$

then the divergence

$$\begin{aligned} \nabla_\mu V^\mu &= G^{-1/2} (G^{1/2} V^\mu)_{,\mu} \\ &= \frac{1}{uv(u^2+v^2)} \left[uv(u^2+v^2) V^\mu \right]_{,\mu} \end{aligned}$$

go term-by-term

$$\begin{aligned} &\left[uv(u^2+v^2) V^u \right]_{,u} \\ &= v(u^2+v^2) V^u + uv(2u) V^u + uv(u^2+v^2) V^u_{,u} \\ &= v(u^2+v^2) V^u + 2u^2 v V^u + uv(u^2+v^2) V^u_{,u} \end{aligned}$$

similarly

$$\begin{aligned} &\left[uv(u^2+v^2) V^v \right]_{,v} \\ &= u(u^2+v^2) V^v + 2uv^2 V^v + uv(u^2+v^2) V^v_{,v} \end{aligned}$$

and

$$\begin{aligned} &\left[uv(u^2+v^2) V^\phi \right]_{,\phi} \\ &= uv(u^2+v^2) V^\phi_{,\phi}, \quad \text{so all together} \end{aligned}$$

~~Therefore~~

we have

$$\sigma_{\mu} X^{\mu} = \left[(u^2 + v^2) [v X^u + u X^v] + 2uv [u X^u + v X^v] + uv (u^2 + v^2) \frac{X^{\mu}}{\partial_{\mu} X^{\mu}} \right] \frac{1}{uv(u^2 + v^2)}$$

$$\begin{aligned} & u^2 v X^u + v^3 X^u + u^3 X^v + u v^2 X^v \\ & + 2u^2 v X^u + 2u v^2 X^v \end{aligned}$$

$$= u^3 X^v + v^3 X^u + 3u^2 v X^u + 3u v^2 X^v + uv(u^2 + v^2) \partial_{\mu} X^{\mu}$$

$$= (u^3 + 3u v^2) X^v + (v^3 + 3u^2 v) X^u + uv(u^2 + v^2) \partial_{\mu} X^{\mu}$$

$$= u(u^2 + 3v^2) X^v + v(v^2 + 3u^2) X^u + uv(u^2 + v^2) \partial_{\mu} X^{\mu}$$

simpler I could make it!

$$= u(u^2 + v^2) X^v + 2u v^2 X^v + v(u^2 + v^2) X^u + 2u^2 v X^u + uv(u^2 + v^2) \partial_{\mu} X^{\mu}$$

$$\nabla^2 f = g^{ij} \nabla_i \nabla_j f$$

$$= \nabla_i \nabla^i f$$

$$= g^{-1/2} (g^{1/2} \nabla^i f)_{,i} \quad \text{— divergence of } \nabla^i f.$$

Now

$$\partial^u f = \cancel{g^{uu}} g^{u\lambda} \nabla_\lambda f$$

$$= g^{uu} \partial_u f$$

and so on, so that

$$\partial^u f = \frac{1}{u^2 + v^2} \partial_u f$$

$$\partial^v f = \frac{1}{u^2 + v^2} \partial_v f$$

$$\partial^\phi f = \frac{1}{u^2 v^2} \partial_\phi f.$$

Then by the previous,

$$\begin{aligned} \nabla^2 f &= (u^2 + v^2) [v \partial^u f + u \partial^v f] \\ &\quad + 2uv [u \partial^u f + v \partial^v f] \\ &\quad + uv (u^2 + v^2) \partial_\mu (\nabla^\mu f). \end{aligned}$$

$$= \cancel{(u^2 + v^2)} \frac{1}{\cancel{u^2 + v^2}} \left[v \partial_u f + u \partial_v f \right]$$

$$= \frac{2uv}{u^2 + v^2} \left[u \partial_u f + v \partial_v f \right]$$

$$+ uv(u^2 + v^2) \cancel{\frac{\partial}{\partial u}} \cancel{\frac{\partial}{\partial v}} \left\{ \partial_u \left(\frac{1}{u^2 + v^2} \partial_u f \right) \right.$$

$$\left. + \partial_v \left(\frac{1}{u^2 + v^2} \partial_v f \right) + \partial_\phi \left(\frac{1}{u^2 + v^2} \partial_\phi f \right) \right\}$$

$$= v \partial_u f + u \partial_v f + \frac{2u^2 v}{u^2 + v^2} \partial_u f + \frac{2uv^2}{u^2 + v^2} \partial_v f$$

$$+ uv(u^2 + v^2) \left\{ \right. \left. \right\},$$

tedious, but can be expanded.

$$\cancel{uv(u^2 + v^2)}$$

$$\{uv\} = - \frac{1}{(u^2 + v^2)^2} 2u \frac{\partial f}{\partial u} + \frac{1}{u^2 + v^2} \frac{\partial^2 f}{\partial u^2}$$

$$- \frac{1}{(u^2 + v^2)^2} 2v \frac{\partial f}{\partial v} + \frac{1}{u^2 + v^2} \frac{\partial^2 f}{\partial v^2}$$

$$+ \frac{1}{u^2 v^2} \frac{\partial^2 f}{\partial \phi^2}$$

$$= - \frac{uv}{(u^2 + v^2)} 2u \frac{\partial f}{\partial u} + uv \frac{\partial^2 f}{\partial u^2} - \frac{uv}{u^2 + v^2} 2v \frac{\partial f}{\partial v} + uv \frac{\partial^2 f}{\partial v^2} \quad (13)$$

As usual, your mistake was sloppy collation of terms, you really have

$$\partial_\mu X^\mu = \frac{1}{uv} (\cancel{vX^u + uX^v})$$

$$= \frac{1}{uv(u^2+v^2)} \left[u(u^2+v^2)X^u + v(u^2+v^2)X^v \right. \\ \left. + 2uv^2X^v + 2u^2vX^u \right. \\ \left. + uv(u^2+v^2)\partial_\mu X^\mu \right]$$

$$= \frac{1}{v} X^u + \frac{1}{u} X^v + \frac{2v}{u^2+v^2} X^v + \frac{2u}{u^2+v^2} X^u \\ + \partial_\mu X^\mu$$

$$= \cancel{1/v X^u + 1/u X^v} + \frac{(u+v)^2}{v(u^2+v^2)} X^u$$

$$+ \frac{u^2+v^2+2uv}{u(u^2+v^2)} X^v + \partial_\mu X^\mu$$

$$\partial_\mu X^\mu = \frac{(u+v)^2}{u^2+v^2} \left(\frac{1}{v} X^u + \frac{1}{u} X^v \right) + \partial_\mu X^\mu$$

This is nice ~~manifestly invariant~~
under $u \leftrightarrow v$.

then

$$\nabla_\mu \nabla^\mu f = \frac{(u+v)^2}{u^2+v^2} \left(\frac{1}{v} \partial^u f + \frac{1}{u} \partial^v f \right) + \partial_\mu \partial^\mu f$$

$$= \frac{(u+v)^2}{u^2+v^2} \left(\frac{1}{v} \frac{1}{u^2+v^2} \frac{\partial f}{\partial u} + \frac{1}{u} \frac{1}{u^2+v^2} \frac{\partial f}{\partial v} \right)$$

$$= \text{now } \frac{(u+v)^2}{v(u^2+v^2)^2} \frac{\partial f}{\partial u} + \frac{(u+v)^2}{u(u^2+v^2)^2} \frac{\partial f}{\partial v}$$

$$\partial_\mu \partial^\mu f = \partial_u \partial^u f + \partial_v \partial^v f + \partial_\phi \partial^\phi f$$

$$= \partial_u \left(\frac{1}{u^2+v^2} \partial_u f \right) + \partial_v \left(\frac{1}{u^2+v^2} \partial_v f \right)$$

$$+ \partial_\phi \left(\frac{1}{u^2+v^2} \partial_\phi f \right)$$

$$= - \frac{1}{(u^2+v^2)^2} 2u \frac{\partial f}{\partial u} + \frac{1}{u^2+v^2} \frac{\partial^2 f}{\partial u^2}$$

$$- \frac{1}{(u^2+v^2)^2} 2v \frac{\partial f}{\partial v} + \frac{1}{u^2+v^2} \frac{\partial^2 f}{\partial v^2}$$

$$+ \frac{1}{u^2+v^2} \frac{\partial^2 f}{\partial \phi^2}$$

we are grinding through this calculation.

$$\begin{aligned}
 & \frac{(u+v)^2}{v(u^2+v^2)^2} \frac{\partial f}{\partial u} + \frac{(u+v)^2}{u(u^2+v^2)^2} \frac{\partial f}{\partial v} \\
 & - \frac{2u}{(u^2+v^2)^2} \frac{\partial f}{\partial u} - \frac{2v}{(u^2+v^2)^2} \frac{\partial f}{\partial v} \\
 & + \frac{1}{u^2+v^2} \frac{\partial^2 f}{\partial u^2} + \frac{1}{u^2+v^2} \frac{\partial^2 f}{\partial v^2} + \frac{1}{u^2 v^2} \frac{\partial^2 f}{\partial \phi^2}
 \end{aligned}$$

Return:

$$\begin{aligned}
 & \frac{(u+v)^2}{v(u^2+v^2)^2} - \frac{2u}{(u^2+v^2)^2} \\
 & = \frac{1}{(u^2+v^2)^2} \left[\frac{(u+v)^2}{v} - 2u \right] \\
 & = \frac{1}{(u^2+v^2)^2} \left[\frac{u^2 + 2uv + v^2 - 2uv}{v} \right] \\
 & = \frac{1}{v(u^2+v^2)}
 \end{aligned}$$

similarly the other combination gives

$$\frac{1}{u(u^2+v^2)}$$

Then,

$$\begin{aligned}
 \nabla_\mu \nabla^\mu f &= \frac{1}{v(u^2+v^2)} \frac{\partial f}{\partial u} + \frac{1}{u(u^2+v^2)} \frac{\partial f}{\partial v} \\
 &+ \frac{1}{u^2+v^2} \frac{\partial^2 f}{\partial u^2} + \frac{1}{u^2+v^2} \frac{\partial^2 f}{\partial v^2} + \frac{1}{u^2 v^2} \frac{\partial^2 f}{\partial \phi^2}
 \end{aligned}$$

(18)

that is, after factoring we get

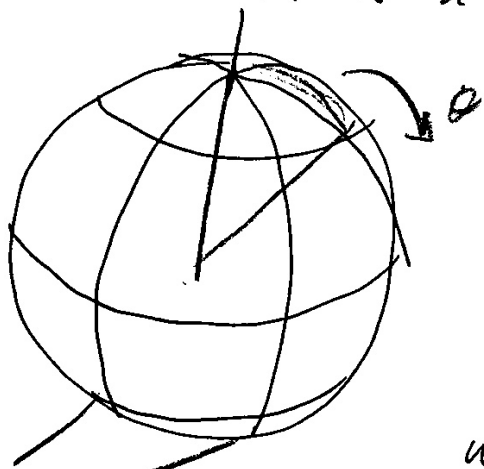
$$\nabla_\mu \nabla^\mu f = \frac{1}{u^2 + v^2} \left[\frac{1}{v} \frac{\partial f}{\partial u} + \frac{1}{u} \frac{\partial f}{\partial v} + \frac{\partial^2 f}{\partial u^2} + \frac{\partial^2 f}{\partial v^2} \right] + \frac{1}{u^2 v^2} \frac{\partial^2 f}{\partial \phi^2},$$

as we wanted to show. $\square$

3.5

$$ds^2 = d\theta^2 + \sin^2 \theta d\phi^2.$$

(a) show that lines of constant longitude are geodesics, and the only line of constant latitude that is a geodesic is the equator.



geodesic equation is

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma^\mu_{\rho\sigma} \frac{dx^\rho}{d\lambda} \frac{dx^\sigma}{d\lambda} = 0.$$

longitude
lines

we need the Christoffels for
this metric. By 3.3 we have

$$\Gamma^\theta_{\phi\phi} = -\frac{1}{2} (g_{\theta\theta})^{-1} \partial_\theta g_{\phi\phi} = -\frac{1}{2(1)} 2 \sin \theta \cos \theta$$

$$\Gamma^\theta_{\theta\theta} = \partial_\theta (\ln \sqrt{g_{\theta\theta}}) = -\sin \theta \cos \theta$$

$$= 2 \frac{1}{\sqrt{g_{\theta\theta}}} \frac{1}{2} \frac{1}{\sqrt{g_{\theta\theta}}} \partial_\theta (1)$$

$$= 0$$

$$\Gamma^{\theta}_{\theta\phi} = \Gamma^{\theta}_{\phi\theta} = \partial_{\phi} (\ln \sqrt{g_{\theta\theta}}) = 0$$

$$\Gamma^{\phi}_{\phi\phi} = \partial_{\phi} () = 0$$

$$\begin{aligned} \Gamma^{\phi}_{\theta\phi} &= \Gamma^{\phi}_{\phi\theta} = \partial_{\theta} (\ln \sqrt{g_{\phi\phi}}) \\ &= \frac{1}{\sqrt{g_{\phi\phi}}} \frac{1}{2\sqrt{g_{\phi\phi}}} \partial_{\theta} (g_{\phi\phi}) \\ &= \frac{1}{2g_{\phi\phi}} \partial_{\theta} (g_{\phi\phi}) \\ &= \cot \theta \end{aligned}$$

$$\Gamma^{\phi}_{\theta\theta} = () \partial_{\phi} () = 0$$

we only have

$$\begin{cases} \Gamma^{\theta}_{\phi\phi} = -\sin \theta \cos \theta \\ \Gamma^{\phi}_{\theta\phi} = \Gamma^{\phi}_{\phi\theta} = \cot \theta \end{cases}$$

then the geodesic equation reads

$$\frac{d^2 x^{\theta}}{d\lambda^2} + \Gamma^{\theta}_{\phi\phi} \left(\frac{dx^{\phi}}{d\lambda} \right)^2 = 0$$

$$\begin{aligned} \frac{d^2 x^{\phi}}{d\lambda^2} + \Gamma^{\phi}_{\theta\phi} \frac{dx^{\theta}}{d\lambda} \frac{dx^{\phi}}{d\lambda} \\ + \Gamma^{\phi}_{\phi\theta} \frac{dx^{\phi}}{d\lambda} \frac{dx^{\theta}}{d\lambda} = 0 \end{aligned}$$

or by using our expressions,

$$\frac{d^2 x^\theta}{d\lambda^2} - \sin\theta \cos\theta \left(\frac{dx^\phi}{d\lambda} \right)^2 = 0$$

$$\frac{d^2 x^\phi}{d\lambda^2} + 2\cot\theta \frac{dx^\theta}{d\lambda} \frac{dx^\phi}{d\lambda} = 0$$

(a)

say we have a line of constant longitude,

i.e. $x^\mu = (\theta(\lambda) \text{ const.})$. then ~~the~~

~~second eq. reads $0 = 0$, and the~~

$$\frac{dx^\phi}{d\lambda} = 0,$$

so the second eq. is trivially satisfied and the first reads

$$\frac{d^2 x^\theta}{d\lambda^2} = 0,$$

which is easily satisfied by putting

$$x^\mu = (\theta_0 + a\lambda, b)$$

with a, b constants.

thus $x^\mu(\lambda)$ is a geodesic, as claimed. amounts to
arc length param.

now suppose we choose

$$x^\mu(\lambda) = (\theta_0, \phi(\lambda)).$$

then

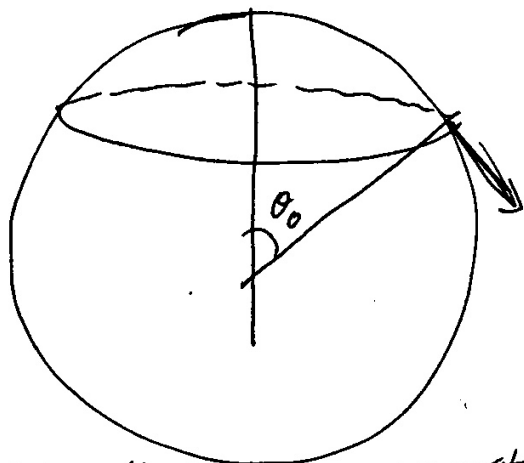
the geodesic eq. reduces to

$$-\sin\theta\cos\theta\left(\frac{dx^\phi}{d\lambda}\right)^2 = 0$$

$$\frac{d^2 x^\phi}{d\lambda^2} = 0$$

The first equation can only be satisfied if $\theta = \frac{\pi}{2}$ (recall that $\theta \in (0, \pi)$), i.e., the line is along the equator, then we just take $x^\phi = \phi_0 + a\lambda$ and we have a geodesic.

(b)



parallel-transport around a circle of constant latitude.

$$V^\mu = (1, 0)$$

parallel transport:

$$\frac{dV^\mu}{d\lambda} + \Gamma^\mu_{\sigma\rho} \frac{dx^\sigma}{d\lambda} V^\rho = 0.$$

we have

$$x^\mu = (\theta_0, \phi_0 + a\lambda),$$

for simplicity,

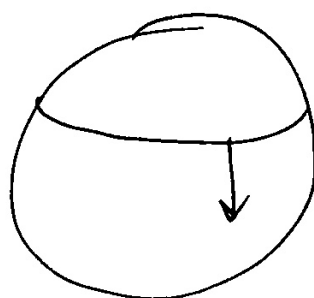
2π -periodic?

then we get

$$\begin{cases} \frac{dV^\theta}{d\lambda} + \Gamma_{\phi\phi}^\theta \frac{dx^\phi}{d\lambda} V^\phi = 0, \\ \frac{dV^\phi}{d\lambda} + \Gamma_{\theta\phi}^\phi \left(\frac{dx^\theta}{d\lambda} V^\phi + \frac{dx^\phi}{d\lambda} V^\theta \right) = 0. \end{cases}$$

now by our parametrization

$$\frac{dx^\theta}{d\lambda} = 0 \quad \frac{dx^\phi}{d\lambda} = a.$$



$$\begin{cases} \frac{dV^\theta}{d\lambda} - a \sin\theta_0 \cos\theta_0 V^\phi = 0 \\ \frac{dV^\phi}{d\lambda} + a \cot\theta_0 V^\theta = 0 \end{cases}$$

now we have

$$\frac{d^2 V^\theta}{d\lambda^2} - a \sin\theta_0 \cos\theta_0 \frac{dV^\phi}{d\lambda} = 0$$

$$\Rightarrow \frac{d^2 V^\theta}{d\lambda^2} - a \sin\theta_0 \cos\theta_0 (-a \cot\theta_0 V^\theta) = 0$$

$$\Rightarrow \frac{d^2 V^\theta}{d\lambda^2} + a^2 \frac{\sin\theta_0 \cos\theta_0}{\sin\theta_0} V^\theta = 0$$

$$\Rightarrow \frac{d^2 V^\theta}{d\lambda^2} + a^2 \cos^2\theta_0 V^\theta = 0$$

$$\frac{d^2 V^\phi}{d\lambda^2} + a \cot\theta_0 \frac{dV^\theta}{d\lambda} \quad \frac{d^2 V^\phi}{d\lambda^2} + a^2 \cos^2\theta_0 V^\phi = 0$$

$$\Rightarrow \frac{d^2 V^\phi}{d\lambda^2} + a \cot\theta_0 (a \sin\theta_0 \cos\theta_0 V^\phi) = 0 \quad (23)$$

thus

$$V^\theta(\lambda) = A \cos(\omega\lambda) + B \sin(\omega\lambda)$$

$$V^\phi(\lambda) = C \cos(\omega\lambda) + D \sin(\omega\lambda)$$

now

$$V^\theta(0) = 1 \text{ hence } A = 1$$

$$V^\phi(0) = 0 \text{ hence } C = 0,$$

thus

$$V^\theta(\lambda) = \cos(\omega\lambda) + B \sin(\omega\lambda)$$

$$V^\phi(\lambda) = D \sin(\omega\lambda)$$

but since

$$\frac{dV^\theta}{d\lambda} = a \sin\theta \cos\theta V^\phi, \quad \frac{dV^\phi}{d\lambda} = -a \cot\theta V^\theta$$

we have

$$\left. \frac{dV^\theta}{d\lambda} \right|_{\lambda=0} = a \sin\theta \cos\theta V^\phi(0) = 0$$

$$\left. \frac{dV^\phi}{d\lambda} \right|_{\lambda=0} = -a \cot\theta V^\theta(0) = -a \cot\theta$$

we get from the prev.

$$\frac{dV^\theta}{d\lambda} = -\omega \sin(\omega\lambda) + \omega B \cos(\omega\lambda)$$

$$\frac{dV^\phi}{d\lambda} = \omega D \cos(\omega\lambda)$$

thus in merid $B=0$ and $WD = -a \cot \theta$,
or

$$D = - \frac{a \cot \theta}{w} = - \frac{d \cot \theta}{d \cos \theta} \\ = - \frac{1}{\sin \theta}$$

thus

$$V^{\theta}(\lambda) = \cos(w\lambda)$$

$$V^{\phi}(\lambda) = - \frac{\sin(w\lambda)}{\sin \theta_0}$$

that is,

$$V^{\theta}(\lambda) = \cos(a \cos \theta_0 \lambda)$$

$$V^{\phi}(\lambda) = - \frac{\sin(a \cos \theta_0 \lambda)}{\sin \theta_0}$$

when $a\lambda = 2\pi$ we have the same?
length along a constant - latitude line.

$$L = \int \left(\sin^2 \theta a \cdot a \right)^{1/2} d\lambda$$

$$a\lambda = 2\pi$$

↓

$$= \int_0^{\frac{2\pi}{a}} a \sin \theta d\lambda$$

$$\lambda = \frac{2\pi}{a}$$

$$= 2\pi \sin \theta_0$$

$$a\lambda = \phi$$

total length,

$$\Rightarrow \lambda = \frac{\phi}{a}$$

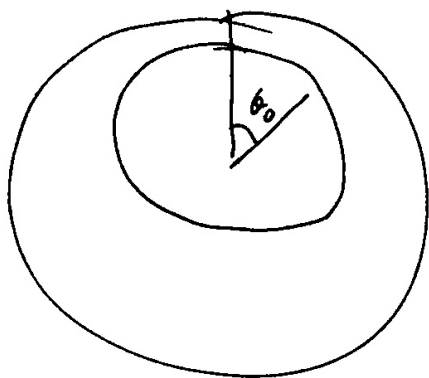
$$= \phi \sin \theta_0$$

we have

$$\frac{dV^\theta}{dt} - \sin \theta_0 \cos \theta_0 V^\phi = 0$$

$$\frac{dV^\phi}{dt} + \cot \theta_0 V^\theta = 0$$

$$x^M(t) = (\theta_0, \lambda)$$



length of this circle is

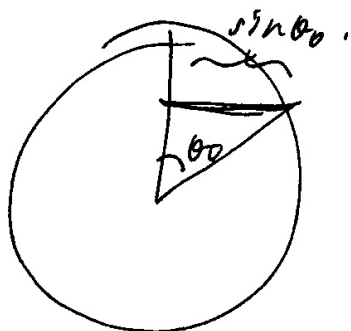
$$x^M(t) = (\theta_0, \lambda)$$

$$\gamma: [0, 2\pi) \rightarrow M$$

$$\int_\gamma ds = \int_0^{2\pi} \sqrt{g_{\phi\phi}} dt$$

$$= \int_0^{2\pi} \sin \theta_0 dt$$

$$= 2\pi \sin \theta_0.$$



length circumference: $2\pi r$,

thus the total length is

$$2\pi \sin \theta_0, \text{ and}$$

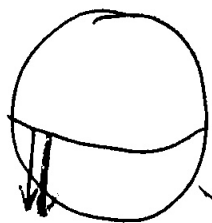
corresponds to $t \rightarrow 2\pi$,

Answer :

$$(1, 0) \rightarrow (\cos(2\pi \cos \theta_0), -\csc \theta_0 \sin(2\pi \cos \theta_0)) \quad (20)$$

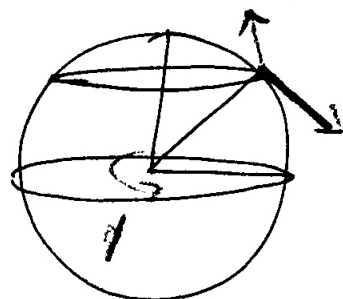
special cases:

$$\theta_0 = \frac{\pi}{2} \rightarrow (1, 0)$$



$$\theta_0 = \frac{\pi}{4} \rightarrow (\cos(\sqrt{2}\pi), -\sqrt{2}\sin(\sqrt{2}\pi))$$

$$\approx (-0.27, 1.36)$$



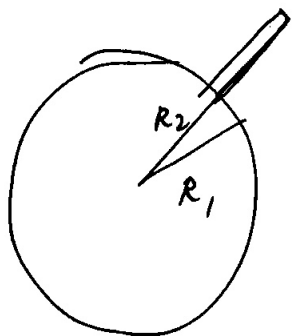
initial

3.6

$$ds^2 = -(1 + 2\Phi) dt^2 + (1 - 2\Phi) dr^2$$

$$+ r^2 (d\theta^2 + \sin^2\theta d\phi^2)$$

when $\Phi = -\frac{GM}{r}$



two clocks

Exercise 3.6

A good approximation to the metric outside the surface of the Earth is provided by

$$ds^2 = -(1 + 2\Phi)dt^2 + (1 - 2\Phi)dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2), \quad (3.218)$$

where

$$\Phi = -\frac{GM}{r} \quad (3.219)$$

may be thought of as the familiar Newtonian gravitational potential. Here G is Newton's constant and M is the mass of the earth. For this problem Φ may be assumed to be small.

- (a) Imagine a clock on the surface of the Earth at distance R_1 from the Earth's center, and another clock on a tall building at distance R_2 from the Earth's center. Calculate the time elapsed on each clock as a function of the coordinate time t . Which clock moves faster?
- (b) Solve for a geodesic corresponding to a circular orbit around the equator of the Earth ($\theta = \pi/2$). What is $d\phi/dt$?
- (c) How much proper time elapses while a satellite at radius R_1 (skimming along the surface of the earth, neglecting air resistance) completes one orbit? You can work to first order in Φ if you like. Plug in the actual numbers for the radius of the Earth and so on (don't forget to restore the speed of light) to get an answer in seconds. How does this number compare to the proper time elapsed on the clock stationary on the surface?

Solution. (a) These two clocks follow the trajectories

$$x_1^\mu(t) = (t, R_1, \theta_1, \phi_1), \quad x_2^\mu(t) = (t, R_2, \theta_2, \phi_2),$$

where t is the coordinate time. The proper time experienced by these clocks is then

$$\begin{aligned} \Delta\tau &= \int \sqrt{-g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}} dt \\ &= \int \sqrt{1 + 2\Phi} dt \\ &= \sqrt{1 + 2\Phi} \Delta t. \end{aligned}$$

For the first clock, the proper time elapsed for an advance in coordinate time Δt is

$$\Delta\tau_1 = \sqrt{1 - \frac{2GM}{R_1}} \Delta t,$$

and the proper time elapsed for the second clock, for the same advance Δt , is

$$\Delta\tau_2 = \sqrt{1 - \frac{2GM}{R_2}} \Delta t > \Delta\tau_1.$$

Thus the clock farther from the center of the Earth moves faster. Incidentally, the advance in coordinate time Δt is also the time elapsed on a clock fixed at infinity.

(b) We want to solve the geodesic equation

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\sigma\rho} \frac{dx^\sigma}{d\tau} \frac{dx^\rho}{d\tau} = 0.$$

A straightforward calculation shows that the nonzero Christoffel symbols are

$$\begin{aligned} \Gamma^t_{tr} &= \frac{\Phi'}{1+2\Phi} & \Gamma^\theta_{r\theta} &= \frac{1}{r} \\ \Gamma^r_{tt} &= \frac{\Phi'}{1-2\Phi} & \Gamma^\theta_{\phi\phi} &= -\sin\theta \cos\theta \\ \Gamma^r_{rr} &= -\frac{\Phi'}{1-2\Phi} & \Gamma^\phi_{r\phi} &= \frac{1}{r} \\ \Gamma^r_{\theta\theta} &= -\frac{r}{1-2\Phi} & \Gamma^\phi_{\theta\phi} &= \cot\theta. \\ \Gamma^r_{\phi\phi} &= -\frac{r \sin^2\theta}{1-2\Phi} \end{aligned}$$

We want a circular orbit around the equator of the Earth ($\theta = \pi/2$), so that only t and ϕ change. So the geodesic equation reduces to

$$\begin{aligned} \ddot{t} &= 0 \\ \frac{\Phi'}{1-2\Phi} \dot{t}^2 - \frac{R \sin^2\theta}{1-2\Phi} \dot{\phi}^2 &= 0 \\ -\sin\theta \cos\theta \dot{\phi}^2 &= 0 \\ \ddot{\phi} &= 0. \end{aligned}$$

The first and fourth equations give immediately

$$t = a\tau + b, \quad \phi = c\tau + d,$$

where a, b, c, d are constants. The third equation is trivially satisfied, since $\theta = \pi/2$ for this trajectory. Thus we have

$$x^\mu(\tau) = \left(a\tau + b, R, \frac{\pi}{2}, c\tau + d \right).$$

We require normalization of the four-velocity. Note that

$$u^\mu = (a, 0, 0, c).$$

Thus, using the given metric, we have

$$u^\mu u_\mu = -\left(1 - \frac{2GM}{R}\right) a^2 + R^2 c^2 = -1,$$

which can be solved to give (we choose the positive root for concreteness)

$$c = \frac{1}{R} \sqrt{-1 + a^2(1 + 2\Phi)}.$$

Finally, we impose the constraint given by the r component of the geodesic equation. We need

$$\Phi' \dot{t}^2 - R \dot{\phi}^2 = 0 \quad \implies \quad \frac{GM}{R^2} \dot{t}^2 - R \dot{\phi}^2 = 0,$$

which fixes

$$\boxed{\left(\frac{d\phi}{dt}\right)^2 = \frac{GM}{R^3}},$$

or, equivalently

$$c^2 = \frac{GM}{R_1^3} a^2.$$

Interestingly, the boxed equation is formally identical to Kepler's law $\Omega^2 = GM/R^3$. Substituting our normalization condition into this, then solving, we get

$$a = (1 + 3\Phi)^{-1/2}, \quad c = \left(\frac{GM}{R^3}\right)^{1/2} (1 + 3\Phi)^{-1/2}.$$

Thus, we conclude that

$$\boxed{x^\mu(\tau) = \left((1 + 3\Phi)^{-1/2} \tau, R, \frac{\pi}{2}, (1 + 3\Phi)^{-1/2} \left(\frac{GM}{R^3}\right)^{1/2} \tau \right)},$$

where we've dropped the additive constants of integration, which only serve to fix an initial coordinate time t and angle ϕ .

(c) For a satellite that orbits around the Earth at radius R_1 , the proper time elapsed in one orbit is given by

$$\Delta\tau = 2\pi \left(1 - \frac{3GM}{R_1}\right)^{1/2} \left(\frac{GM}{R_1^3}\right)^{-1/2}.$$

To restore units, note that $(GM/R^3)^{-1/2}$ already has the units of seconds, and for the other term we note that GM/Rc^2 is dimensionless. Hence

$$\Delta\tau_{\text{orb}} = 2\pi \left(1 - \frac{3GM}{R_1 c^2}\right)^{1/2} \left(\frac{GM}{R_1^3}\right)^{-1/2} \approx \boxed{5060.9088348 \text{ s}}.$$

Compare to a clock stationary on the surface: the corresponding advance in coordinate time is

$$\Delta t = (1 + 3\Phi)^{-1/2} \Delta\tau = 2\pi \left(\frac{GM}{R_1^3}\right)^{-1/2},$$

so by the result of part (a) we have, again restoring the speed of light,

$$\Delta\tau_{\text{stationary}} = \left(1 - \frac{2GM}{R_1 c^2}\right)^{1/2} 2\pi \left(\frac{GM}{R_1^3}\right)^{-1/2} \approx \boxed{5060.9088365 \text{ s}}.$$

Thus the orbiting clock runs slightly slower; every complete orbit accumulates a difference of around $1.76 \mu\text{s}$ between the two. Over the course of 24 hours, which is around 17.07 orbits, this adds up to a difference of around $30.1 \mu\text{s}$. This is small but not negligible, especially for precision applications – if you were to neglect this correction, then over a year, the two clocks would drift apart by around 11.0 milliseconds.

We also consider a satellite in a geostationary orbit, which corresponds to $R_2 = 6.2164 \times 10^7 \text{ m}$. In this case we have

$$\Delta\tau_{\text{orb}} = 2\pi \left(1 - \frac{3GM}{R_2 c^2}\right)^{1/2} \left(\frac{GM}{R_2^3}\right)^{-1/2} \approx \boxed{86164.78603 \text{ s}},$$

while for a clock stationary on the surface of the Earth,

$$\Delta\tau_{\text{stationary}} = \left(1 - \frac{2GM}{R_1 c^2}\right)^{1/2} 2\pi \left(\frac{GM}{R_2^3}\right)^{-1/2} \approx \boxed{86164.78599 \text{ s}}.$$

This time it is the clock on the Earth that runs slow, and every complete orbit accumulates a difference of around $46.4 \mu\text{s}$ between the two. Over the course of a year, the two clocks would drift apart by around 17.0 milliseconds.

Notice that for an Earth-radius orbit, the satellite clock runs slow relative to the stationary surface clock, but for the geosynchronous orbit, the satellite clock runs fast. This is because in the first case, the two clocks are subject to the same gravitational field, so there is no relative dilation as in (a). But since the satellite is moving relative to the stationary observer, its clock runs slower. In the second case, there are two competing effects – the satellite clock tends to run slow due to the motion, but also runs faster because it is in a weaker field. The latter effect wins out. Interestingly, this implies that there is some intermediate orbital radius where there is no relative dilation between the satellite and stationary surface-clock times.

One can show, in the same way, that a satellite orbiting at the radius of a GPS satellite would run about $38 \mu\text{s}$ faster than a surface clock. This is a result you’ll see quoted online, but somewhat confusingly, many sources will talk about competing “gravitational time dilation” and “special-relativistic time dilation” effects, as though these are distinct physical phenomena which need to be computed separately and then combined. The beauty of GR is that it contains all of SR, so there’s no need to worry about whether the latter has been accounted for. $\square$

Exercise 3.8

The metric for the three-sphere in coordinates $x^\mu = (\psi, \theta, \phi)$ can be written

$$ds^2 = d\psi^2 + \sin^2 \psi (d\theta^2 + \sin^2 \theta d\phi^2).$$

- (a) Calculate the Christoffel connection coefficients. Use whatever method you like, but it is good practice to get the connection coefficients by varying the integral (3.49).
- (b) Calculate the Riemann tensor, Ricci tensor, and Ricci scalar.
- (c) Show that (3.191) is obeyed by this metric, confirming that the three-sphere is a maximally

symmetric space (as you would expect).

Solution.

□

Exercise 3.12

Show that any Killing vector K^μ satisfies the relations mentioned in the text:

$$\begin{aligned}\nabla_\mu \nabla_\sigma K^\rho &= R^\rho_{\sigma\mu\nu} K^\nu \\ K^\lambda \nabla_\lambda R &= 0.\end{aligned}\tag{3.223}$$

Solution. We proved the first relation in Exercise 7.15 of d’Inverno. The second relation can be proven as follows. First, note that by the contracted Bianchi identity,

$$K^\lambda \nabla_\lambda R = 2K^\lambda \nabla^\mu R_{\lambda\mu}.$$

Thus it suffices to show that the quantity on the right vanishes. We have

$$\begin{aligned}K^\lambda \nabla^\mu R_{\lambda\mu} &= \nabla^\mu (K^\lambda R_{\lambda\mu}) - \cancel{R_{\lambda\mu} \nabla^\mu K^\lambda} \\ &= \nabla^\mu (\nabla_\alpha \nabla_\mu K^\alpha) \\ &= \nabla^\mu \nabla^\alpha \nabla_\mu K_\alpha \\ &= [\nabla^\mu, \nabla^\alpha] \nabla_\mu K_\alpha,\end{aligned}$$

where the second line follows from the product rule, and the term $R_{\lambda\mu} \nabla^\mu K^\lambda$ vanishes because we are contracting a pair of symmetric indices against a pair of antisymmetric ones. The third line follows from (3.177), the fourth line simultaneously raises and lowers dummy indices, and the fifth line follows because $\nabla_\mu K_\alpha$ is antisymmetric (by Killing’s equation), so we are free to take only the antisymmetric part of $\nabla^\mu \nabla^\alpha$. (This is the same reasoning used in line 2.)

From here we apply the product rule for the commutator of covariant derivatives, which we established in Exercise 6.10 of d’Inverno. In particular, since

$$[\nabla_\mu, \nabla_\nu](t^\alpha \omega_\beta) = R^\alpha_{\sigma\mu\nu} t^\sigma \omega_\beta - R^\sigma_{\beta\mu\nu} t^\alpha \omega_\sigma,$$

we have (raising and lowering indices appropriately)

$$\begin{aligned}[\nabla^\mu, \nabla^\alpha] \nabla_\mu K_\alpha &= R_{\mu\sigma}{}^{\mu\alpha} \nabla^\sigma K_\alpha - R^\sigma{}_\alpha{}^{\mu\alpha} \nabla_\mu K_\sigma \\ &= R_\sigma{}^\alpha \nabla^\sigma K_\alpha - R^{\sigma\mu} \nabla_\mu K_\sigma \\ &= 0,\end{aligned}$$

where both terms on the second line vanish. This shows that $K^\lambda \nabla^\mu R_{\lambda\mu} = 0$, and therefore

$$\boxed{K^\lambda \nabla_\lambda R = 0}$$

as we wanted to show. That is, the directional derivative of the Ricci scalar along a Killing vector field is zero. $\square$

Exercise 3.13

Find explicit expressions for a complete set of Killing vector fields for the following spaces:

- (a) Minkowski space, with metric $ds^2 = -dt^2 + dx^2 + dy^2 + dz^2$.
- (b) A spacetime with coordinates $\{u, v, x, y\}$ and metric

$$ds^2 = -(du\,dv + dv\,du) + a^2(u)\,dx^2 + b^2(u)\,dy^2,$$

where a and b are unspecified functions of u . This represents a gravitational wave spacetime. (*Hints*, which you need not show: there are five Killing vectors in all, and all of them have a vanishing u component K^u .)

Be careful, in all of these cases, about the distinction between upper and lower indices.

Solution. $\square$

Exercise 3.14

Consider the three Killing vectors of the two-sphere, (3.188). Show that their commutators satisfy the following algebra:

$$[R, S] = T$$

$$[S, T] = R$$

$$[T, R] = S.$$

Solution. $\square$

Exercise 3.15

Use Raychaudhuri's equation, discussed in Appendix F, to show that, if a fluid is flowing on geodesics through spacetime with zero shear and expansion, then spacetime must have a timelike Killing vector.

Solution. $\square$

Exercise 3.16

Consider again the metric on a three-sphere,

$$ds^2 = d\psi^2 + \sin^2 \psi (d\theta^2 + \sin^2 \theta d\phi^2).$$

In this problem we make use of noncoordinate bases, discussed in Appendix J. In an orthonormal frame of one-forms $\hat{\theta}^{(a)}$ the metric would become

$$ds^2 = \hat{\theta}^{(1)} \otimes \hat{\theta}^{(1)} + \hat{\theta}^{(2)} \otimes \hat{\theta}^{(2)} + \hat{\theta}^{(3)} \otimes \hat{\theta}^{(3)}.$$

- (a) Find such an orthonormal frame of one-forms, such that the matrix e_a^μ is diagonal. Don't worry about covering the entire manifold.
- (b) Compute the components of the spin connection by solving $de^a + \omega^a_b \wedge e^b = 0$.
- (c) Compute the components of the Riemann tensor $R^\rho_{\sigma\mu\nu}$ in the coordinate basis adapted to x^μ by computing the components of the curvature two-form $R^a_{b\mu\nu}$ and then converting.

Solution.

□

Exercise 4.1

The Lagrange density for electromagnetism in curved space is

$$\mathcal{L} = \sqrt{-g} \left(-\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + A_\mu J^\mu \right),$$

where J^μ is the conserved current.

- (a) Derive the energy-momentum tensor by functional differentiation with respect to the metric.
- (b) Consider adding a new term to the Lagrangian,

$$\mathcal{L}' = \beta R^{\mu\nu} g^{\rho\sigma} F_{\mu\rho} F_{\nu\sigma}.$$

How are Maxwell's equations altered in the presence of this term? Einstein's equation? Is the current still conserved?

Solution.

□

Exercise 4.3

The four-dimensional δ -function on a manifold M is defined by

$$\int_M F(x^\mu) \left[\frac{\delta^{(4)}(x^\sigma - y^\sigma)}{\sqrt{-g}} \right] \sqrt{-g} d^4x = F(y^\sigma),$$

for an arbitrary function $F(x^\mu)$. Meanwhile, the energy-momentum tensor for a pressureless perfect fluid (dust) is

$$T^{\mu\nu} = \rho U^\mu U^\nu,$$

where ρ is the energy density and U^μ is the four-velocity. Consider such a fluid that consists of a single particle traveling on a world line $x^\mu(\tau)$, with τ the proper time. The energy-momentum tensor for this fluid is then given by

$$T^{\mu\nu}(y^\sigma) = m \int_M \left[\frac{\delta^{(4)}(y^\sigma - x^\sigma(\tau))}{\sqrt{-g}} \right] \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} d\tau,$$

where m is the rest mass of the particle. Show that covariant conservation of the energy-momentum tensor, $\nabla_\mu T^{\mu\nu} = 0$, implies that $x^\mu(\tau)$ satisfies the geodesic equation.

Solution.

□

Exercise 4.4

Show that the energy-momentum tensors for electromagnetism and for scalar field theory satisfy the dominant energy condition, and thus also the weak, null, and null dominant conditions. Show that they also satisfy $w \geq -1$.

Solution.

□

Baumgarte and Shapiro, Numerical Relativity: SfS

Exercise 1.1

Consider a point mass M located at a position $\mathbf{r}_M$. At a position $\mathbf{r}$, the Newtonian potential Φ created by the point mass is

$$\Phi(\mathbf{r}) = -\frac{GM}{s}, \quad (1.16)$$

where we have defined $\mathbf{s} = \mathbf{r} - \mathbf{r}_M$ and $s = |\mathbf{s}|$.

(a) Show that the tidal tensor can be written in the compact form

$$\mathcal{R}_{ij} = \frac{GM}{s^5} (\delta_{ij}s^2 - 3s_i s_j). \quad (1.17)$$

(b) Choose a (Cartesian) coordinate system in which $\mathbf{r}_M = (0, 0, z_M)$ and find all non-zero components of the tidal tensor at the origin, $\mathbf{r} = (0, 0, 0)$.

(c) Consider two particles close to the origin that are separated by a distance $\Delta z \ll z_M$ along the direction vector towards M , i.e. $\Delta x^i = (0, 0, \Delta z)$, and find $d^2 \Delta z / dt^2$. Then consider two particles close to the origin that are separated by a distance $\Delta x \ll z_M$ in a direction orthogonal to the direction towards M , i.e. $\Delta x^i = (\Delta x, 0, 0)$, and find $d^2 \Delta x / dt^2$. If all goes well, this should explain why objects in the gravitational field of a companion, e.g. the Earth in the gravitational field of the Moon, are elongated in the direction of the companion and compressed in the orthogonal directions.

(d) Show that the trace $\mathcal{R} = \mathcal{R}^i_i$ of the tidal tensor (1.17) vanishes for all $s > 0$, as expected from the field equation (1.15).

Solution. (a) From the definition,

$$\mathcal{R}_{ij} = \partial_i \partial_j \left(-\frac{GM}{s} \right).$$

To compute the derivatives, we note that

$$\begin{aligned} \partial_j \left(\frac{1}{s} \right) &= -\frac{1}{s^2} \partial_j s, \\ \partial_i \partial_j \left(\frac{1}{s} \right) &= \frac{2}{s^3} (\partial_i s) (\partial_j s) - \frac{1}{s^2} \partial_i \partial_j s. \end{aligned}$$

Now since

$$s = \sqrt{(r - r_M)_i (r - r_M)^i},$$

we have

$$\begin{aligned}
\partial_j s &= \frac{1}{2\sqrt{(r-r_M)_i(r-r_M)^i}} 2\delta_{jk}(r-r_M)^k \\
&= \frac{s_j}{s}, \\
\partial_i \partial_j s &= -\frac{1}{s^2} s_j (\partial_i s) + \frac{1}{s} \partial_i s_j \\
&= -\frac{1}{s^2} s_j \left(\frac{s_i}{s} \right) + \frac{1}{s} \delta_{ij} \\
&= -\frac{s_i s_j}{s^3} + \frac{\delta_{ij}}{s}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
\partial_i \partial_j \left(\frac{1}{s} \right) &= \frac{2}{s^3} (\partial_i s) (\partial_j s) - \frac{1}{s^2} \partial_i \partial_j s \\
&= \frac{2}{s^3} \frac{s_i s_j}{s^2} - \frac{1}{s^2} \left(-\frac{s_i s_j}{s^3} + \frac{\delta_{ij}}{s} \right) \\
&= \frac{2s_i s_j}{s^5} + \frac{s_i s_j}{s^5} - \frac{\delta_{ij}}{s^3} \\
&= \frac{3s_i s_j - s^2 \delta_{ij}}{s^5}.
\end{aligned}$$

This shows that

$$\boxed{\mathcal{R}_{ij} = \frac{GM}{s^5} (\delta_{ij} s^2 - 3s_i s_j)},$$

as we wanted.

(b) For $\mathbf{r}_M = (0, 0, z_M)$ and $\mathbf{r} = (0, 0, 0)$ we have

$$\mathbf{s} = \mathbf{r} - \mathbf{r}_M = (0, 0, -z_M).$$

Thus $s_i = -z_M \delta_{iz}$, and $s^2 = z_M^2$. The tidal tensor is

$$\mathcal{R}_{ij} = \frac{GM}{z_M^5} (\delta_{ij} z_M^2 - 3z_M^2 \delta_{iz} \delta_{jz}).$$

The only nonzero components are

$$\boxed{
\begin{aligned}
\mathcal{R}_{xx} &= \frac{GM}{z_M^3}, \\
\mathcal{R}_{yy} &= \frac{GM}{z_M^3}, \\
\mathcal{R}_{zz} &= -\frac{2GM}{z_M^3}.
\end{aligned}
}$$

(c) Using the Newtonian deviation equation (1.12), we have for the particles separated along the z axis, with $\Delta x^i = (0, 0, \Delta z)$

$$\begin{aligned}\frac{d^2 \Delta z}{dt^2} &= -\mathcal{R}_{iz} \Delta x^i \\ &= -\mathcal{R}_{zz} \Delta z \\ &= \frac{2GM}{z_M^3} \Delta z.\end{aligned}$$

To leading order, the separation between the particles grows exponentially; that is, an object would be elongated in the z direction. For particles separated by $\Delta x^i = (\Delta x, 0, 0)$, we have

$$\begin{aligned}\frac{d^2 \Delta x}{dt^2} &= -\mathcal{R}_{ix} \Delta x^i \\ &= -\mathcal{R}_{xx} \Delta x \\ &= -\frac{GM}{z_M^3} \Delta x.\end{aligned}$$

To leading order, the separation decays; an object would be compressed in the x direction. (We've also considered our initial conditions; since the particles are initially stationary and have nonzero separation, only the cosine term survives in the solution.) The same argument applies for particles separated along the y direction. This explains why we get elongation in the direction of the companion, and compression in the orthogonal directions.

(d) This follows because $\delta^i_i = 3$ in three dimensions, hence

$$\mathcal{R}_i^i = \frac{GM}{s^5} (\delta_i^i s^2 - 3s_i s^i) = 0.$$

Note, however, that the denominator goes to zero when evaluated at $s = 0$, which is the location of the companion. $\square$

Exercise 1.5

Consider a new “isotropic” radius r that satisfies

$$R = r \left(1 + \frac{M}{2r} \right)^2. \quad (1.44)$$

(a) Show that under this transformation the Schwarzschild metric (1.38) takes the form

$$ds^2 = - \left(\frac{1 - M/2r}{1 + M/2r} \right)^2 dt^2 + \psi^4 (dr^2 + r^2 d\Omega^2) \quad (1.45)$$

and find the “conformal factor” ψ . Note that the spatial part of this metric takes the form $\psi^4 \eta_{ij}$, where η_{ij} is the flat metric in any coordinate system (here, in spherical coordinates; in Cartesian coordinates $\eta_{ij} = \delta_{ij}$). This form of the spacetime metric is called “isotropic”

since its spatial part does not single out any direction.

- (b) Find the inverse expression for r in terms of R . Show that real solutions for r exist only for $R \geq 2M$.

Solution. (a) Since

$$R = r \left(1 + \frac{M}{2r} \right)^2,$$

we have

$$\begin{aligned} dR &= dr \left(1 + \frac{M}{2r} \right)^2 + 2r \left(1 + \frac{M}{2r} \right) \left(-\frac{M}{2r^2} \right) dr \\ &= \left[\left(1 + \frac{M}{r} + \frac{M^2}{4r^2} \right) - \frac{M}{r} \left(1 + \frac{M}{2r} \right) \right] dr \\ &= \left[1 + \frac{M^2}{4r^2} - \frac{M^2}{2r^2} \right] dr \\ &= \left[1 - \frac{M^2}{4r^2} \right] dr. \end{aligned}$$

Thus

$$dR^2 = \left(1 - \frac{M^2}{4r^2} \right)^2 dr^2.$$

We also have

$$\begin{aligned} 1 - \frac{2M}{R} &= 1 - \frac{2M}{r} \frac{1}{(1 + M/2r)^2} \\ &= 1 - \frac{2M}{r} \frac{4r^2}{(2r + M)^2} \\ &= 1 - \frac{8Mr}{(2r + M)^2} \\ &= \frac{4r^2 + 4Mr + M^2 - 8Mr}{(M + 2r)^2} \\ &= \frac{4r^2 - 4Mr + M^2}{(M + 2r)^2} \\ &= \frac{(M - 2r)^2}{(M + 2r)^2} \\ &= \left(\frac{1 - M/2r}{1 + M/2r} \right)^2. \end{aligned}$$

Putting these together, we get

$$\left(1 - \frac{2M}{R} \right)^{-1} dR^2 = \left(\frac{1 + M/2r}{1 - M/2r} \right)^2 (1 - M^2/4r^2)^2 dr^2 = \left(1 + \frac{M}{2r} \right)^4 dr^2.$$

Finally, we have from the definition

$$R^2 d\Omega^2 = r^2 \left(1 + \frac{M}{2r}\right)^4 d\Omega^2.$$

Putting everything together, we have in isotropic coordinates

$$ds^2 = -\left(\frac{1 - M/2r}{1 + M/2r}\right)^2 dt^2 + \left(1 + \frac{M}{2r}\right)^4 (dr^2 + r^2 d\Omega^2),$$

which matches (1.45) with conformal factor

$$\boxed{\psi = 1 + \frac{M}{2r}}.$$

This is what we wanted to show.

(b) To invert the transformation, we note from the definition that

$$\begin{aligned} 4Rr^2 &= r(2r + M)^2 \\ &= r(4r^2 + 4Mr + M^2) \\ &= 4r^3 + 4Mr^2 + M^2r, \end{aligned}$$

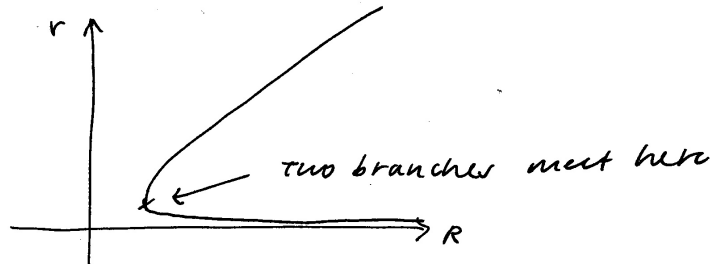
which can be regrouped and multiplied through by 1/4 to give

$$0 = r \left[r^2 + (M - R)r + \frac{M^2}{4} \right].$$

Equating the term in brackets to zero and solving, we get

$$\begin{aligned} r &= \frac{R - M \pm \sqrt{(R - M)^2 - 4(1)(M^2/4)}}{2(1)} \\ &= \frac{R - M \pm \sqrt{R^2 - 2RM + M^2 - M^2}}{2} \\ &= \frac{R - M \pm \sqrt{R(R - 2M)}}{2}. \end{aligned}$$

Thus real solutions exist only for $R > 2M$. There are two solutions; one that increases and another that decreases with R . The two meet at $R = 2M$ (see drawing).



□

Exercise 1.6

- (a) Show that under the coordinate transformation

$$dt = d\bar{t} - \frac{1}{f_0} \frac{M}{R-M} dR \quad (1.46)$$

the Schwarzschild metric (1.38) takes the new form

$$ds^2 = -f_0 d\bar{t}^2 + \frac{2M}{R-M} d\bar{t} dR + \left(\frac{R}{R-M} \right)^2 dR^2 + R^2 d\Omega^2, \quad (1.47)$$

where $f_0 = 1 - 2M/R$.

- (b) Now transform to a new spatial coordinate $\bar{r} = R - M$ and show that the new metric takes the form

$$ds^2 = -\frac{\bar{r} - M}{\bar{r} + M} d\bar{t}^2 + \frac{2M}{\bar{r}} d\bar{t} d\bar{r} + \psi^4 (d\bar{r}^2 + \bar{r}^2 d\Omega^2) \quad (1.48)$$

with

$$\psi = \left(1 + \frac{M}{\bar{r}} \right)^{1/2}. \quad (1.49)$$

Note that the spatial part of the metric is again isotropic, so $\bar{r}$ is again an isotropic radius, as in exercise 1.5. Here, however, r is measured on a different $\bar{t} = \text{const}$ slice; therefore it is different from the isotropic radius r in exercise 1.5.

Solution. (a) From the definition,

$$\begin{aligned} dt^2 &= \left(d\bar{t} - \frac{1}{f_0} \frac{M}{R-M} \frac{1}{dR} \right) \left(d\bar{t} - \frac{1}{f_0} \frac{M}{R-M} \frac{1}{dR} \right) \\ &= d\bar{t}^2 - \frac{1}{f_0} \frac{2M}{R-M} d\bar{t} dR + \frac{1}{f_0^2} \frac{M^2}{(R-M)^2} dR^2. \end{aligned}$$

Substituting into the Schwarzschild metric (1.38), we obtain

$$\begin{aligned} ds^2 &= -f_0 \left(d\bar{t}^2 - \frac{1}{f_0} \frac{2M}{R-M} d\bar{t} dR + \frac{1}{f_0^2} \frac{M^2}{(R-M)^2} dR^2 \right) + \frac{1}{f_0} dR^2 + R^2 d\Omega^2 \\ &= -f_0 d\bar{t}^2 + \frac{2M}{R-M} d\bar{t} dR - \frac{1}{f_0} \frac{M^2}{(R-M)^2} dR^2 + \frac{1}{f_0} dR^2 + R^2 d\Omega^2 \\ &= -f_0 d\bar{t}^2 + \frac{2M}{R-M} d\bar{t} dR + \left[-\frac{1}{f_0} \frac{M^2}{(R-M)^2} + \frac{1}{f_0} \right] dR^2 + R^2 d\Omega^2. \end{aligned}$$

But

$$\begin{aligned}
-\frac{1}{f_0} \frac{M^2}{(R-M)^2} + \frac{1}{f_0} &= \frac{1}{f_0} \left(-\frac{M^2}{(R-M)^2} + 1 \right) \\
&= \frac{1}{f_0} \frac{-M^2 + R^2 - 2RM + M^2}{(R-M)^2} \\
&= \frac{1}{f_0} \frac{R(R-2M)}{(R-M)^2} \\
&= \frac{R}{R-2M} \frac{R(R-2M)}{(R-M)^2} \\
&= \left(\frac{R}{R-M} \right)^2,
\end{aligned}$$

so we've recovered (1.47).

(b) Under this coordinate transformation, we have

$$\bar{r} = R - M, \quad d\bar{r} = dR,$$

hence

$$\begin{aligned}
f_0 &= 1 - \frac{2M}{R} \\
&= \frac{R-2M}{R} \\
&= \frac{\bar{r}-M}{\bar{r}+M},
\end{aligned}$$

and we get

$$\begin{aligned}
ds^2 &= -\frac{\bar{r}-m}{\bar{r}+m} d\bar{t}^2 + \frac{2m}{\bar{r}} d\bar{t} d\bar{r} + \left(\frac{\bar{r}+m}{\bar{r}} \right)^2 d\bar{r}^2 + (\bar{r}+m)^2 d\Omega^2 \\
&= -\frac{\bar{r}-m}{\bar{r}+m} d\bar{t}^2 + \frac{2m}{\bar{r}} d\bar{t} d\bar{r} + \left(\frac{\bar{r}+m}{\bar{r}} \right)^2 (\bar{r}^2 + \bar{r}^2 d\Omega^2),
\end{aligned}$$

which takes the form (1.49) with

$$\psi = \left(1 + \frac{M}{\bar{r}} \right)^{1/2},$$

as we wanted to show. □

Exercise 1.7

Find the location of the horizon in terms of the isotropic radius r of Exercise 1.5.

Solution. For Schwarzschild, the apparent and event horizons coincide; further, this horizon is a spherical surface whose location can be identified by locating the radius at which an outgoing radial photon no longer propagates to larger areal radii. So we consider a radial photon; the proper distance along its trajectory, as given by the metric (1.45), is

$$0 = -\left(\frac{1 - M/2r}{1 + M/2r}\right)^2 dt^2 + (1 + M/2r)^4 dr^2.$$

This then gives

$$\frac{dr}{dt} = \pm \frac{(1 - M/2r)}{(1 + M/2r)^3},$$

which shows that the horizon is located at $r = M/2$. As a check, we will see whether this agrees with the result obtained in Schwarzschild coordinates (1.38). Using the inverse transformation of exercise 1.5(b), $r = M/2$ implies that

$$\begin{aligned} M &= R - M \pm \sqrt{R(R - 2M)} \\ \implies 2M - R &= \pm \sqrt{R(R - 2M)} \\ \implies (R - 2M)^2 &= R(R - 2M) \\ \implies 0 &= 2M(R - 2M) \\ \implies R &= 2M, \end{aligned}$$

as expected. Notice, however, that the metric (1.45) doesn't become singular at the horizon, which is a hint that the singularity at $R = 2M$ in (1.38) is merely an artifact of the choice of coordinates. $\square$

Exercise 1.8

Consider the $\bar{t} = \text{const}$, $\theta = \pi/2$ subspace of the line element (1.47), and equate the resulting two-dimensional line element to (1.41). Match the coefficients in the line elements to find a differential equation for z , and then solve this differential equation to obtain

$$z = (4M(2R - M))^{1/2} + M \ln \frac{(2R - M)^{1/2} - M^{1/2}}{(2R - M)^{1/2} + M^{1/2}}, \quad (1.52)$$

where we have chosen the positive root and omitted an irrelevant constant of integration.

Solution. Restricting (1.47) to the $\bar{t} = \text{const}$, $\theta = \pi/2$ subspace gives the 2D line element

$$^{(2)}ds^2 = \left(\frac{R}{R - M}\right)^2 dR^2 + R^2 d\phi^2.$$

Matching this to (1.41), we get

$$1 + \left(\frac{dz}{dR} \right)^2 = \frac{R^2}{(R - M)^2},$$

which implies

$$\begin{aligned} \frac{dz}{dR} &= \pm \left(\frac{R^2}{(R - M)^2} - 1 \right)^{1/2} \\ &= \pm \left(\frac{R^2 - R^2 + 2MR - M^2}{(R - M)^2} \right)^{1/2} \\ &= \pm \left(\frac{M(2R - M)}{(R - M)^2} \right)^{1/2} \\ &= \pm \frac{[M(2R - M)]^{1/2}}{R - M}. \end{aligned}$$

In removing the absolute value sign in the denominator, we put in the assumption that $R > M$. Integrating, we obtain

$$\begin{aligned} z &= \pm \int \frac{[M(2R - M)]^{1/2}}{R - M} dR \\ &= \pm M^{1/2} \int \frac{(2R - M)^{1/2}}{R - M} dR. \end{aligned}$$

We focus on the integral first. The substitution

$$u = (2R - M)^{1/2}, \quad du = \frac{1}{(2R - M)^{1/2}} dR,$$

together with the observation that $R = (u^2 + M)/2$, then gives

$$\begin{aligned} \int \frac{(2R - M)^{1/2}}{R - M} dR &= \int \frac{(2R - M)}{R - M} du \\ &= \int \frac{u^2}{\frac{1}{2}(u^2 + M) - M} du \\ &= 2 \int \frac{u^2}{u^2 - M} du \\ &= 2 \int \left(1 + \frac{M}{u^2 - M} \right) du. \end{aligned}$$

The first term is easily integrated. The second term can be treated as follows:

$$\begin{aligned}
\int \frac{M}{u^2 - M} du &= - \int \frac{1}{1 - \left(\frac{u}{M^{1/2}}\right)^2} du \\
&= -M^{1/2} \int \frac{1}{1 - x^2} dx \\
&= -\frac{1}{2} M^{1/2} \ln \left| \frac{1+x}{1-x} \right| \\
&= -\frac{1}{2} M^{1/2} \ln \frac{x+1}{x-1} \\
&= -\frac{1}{2} M^{1/2} \ln \frac{u + M^{1/2}}{u - M^{1/2}},
\end{aligned}$$

where in the second line we apply the substitution $x = u/M^{1/2}$. The fourth line follows if we assume $R > M$, so that

$$u = (2R - M)^{1/2} > M^{1/2} \implies x > 1.$$

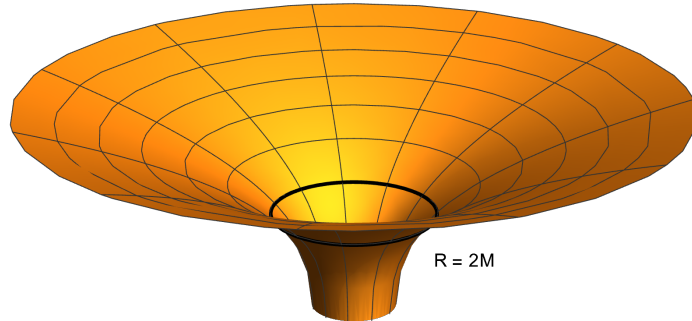
Putting everything together, we obtain

$$\begin{aligned}
\int \frac{(2R - M)^{1/2}}{R - M} dR &= 2u - M^{1/2} \ln \frac{u + M^{1/2}}{u - M^{1/2}} \\
&= 2(2R - M)^{1/2} - M^{1/2} \ln \frac{(2R - M)^{1/2} + M^{1/2}}{(2R - M)^{1/2} - M^{1/2}},
\end{aligned}$$

and finally

$$z = (4M(2R - M))^{1/2} - M \ln \frac{(2R - M)^{1/2} + M^{1/2}}{(2R - M)^{1/2} - M^{1/2}},$$

where we've chosen the positive solution and ignored the overall constant of integration, which only serves to introduce a vertical shift in the surface. The corresponding embedding diagram is shown below, with a black ring to mark the horizon.



□

Exercise 1.9

- (a) Apply a coordinate transformation of the form

$$dt = d\tilde{t} - f(R) dR \quad (1.53)$$

to the Schwarzschild metric (1.38) and find a function $f(R)$ such that the three-dimensional line element of the $\tilde{t} = \text{const}$ slices becomes that of *flat* space, $ds^2 = dR^2 + R^2 d\Omega^2$.

- (b) Show that, under the coordinate transformation of part (a), the four-dimensional line element of the Schwarzschild geometry takes the form

$$ds^2 = -\left(1 - \frac{2M}{R}\right) d\tilde{t}^2 + 2\left(\frac{2M}{R}\right)^{1/2} d\tilde{t} dR + dR^2 + R^2 d\Omega^2. \quad (1.54)$$

These coordinates are called *Painlevé-Gullstrand* coordinates.

Solution. (a) We first note that

$$\begin{aligned} dt^2 &= (d\tilde{t} - f(R) dR)(d\tilde{t} - f(R) dR) \\ &= d\tilde{t}^2 - 2f(R) d\tilde{t} dR + f(R)^2 dR^2. \end{aligned}$$

Substituting into (1.38) and rearranging terms, we obtain

$$\begin{aligned} ds^2 &= -\left(1 - \frac{2M}{R}\right) d\tilde{t}^2 + 2f(R)\left(1 - \frac{2M}{R}\right) d\tilde{t} dR \\ &\quad + \left(1 - \frac{2M}{R}\right)^{-1} \left[1 - \left(1 - \frac{2M}{R}\right)^2 f(R)^2\right] dR^2 + R^2 d\Omega^2. \end{aligned}$$

We want this to reduce to the flat-space metric on $\tilde{t} = \text{const}$ slices. Thus, we require that

$$1 = \left(1 - \frac{2M}{R}\right)^{-1} \left[1 - \left(1 - \frac{2M}{R}\right)^2 f(R)^2\right],$$

or equivalently,

$$1 - \frac{2M}{R} = 1 - f(R)^2 \left(1 - \frac{2M}{R}\right)^2.$$

Let us define $f_0 = 1 - 2M/R$. Then the condition above becomes $f_0 = 1 - f(R)^2 f_0^2$, which has two solutions. Either works; for concreteness we pick the positive root, and get

$$f(R) = \frac{(1 - f_0)^{1/2}}{f_0} = \boxed{\frac{(2M/R)^{1/2}}{1 - 2M/R}}.$$

- (b) Follows immediately upon substituting our choice for $f(R)$ into our expression for ds^2 . $\square$

Exercise 1.10

Consider a *Newtonian* binary consisting of two point masses m_1 and m_2 in circular orbit about their center of mass. Choose coordinates such that the binary system orbits in the xy plane and is aligned with the x axis at $t = 0$.

- (a) Show that the second moment of the mass distribution can be written as

$$I_{ij} = \frac{1}{2}\mu R^2 \begin{pmatrix} 1 + \cos 2\Omega t & \sin 2\Omega t & 0 \\ \sin 2\Omega t & 1 - \cos 2\Omega t & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

where Ω is the orbital angular velocity, $\mu \equiv m_1 m_2 / (m_1 + m_2)$ is the reduced mass, and R is the binary separation.

- (b) Show that the reduced quadrupole moment is

$$\mathcal{I}_{ij} = \frac{1}{2}\mu R^2 \begin{pmatrix} 1/3 + \cos 2\Omega t & \sin 2\Omega t & 0 \\ \sin 2\Omega t & 1/3 - \cos 2\Omega t & 0 \\ 0 & 0 & -2/3 \end{pmatrix}.$$

- (c) Show that the magnitude of the metric perturbations h_{ij} is approximately

$$h \approx \frac{4}{d} \frac{\mu M}{R},$$

where $M = m_1 + m_2$ is the total mass, and where we've used Kepler's third law $\Omega^2 = M/R^3$.

Solution. (a) The center of mass of the binary is located at

$$\mathbf{R}_{\text{CM}} = \frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2}{m_1 + m_2}.$$

Setting this to be the origin of our coordinate system, we obtain the constraint

$$\mathbf{r}_2 = -\frac{m_1}{m_2} \mathbf{r}_1.$$

Furthermore, if we choose coordinates such that the binary orbits in the xy plane and is aligned with the x axis at $t = 0$, then we get

$$\mathbf{r}_1(t) = r_1 \cos \Omega t \hat{\mathbf{i}} + r_1 \sin \Omega t \hat{\mathbf{j}}$$

$$\mathbf{r}_2(t) = -r_2 \cos \Omega t \hat{\mathbf{i}} - r_2 \sin \Omega t \hat{\mathbf{j}}.$$

In these coordinates, the binary separation is given by $R = r_1 + r_2$. Together with our previous

constraint, this implies

$$r_1 + \frac{m_1}{m_2} r_1 = R, \quad \implies \quad r_1 = \frac{m_2 R}{m_1 + m_2}.$$

Therefore, we have

$$\begin{aligned} \mathbf{r}_1(t) &= \frac{m_2 R}{m_1 + m_2} (\cos \Omega t \hat{\mathbf{i}} + \sin \Omega t \hat{\mathbf{j}}) \\ \mathbf{r}_2(t) &= -\frac{m_1 R}{m_1 + m_2} (\cos \Omega t \hat{\mathbf{i}} + \sin \Omega t \hat{\mathbf{j}}), \end{aligned}$$

and the corresponding mass distribution is

$$\rho(\mathbf{r}, t) = m_1 \delta(\mathbf{r} - \mathbf{r}_1(t)) + m_2 \delta(\mathbf{r} - \mathbf{r}_2(t)).$$

Now we compute the second moment of the mass distribution,

$$I_{ij} = \int d^3x \rho x_i x_j.$$

But we have $I_{iz} = 0$, since $\rho x_i z$ is zero when evaluated at the positions of the masses (they lie in the xy plane). So there are only three integrals to compute:

$$\begin{aligned} I_{xx} &= \int d^3x \rho x^2 \\ &= \int d^3x \left[m_1 \delta(\mathbf{r} - \mathbf{r}_1(t)) + m_2 \delta(\mathbf{r} - \mathbf{r}_2(t)) \right] x^2 \\ &= m_1 x_1^2 + m_2 x_2^2 \\ &= \frac{m_1 m_2^2 R^2}{(m_1 + m_2)^2} \cos^2 \Omega t + \frac{m_1^2 m_2}{(m_1 + m_2)^2} \cos^2 \Omega t \\ &= \frac{m_1 m_2}{m_1 + m_2} R^2 \left[\frac{m_2}{m_1 + m_2} + \frac{m_1}{m_1 + m_2} \right] \cos^2 \Omega t \\ &= \mu R^2 \cos^2 \Omega t \\ &= \frac{1}{2} \mu R^2 (1 + \cos 2\Omega t), \\ I_{xy} &= m_1 x_1 y_1 + m_2 x_2 y_2 \\ &= \frac{m_1 m_2 R^2}{(m_1 + m_2)^2} \sin \Omega t \cos \Omega t + \frac{m_1^2 m_2 R^2}{(m_1 + m_2)^2} \sin \Omega t \cos \Omega t \\ &= \frac{m_1 m_2}{m_1 + m_2} R^2 \left[\frac{m_2}{m_1 + m_2} + \frac{m_1}{m_1 + m_2} \right] \sin \Omega t \cos \Omega t \\ &= \mu R^2 \sin \Omega t \cos \Omega t \\ &= \frac{1}{2} \mu R^2 \sin 2\Omega t, \end{aligned}$$

$$\begin{aligned}
I_{zz} &= m_1 y_1^2 + m_2 y_2^2 \\
&= \frac{1}{2} \mu R^2 (1 - \cos 2\Omega t).
\end{aligned}$$

(The steps for computing I_{zz} are similar.) Thus we get

$$I_{ij} = \frac{1}{2} \mu R^2 \begin{pmatrix} 1 + \cos 2\Omega t & \sin 2\Omega t & 0 \\ \sin 2\Omega t & 1 - \cos 2\Omega t & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

as we wanted to show.

(b) The trace of I_{ij} is $I = \mu R^2$, which is easily confirmed by inspection. Thus the reduced quadrupole moment is

$$\begin{aligned}
\mathcal{I}_{ij} &= I_{ij} - \frac{1}{3} \eta_{ij} (\mu R^2) \\
&= I_{ij} - \frac{1}{2} \mu R^2 \left(\frac{2}{3} \eta_{ij} \right),
\end{aligned}$$

and since $\eta_{ij} = \delta_{ij}$ here, we can read off the components:

$$\mathcal{I}_{ij} = \frac{1}{2} \mu R^2 \begin{pmatrix} 1/3 + \cos 2\Omega t & \sin 2\Omega t & 0 \\ \sin 2\Omega t & 1/3 - \cos 2\Omega t & 0 \\ 0 & 0 & -2/3 \end{pmatrix}.$$

(c) Using the weak-field, slow-velocity approximation (1.57),

$$h_{ij}^{\text{TT}} \approx \frac{2}{d} \ddot{\mathcal{I}}_{ij}^{\text{TT}}(t - d),$$

we can estimate the magnitude of the metric perturbations. The two time derivatives of $\ddot{\mathcal{I}}$ bring out a factor of $4\Omega^2$, so we get

$$h \approx \frac{2}{d} \times \frac{1}{2} \mu R^2 \times 4\Omega^2 = \frac{4\mu R^2 \Omega^2}{d}.$$

But if the orbit is approximately Keplerian, then $\Omega^2 = M/R^3$, so this becomes

$$h \approx \frac{4\mu M}{d R},$$

as we wanted to show. This magnitude increases with decreasing binary separation R , and also with increasing total mass M , which makes sense. Interestingly, h also increases if μ increases. Thus an equal-mass binary corresponds to the weakest metric perturbation for a given M . $\square$

Exercise 1.11

- (a) Revisit the *Newtonian* binary of exercise 1.10 and show that gravitational wave emission will result in the binary losing energy E at a rate

$$L_{\text{GW}} = -\frac{dE}{dt} = \frac{32}{5} \frac{M^3 \mu^2}{R^5}. \quad (1.65)$$

- (b) Recall that the equilibrium energy of a binary in circular orbit is given by

$$E_{\text{eq}} = -\frac{1}{2} \frac{M\mu}{R}. \quad (1.66)$$

In response to the energy loss of part (a) the binary separation R will therefore shrink. Assuming that we can consider this inspiral as “adiabatic”, i.e. as a sequence of circular orbits along which the radius changes on timescales much longer than the orbital period, we can compute the inspiral rate from

$$\frac{dR}{dt} = \frac{dE_{\text{eq}}/dt}{dE_{\text{eq}}/dR} = -\frac{L_{\text{GW}}}{dE_{\text{eq}}/dR} = -\frac{64}{5} \frac{M^2 \mu}{R^3}. \quad (1.67)$$

Verify the last equality.

- (c) Integrate equation (1.67) to find the binary separation R as a function of time. Express your result in terms of $T - t$, where T is the time of coalescence, when $R = 0$.
- (d) Combine your result from part (c) with Kepler’s law $\Omega = (M/R^3)^{1/2}$ to compute the orbital angular velocity Ω as a function of time. To leading order, and for quadrupole emission, the frequency of gravitational waves f_{GW} is twice that of the orbit, $f_{\text{GW}} = 2f = \Omega/\pi$. Show that

$$f_{\text{GW}} = \frac{5^{3/8}}{8\pi} \frac{1}{\mathcal{M}^{5/8} (T - t)^{3/8}}, \quad (1.68)$$

where $\mathcal{M} = \mu^{3/5} M^{2/5}$ is the so-called “chirp mass”.

Solution. (a) In the weak-field, slow-velocity regime, we have the quadrupole formula (1.64),

$$\frac{dE}{dt} = -\frac{1}{5} \langle \ddot{\mathcal{I}}_{ij} \ddot{\mathcal{I}}^{ij} \rangle.$$

From our expression in part (b) of the previous exercise, we can read off

$$\ddot{\mathcal{I}}_{ij} = 4\mu\Omega^3 R^2 \begin{pmatrix} \sin 2\Omega t & -\cos 2\Omega t & 0 \\ -\cos 2\Omega t & -\sin 2\Omega t & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Now since the metric is flat, we can also read off

$$\langle \ddot{\mathcal{I}}_{ij} \ddot{\mathcal{I}}^{ij} \rangle = 16\mu^2 \Omega^6 R^4 [2 \cos^2 2\Omega t + 2 \sin^2 2\Omega t] = 32\mu^2 \Omega^4 R^4.$$

Then

$$L_{\text{GW}} = -\frac{dE}{dt} = \frac{32}{5} \mu^2 \Omega^6 R^4 = \boxed{\frac{32}{5} \frac{\mu^2 M^3}{R^5}},$$

where in the last line we used Kepler's law, $\Omega^2 = M/R^3$. This is what we wanted to show.

(b) A straightforward calculation: we have

$$\frac{dE_{\text{eq}}}{dR} = \frac{1}{2} \frac{M\mu}{R^2},$$

so that for an adiabatic inspiral, the radius changes at a rate

$$\frac{dR}{dt} = -\frac{32}{5} \frac{\mu^2 M^3}{R^5} \frac{2R^2}{M\mu} = \boxed{-\frac{64}{5} \frac{M^2 \mu}{R^3}}.$$

(c) By the previous result, we have

$$R^3 dR = -\frac{64}{5} M^2 \mu dt,$$

so that

$$\frac{1}{4} R^4 \Big|_{R(t)}^{R(T)} = -\frac{64}{5} M^2 \mu (T - t).$$

If T is the time of coalescence, with $R(T) = 0$, then this gives

$$R(t)^4 = \frac{256}{5} M^2 \mu (T - t),$$

and then

$$\boxed{R(t) = \left[\frac{256}{5} M^2 \mu (T - t) \right]^{1/4}}.$$

(d) Combining this with Kepler's law gives

$$\begin{aligned} \Omega(t) &= \left\{ \frac{M}{\left[\frac{256}{5} M^2 \mu (T - t) \right]^{3/4}} \right\}^{1/2} \\ &= \frac{M^{1/2}}{\left(\frac{256}{5} \right)^{3/8} M^{3/4} \mu^{3/8} (T - t)^{3/8}} \\ &= \frac{5^{3/8}}{8 \mu^{3/8} M^{1/4} (T - t)^{3/8}}. \end{aligned}$$

Then

$$\boxed{f_{\text{GW}} = \frac{\Omega}{\pi} = \frac{5^{3/8}}{8\pi} \frac{1}{\mathcal{M}^{5/8} (T - t)^{3/8}}},$$

which follows because

$$\mathcal{M}^{5/8} = (\mu^{3/5} M^{2/5})^{5/8} = \mu^{3/8} M^{1/4},$$

by the definition of the chirp mass $\mathcal{M}$. □

Exercise 1.12

- (a) Express τ_{GW} in (1.73) in terms of the gravitational wave frequency f_{GW} and equate your result to P_{orb} to show that the approximation of an adiabatic inspiral breaks down for a critical frequency $f_{\text{GW}}^{\text{crit}}$ given by

$$f_{\text{GW}}^{\text{crit}} = \frac{5^{3/5}}{2^{21/5} \pi^{8/5}} \frac{1}{\mathcal{M}}. \quad (1.74)$$

- (b) Use Kepler's law to show that the approximate binary separation R corresponding to the critical orbital frequency $\Omega^{\text{crit}} = \pi f_{\text{GW}}^{\text{crit}}$ is given by

$$R^{\text{crit}} = \frac{2^{14/5} \pi^{2/5}}{5^{2/5}} M^{3/5} \mu^{2/5} \approx 5.8 M^{3/5} \mu^{2/5}. \quad (1.75)$$

- (c) Use (1.63) to show that a crude estimate for the maximum gravitational wave amplitude, as measured at a distance d , is

$$h^{\text{crit}} \approx \frac{5^{2/5}}{2^{4/5} \pi^{2/5}} \frac{\mathcal{M}}{d} \approx 0.69 \frac{\mathcal{M}}{d}. \quad (1.76)$$

- (d) Show that, in physical units, the critical frequency (1.74) becomes

$$f_{\text{GW}}^{\text{crit}} \approx 4.9 \text{ kHz} \left(\frac{M_{\odot}}{\mathcal{M}} \right),$$

with wave amplitude (1.76) given by

$$h^{\text{crit}} \approx 3.3 \times 10^{-20} \left(\frac{\mathcal{M}}{M_{\odot}} \right) \left(\frac{\text{Mpc}}{d} \right).$$

Solution. (a) Using part (c) from the previous exercise, we have

$$\begin{aligned} \tau_{\text{GW}} &= \frac{5}{64} \frac{R^4}{M^2 \mu} \\ &= \frac{\cancel{5}}{64} \frac{1}{\cancel{M^2} \mu} \frac{256}{\cancel{5}} \cancel{M^2} \mu (T - t) \\ &= 4(T - t) \\ &= 4 \frac{5}{(8\pi)^{8/3}} \frac{1}{M^{2/3} \mu} \frac{1}{f_{\text{GW}}^{8/3}}, \end{aligned}$$

where we used the result from part (d) in the last line. Equating this to the orbital frequency $P_{\text{orb}} = 2/f_{\text{GW}}$ gives

$$(f_{\text{GW}}^{\text{crit}})^{5/3} = \frac{10}{(8\pi)^{8/3}} \frac{1}{M^{2/3} \mu},$$

and simplifying gives

$$f_{\text{GW}}^{\text{crit}} = \frac{5^{3/2}}{2^{21/5} \pi^{8/5}} \frac{1}{\mathcal{M}},$$

as we wanted to show.

(b) We have

$$\begin{aligned} \Omega^{\text{crit}} &= \pi f_{\text{GW}}^{\text{crit}} \\ &= \left[\frac{M}{(R^{\text{crit}})^3} \right]^{1/2}, \end{aligned}$$

which follows from Kepler's law. Thus

$$\begin{aligned} R^{\text{crit}} &= \frac{M^{1/3}}{\pi^{2/3} (f_{\text{GW}}^{\text{crit}})^{2/3}} \\ &= \frac{2^{14/5} \pi^{2/5}}{5^{2/5}} M^{3/5} \mu^{2/5}, \end{aligned}$$

where the last line follows from part (a), after some algebra.

(c) Similarly, using (1.63) we have

$$\begin{aligned} h^{\text{crit}} &\approx \frac{4}{d} \frac{\mu M}{R^{\text{crit}}} \\ &= \frac{2^{10/5}}{d} \frac{5^{2/5}}{2^{14/5} \pi^{2/5}} \frac{1}{M^{3/5} \mu^{2/5}} \mu M \\ &= \frac{5^{2/5}}{2^{4/5} \pi^{2/5}} \frac{\mathcal{M}}{d}. \end{aligned}$$

(d) We note that

$$\frac{[G][M]}{[c]^3} = \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2} \times \text{kg} \times \frac{\text{s}^3}{\text{m}^3} = \text{s},$$

so that to restore the units to $f_{\text{GW}}^{\text{crit}}$ we put

$$\begin{aligned} f_{\text{GW}}^{\text{crit}} &= \frac{5^{3/5}}{2^{21/5} \pi^{8/5}} \left(\frac{c^3}{GM_{\odot}} \right) \left(\frac{M_{\odot}}{\mathcal{M}} \right) \\ &= 4.64 \text{ kHz} \left(\frac{M_{\odot}}{\mathcal{M}} \right). \end{aligned}$$

Not sure why this differs from the text, since values for the physical quantities were pulled from Mathematica. You can also check (1.72) in the text; the number there is a bit different from what I get. In any case, this is more than acceptable for an order of magnitude estimate. Similarly, since h^{crit} is dimensionless, we note that

$$\frac{[G][M]}{[c]^2[d]} = \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2} \times \text{kg} \times \frac{\text{s}^2}{\text{m}^2} \times \frac{1}{\text{m}} = 1,$$

which means that to restore units we have

$$h^{\text{crit}} \approx \left[\frac{5^{2/5}}{2^{4/5} \pi^{2/5}} \frac{GM_{\odot}}{c^2} \frac{\text{Mpc}}{3.086 \times 10^{22} \text{ m}} \right] \left(\frac{\mathcal{M}}{M_{\odot}} \right) \left(\frac{1}{d \text{ (in Mpc)}} \right),$$

that is,

$$\boxed{h^{\text{crit}} = 3.3 \times 10^{-20} \left(\frac{\mathcal{M}}{M_{\odot}} \right) \left(\frac{\text{Mpc}}{d} \right)},$$

which was what we wanted to show. □

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Exercise 5.2

Write down the change of coordinates from Cartesian coordinates $(x^a) = (x, y, z)$ to spherical polar coordinates $(x'^a) = (r, \theta, \phi)$ in $\mathbb{R}^3$. Obtain the transformation matrices and express them both in terms of the primed coordinates. Obtain the Jacobians J and J' . Where is J' zero or infinite?

Solution. To go from spherical to Cartesian coordinates, we have

$$(x'^a) \rightarrow (x^a) : \quad \begin{cases} x = r \sin \theta \cos \phi \\ y = r \sin \theta \sin \phi \\ z = r \cos \theta \end{cases}$$

and the reverse direction is given by

$$(x^a) \rightarrow (x'^a) : \quad \begin{cases} r = \sqrt{x^2 + y^2 + z^2} \\ \theta = \arccos\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right) \\ \phi = \arctan(y/x) \end{cases}$$

Strictly speaking, the expression for ϕ is not exactly correct, but the necessary modification involves adding on constants, and this doesn't affect the derivatives. From here the transformation matrices can be obtained by a direct computation:

$$\begin{aligned} \left[\frac{\partial x'^a}{\partial x^b} \right] &= \begin{bmatrix} \frac{\partial x'^1}{\partial x^1} & \frac{\partial x'^1}{\partial x^2} & \frac{\partial x'^1}{\partial x^3} \\ \frac{\partial x'^2}{\partial x^1} & \frac{\partial x'^2}{\partial x^2} & \frac{\partial x'^2}{\partial x^3} \\ \frac{\partial x'^3}{\partial x^1} & \frac{\partial x'^3}{\partial x^2} & \frac{\partial x'^3}{\partial x^3} \end{bmatrix} \\ &= \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \frac{1}{r} \cos \theta \cos \phi & \frac{1}{r} \cos \theta \sin \phi & -\frac{1}{r} \sin \theta \\ -\frac{1}{r} \frac{\sin \phi}{\sin \theta} & \frac{1}{r} \frac{\cos \phi}{\sin \theta} & 0 \end{bmatrix}. \end{aligned}$$

Similarly, one finds that

$$\left[\frac{\partial x^a}{\partial x'^b} \right] = \begin{bmatrix} \sin \theta \cos \phi & r \cos \theta \cos \phi & -r \sin \theta \sin \phi \\ \sin \theta \sin \phi & r \cos \theta \sin \phi & r \sin \theta \cos \phi \\ \cos \theta & -r \sin \theta & 0 \end{bmatrix}.$$

Another straightforward computation shows

$$J' = \left| \frac{\partial x'^a}{\partial x^b} \right| = \frac{1}{r^2 \sin \theta} \quad \text{and} \quad J = \left| \frac{\partial x^a}{\partial x'^b} \right| = r^2 \sin \theta.$$

Thus J' is infinite when $\theta = 0$ or $\theta = \pi$, and goes to zero as $r \rightarrow \infty$. Otherwise we have $JJ' = 1$, as we expect. $\square$

Exercise 5.9

Show that $\frac{\partial^2 \phi}{\partial x^a \partial x^b}$ is not a tensor.

Solution. Since

$$\frac{\partial \phi}{\partial x'^a} = \frac{\partial x^b}{\partial x'^a} \frac{\partial \phi}{\partial x^b},$$

we have (taking care to differentiate with respect to the correct functions)

$$\begin{aligned} \frac{\partial^2 \phi}{\partial x'^c \partial x'^a} &= \frac{\partial}{\partial x'^c} \left(\frac{\partial x^b}{\partial x'^a} \frac{\partial \phi}{\partial x^b} \right) \\ &= \frac{\partial x^d}{\partial x'^c} \frac{\partial}{\partial x^d} \left(\frac{\partial x^b}{\partial x'^a} \frac{\partial \phi}{\partial x^b} \right) \\ &= \frac{\partial x^d}{\partial x'^c} \frac{\partial x^b}{\partial x'^a} \frac{\partial^2 \phi}{\partial x^d \partial x^b} + \frac{\partial x^d}{\partial x'^c} \frac{\partial^2 x^b}{\partial x^d \partial x'^a} \frac{\partial \phi}{\partial x^b}. \end{aligned}$$

The second term is what prevents this from transforming like a tensor. $\square$

Exercise 6.3

Establish (6.23) by assuming that the quantity defined by (6.22) has the tensor character indicated. Take the partial derivative of

$$\delta_c'^a = \frac{\partial x'^a}{\partial x'^c} = \frac{\partial x'^a}{\partial x^d} \frac{\partial x^d}{\partial x'^c}$$

with respect to x'^b to establish the alternative form, (6.24).

Solution. Some texts prefer to put the primes on the indices, but we'll follow d'Inverno's notation. Since X^a is a $(1, 0)$ tensor, we have

$$\begin{aligned} \nabla'_c X'^a &= \partial'_c X'^a + \Gamma'^a_{bc} X'^b \\ &= \frac{\partial}{\partial x'^c} \left(\frac{\partial x'^a}{\partial x^e} X^e \right) + \Gamma'^a_{bc} \frac{\partial x'^b}{\partial x^e} X^e \\ &= \frac{\partial x^d}{\partial x'^c} \frac{\partial}{\partial x^d} \left(\frac{\partial x'^a}{\partial x^e} X^e \right) + \Gamma'^a_{bc} \frac{\partial x'^b}{\partial x^e} X^e \\ &= \frac{\partial x^d}{\partial x'^c} \frac{\partial^2 x'^a}{\partial x^d \partial x^e} X^e + \frac{\partial x^d}{\partial x'^c} \frac{\partial x'^a}{\partial x^e} \partial_d X^e + \Gamma'^a_{bc} \frac{\partial x'^b}{\partial x^e} X^e. \end{aligned}$$

At the same time, if we assume $\nabla'_c X'^a$ is a $(1, 1)$ tensor, then

$$\begin{aligned} \nabla'_c X'^a &= \frac{\partial x'^a}{\partial x^d} \frac{\partial x^f}{\partial x'^c} \nabla_f X^d \\ &= \frac{\partial x'^a}{\partial x^d} \frac{\partial x^f}{\partial x'^c} \left(\partial_f X^d + \Gamma^d_{ef} X^e \right) \\ &= \frac{\partial x'^a}{\partial x^e} \frac{\partial x^d}{\partial x'^c} \partial_d X^e + \frac{\partial x'^a}{\partial x^d} \frac{\partial x^f}{\partial x'^c} \Gamma^d_{ef} X^e, \end{aligned}$$

where we relabeled a couple of bound indices in the third line. Since X^e is arbitrary, these two expressions for $\nabla'_c X'^a$ will be equal if

$$\Gamma'^a_{bc} \frac{\partial x'^b}{\partial x^e} = \frac{\partial x'^a}{\partial x^d} \frac{\partial x^f}{\partial x'^c} \Gamma^d_{ef} - \frac{\partial x^d}{\partial x'^c} \frac{\partial^2 x'^a}{\partial x^d \partial x^e}.$$

Multiplying through by $\partial x^e / \partial x'^g$, relabeling bound indices, and swapping the order of partial derivatives in the second term, we obtain

$$\Gamma'^a_{bc} = \frac{\partial x'^a}{\partial x^d} \frac{\partial x'^b}{\partial x^e} \frac{\partial x^f}{\partial x'^c} \Gamma^d_{ef} - \frac{\partial x^d}{\partial x'^b} \frac{\partial x^e}{\partial x'^c} \frac{\partial^2 x'^a}{\partial x^d \partial x^e},$$

which is precisely (6.23). Alternatively, then, you can think of Γ^a_{bc} as a sort of ‘correction term’ that you add on to the ordinary derivative of a tensor in order to make the result a tensor.

To establish the alternative form (6.24), we note that (again being careful to differentiate with respect to the correct functions)

$$\begin{aligned}\frac{\partial}{\partial x'^b} \delta_c'^a &= \frac{\partial}{\partial x'^b} \left(\frac{\partial x'^a}{\partial x^d} \frac{\partial x^d}{\partial x'^c} \right) \\ &= \frac{\partial x'^a}{\partial x^d} \frac{\partial^2 x^d}{\partial x'^b \partial x'^c} + \frac{\partial x^e}{\partial x'^b} \frac{\partial x^d}{\partial x'^c} \frac{\partial^2 x'^a}{\partial x^e \partial x^d} \\ &= 0,\end{aligned}$$

where the last line follows because the components $\delta_c'^a$ are constant. Thus

$$\Gamma'^a_{bc} = \frac{\partial x'^a}{\partial x^d} \frac{\partial x'^b}{\partial x^e} \frac{\partial x^f}{\partial x'^c} \Gamma^d_{ef} + \frac{\partial x'^a}{\partial x^d} \frac{\partial^2 x^d}{\partial x'^b \partial x'^c},$$

which is (6.24). □

Exercise 6.4

Show that covariant differentiation commutes with contraction by checking that $\nabla_c \delta_b^a = 0$.

Solution. Alternatively, you can *define* ∇ to be a mapping that (among other things) commutes with contractions, then after some work *derive* the component expression for $\nabla_c X^a$. Here we're working in the opposite direction, so all these properties seem to come out of nowhere.

Since

$$\begin{aligned}\nabla_c \delta_b^a &= \cancel{\partial_c \delta_b^a} + \Gamma^a_{dc} \delta_b^d - \Gamma^d_{bc} \delta_d^a \\ &= \Gamma^a_{bc} - \Gamma^a_{bc} \\ &= 0,\end{aligned}$$

it follows that

$$\begin{aligned}\nabla_c (\delta_d^e T^{a\cdots d\cdots}_{b\cdots e\cdots}) &= \delta_d^e \nabla_c T^{a\cdots d\cdots}_{b\cdots e\cdots} + \cancel{T^{a\cdots d\cdots}_{b\cdots e\cdots} \nabla_c \delta_d^e} \\ &= \delta_d^e \nabla_c T^{a\cdots d\cdots}_{b\cdots e\cdots},\end{aligned}$$

so covariant differentiation commutes with contraction. This holds even for connections that are not torsion-free. □

Exercise 6.10

Show that

$$\nabla_c \nabla_d X^a_b - \nabla_d \nabla_c X^a_b = R^a_{ecd} X^e_b - R^e_{bcd} X^a_e.$$

Solution. This solution is way, way too much work – better to use the Ricci identity, which has the torsion-free property built in. The result can be summarized informally by saying that $[\nabla_c, \nabla_d]$ obeys a product rule on the tensor indices. And one can show that this is true for general (r, s) tensors, so this is mostly an exercise in index gymnastics.

$$\begin{aligned} X^a_{b;dc} &= (X^a_{b,d} + \Gamma^a_{ed} X^e_b - \Gamma^e_{bd} X^a_e)_{;c} \\ &= (X^a_{b,d} + \Gamma^a_{ed} X^e_b - \Gamma^e_{bd} X^a_e)_{,c} \\ &\quad + \Gamma^a_{fc} (X^f_{b,d} + \Gamma^f_{ed} X^e_b - \Gamma^e_{bd} X^f_e) \\ &\quad - \Gamma^f_{dc} (X^a_{b,f} + \Gamma^a_{ef} X^e_b - \Gamma^e_{bf} X^a_e) \\ &\quad - \Gamma^f_{bc} (X^a_{f,d} + \Gamma^a_{ed} X^e_f - \Gamma^e_{fd} X^a_e) \\ &= X^a_{b,dc} + \Gamma^a_{ed,c} X^e_b + \Gamma^a_{ed} X^e_{b,c} - \Gamma^e_{bd,c} X^a_e - \Gamma^e_{bd} X^a_{e,c} \\ &\quad + \Gamma^a_{fc} X^f_{b,d} + \Gamma^a_{fc} \Gamma^f_{ed} X^e_b - \Gamma^a_{fc} \Gamma^e_{bd} X^f_e \\ &\quad - \Gamma^f_{dc} X^a_{b,f} - \Gamma^f_{dc} \Gamma^a_{ef} X^e_b + \Gamma^f_{dc} \Gamma^e_{bf} X^a_e \\ &\quad - \Gamma^f_{bc} X^a_{f,d} - \Gamma^e_{bc} \Gamma^a_{fd} X^f_e + \Gamma^f_{bc} \Gamma^e_{fd} X^a_e \\ &= X^a_{b,dc} + \Gamma^a_{ed} X^e_{b,c} - \Gamma^e_{bd} X^a_{e,c} + \Gamma^a_{ec} X^e_{b,d} - \Gamma^e_{dc} X^a_{b,e} - \Gamma^e_{bc} X^a_{e,d} \\ &\quad + (\Gamma^a_{ed,c} + \Gamma^a_{fc} \Gamma^f_{ed} - \Gamma^f_{dc} \Gamma^a_{ef}) X^e_b \\ &\quad - (\Gamma^e_{bd,c} - \Gamma^f_{dc} \Gamma^e_{bf} - \Gamma^f_{bc} \Gamma^e_{fd}) X^a_e \\ &\quad - (\Gamma^a_{fc} \Gamma^e_{bd} + \Gamma^e_{bc} \Gamma^a_{fd}) X^f_e. \end{aligned}$$

Swapping the c and d indices and subtracting causes most of these terms to cancel out, and after a bit more work, we end up with (possibly incorrect sign on torsion)

$$X^a_{b;dc} - X^a_{b;cd} = T^e_{cd} X^a_{b;e} + R^a_{ecd} X^e_b - R^e_{bcd} X^a_e.$$

If the connection is torsion-free, then the first term on the right-hand side is zero, and we obtain the desired identity. $\square$

Exercise 6.12

Prove that if a manifold is affine flat, then the connection is necessarily symmetric and integrable.

Proof. Affine flat means that there exists a coordinate system such that $\Gamma^a_{bc} = 0$ everywhere. In particular $\Gamma^a_{bc} = 0 = \Gamma^a_{cb}$, so the connection is symmetric. Now pick a vector X^a at some point on the manifold, and pick some curve $x^a(u)$. The equation for the vector field obtained via parallel transport along this curve is

$$\begin{aligned}\frac{DX^a}{Du} &= \frac{dx^c}{du} \nabla_c X^a \\ &= \frac{dx^c}{du} (\partial_c X^a + \Gamma^a_{bc} X^b) \\ &= \frac{dx^c}{du} \partial_c X^a \\ &= 0.\end{aligned}$$

Recall that the vector field specified by parallel transport is unique. At the same time, a solution is clearly $X^a(x) = X^a = \text{const.}$ everywhere along the curve, so this must be it. Then since $x^a(u)$ was arbitrary, we conclude that parallel transport leaves vectors unchanged; that is, the connection is integrable. $\square$

Exercise 6.24

Establish the identities (6.79) and (6.80). Show that (6.79) is equivalent to $R^a_{[bcd]} \equiv 0$.

Solution. Pick an arbitrary point on the manifold and introduce normal coordinates there. Then at this point

$$\Gamma^a_{bc} \stackrel{*}{=} 0 \implies R^a_{bcd} \stackrel{*}{=} \partial_c \Gamma^a_{bd} - \partial_d \Gamma^a_{bc},$$

which implies (6.79), provided the connection is torsion-free:

$$\begin{aligned} R^a_{bcd} + R^a_{dbc} + R^a_{cdb} &\stackrel{*}{=} \partial_c \Gamma^a_{bd} - \partial_d \Gamma^a_{bc} + \partial_b \Gamma^a_{dc} - \partial_c \Gamma^a_{db} + \partial_d \Gamma^a_{cb} - \partial_b \Gamma^a_{cd} \\ &= \cancel{\partial_c \Gamma^a_{bd}} - \cancel{\partial_c \Gamma^a_{db}} + \cancel{\partial_b \Gamma^a_{dc}} - \cancel{\partial_b \Gamma^a_{cd}} + \cancel{\partial_d \Gamma^a_{cb}} - \cancel{\partial_d \Gamma^a_{bc}} \\ &= 0. \end{aligned}$$

This is an equality between tensors, so it is true in any coordinate system. And since the point we chose was arbitrary, this identity holds everywhere – that is, we can replace the $=$ with a $\equiv$. Next, we note that

$$\begin{aligned} R^a_{[bcd]} &= \frac{1}{3!} (R^a_{bcd} - R^a_{bdc} + R^a_{dbc} - R^a_{dcb} + R^a_{cdb} - R^a_{cbd}) \\ &= \frac{1}{3} (R^a_{bcd} + R^a_{dbc} + R^a_{cdb}), \end{aligned}$$

since R^a_{bcd} is antisymmetric in its last pair of indices. This shows that the vanishing of the left-hand side is equivalent to the vanishing of the right-hand side.

To establish (6.80), we again pick an arbitrary point and set up normal coordinates. Then

$$\begin{aligned} R_{abcd} &= g_{ae} R^e_{bcd} \\ &\stackrel{*}{=} g_{ae} (\partial_c \Gamma^e_{bd} - \partial_d \Gamma^e_{bc}). \end{aligned}$$

Now, if our connection is torsion-free and metric-compatible, then

$$\begin{aligned} \partial_c \Gamma^e_{bd} &= \partial_c \left[\frac{1}{2} g^{ef} (\partial_b g_{fd} + \partial_d g_{fb} - \partial_f g_{bd}) \right] \\ &\stackrel{*}{=} \frac{1}{2} g^{ef} (\partial_c \partial_b g_{fd} + \partial_c \partial_d g_{fb} - \partial_c \partial_f g_{bd}), \end{aligned}$$

where we've again used metric compatibility and the fact that $\partial_c = \nabla_c$ at this point. Thus

$$g_{ae} \partial_c \Gamma^e_{bd} \stackrel{*}{=} \frac{1}{2} (\partial_c \partial_b g_{ad} + \partial_c \partial_d g_{ab} - \partial_c \partial_a g_{bd}).$$

Repeating the process for the remaining term, adding, and then using the equality of mixed partials, together with the fact that $g_{ab} = g_{ba}$, we get

$$R_{abcd} \stackrel{*}{=} \frac{1}{2} (\partial_c \partial_b g_{ad} - \partial_d \partial_b g_{ac} + \partial_a \partial_d g_{cb} - \partial_c \partial_a g_{bd}),$$

which is easily seen to be invariant under the exchange of indices $a \leftrightarrow c$, $b \leftrightarrow d$. This proves that $R_{abcd} = R_{cdab}$, as we wanted. It's worth noting that (6.79) only requires a torsion-free connection, whereas (6.80) requires metric compatibility as well. $\square$

Exercise 6.25

Establish the identity (6.83). Show that (6.83) is equivalent to $R_{de[ab;c]} \equiv 0$. Deduce (6.87).

Solution. A straightforward way to establish (6.83) is to set up normal coordinates, then use the expression for R_{abcd} derived in the previous exercise. If you don't have that expression at hand, or if you don't want to introduce a coordinate system, then you can show that $\omega_e \nabla_{[a} R_{bc]d}{}^e$ vanishes for arbitrary one-forms ω_e ; see Wald p. 39 for details. Then since R_{abcd} is symmetric in its last two indices, we have

$$\begin{aligned} R_{de[ab;c]} &= \frac{1}{3!} (R_{deab;c} - R_{deac;b} + R_{deca;b} - R_{dec b;a} + R_{debc;a} - R_{deba;c}) \\ &= \frac{1}{3} (R_{debc;a} + R_{deab;c} + R_{deca;b}). \end{aligned}$$

Thus, the left-hand side vanishes if and only if the right-hand side does. From here, we can deduce the contracted Bianchi identities (6.87). Since

$$\nabla_a R_{debc} + \nabla_c R_{deab} + \nabla_b R_{deca} = 0,$$

we can contract with g^{dc} to get

$$\nabla_a R^c{}_{ebc} + \nabla_c R^c{}_{eab} + \nabla_b R_{ea} = 0.$$

By the symmetries of the Riemann tensor, this is equivalent to

$$\nabla_b R_{ea} - \nabla_a R_{eb} - \nabla_c R_e{}^c{}_{ab} = 0.$$

Finally, contracting with g^{ea} , relabeling bound indices, and multiplying both sides by -1 gives

$$\begin{aligned} 2\nabla_a R^a{}_b - \nabla_b R &= 0 \\ \iff \nabla_a (R^a{}_b - \tfrac{1}{2}\delta^a_b R) &= 0 \\ \iff \nabla_a (g^{ac} G_{cb}) &= 0 \\ \iff \nabla_b G_a{}^b &= 0, \end{aligned}$$

where in the last line we relabel indices and use the fact that $G_{ab} = G_{ba}$. $\square$

Algebraic symmetries of the Riemann tensor

Deduce the following properties of the Riemann tensor, assuming a Levi-Civita connection:

- (1) $R^d_{cab} = -R^d_{cba},$
- (2) $R_{cdab} = -R_{dcab},$
- (3) $R_{a[bcd]} = 0.$

Using these, show that

- (4) $R_{abcd} = R_{cdab},$
- (5) $R_{[abcd]} = 0.$

Finally, show that properties (1), (2), (4), and (5) are equivalent to properties (1)–(3).

Solution. Property (1) follows from the definition, (2) can be derived using the Ricci identity (see the lecture notes for details) (fill in!), and (3) was derived in Exercise 6.24. Then

$$\begin{aligned}
 R_{abcd} &\stackrel{(3)}{=} -R_{adbc} - R_{acdb} \\
 &\stackrel{(1)}{=} -R_{adbc} + R_{cadb} \\
 &\stackrel{(3)}{=} -R_{adbc} - R_{cbad} - R_{cdba} \\
 &\stackrel{(2)}{=} -R_{adbc} - R_{cbad} + R_{cdab}.
 \end{aligned}$$

We also have

$$\begin{aligned}
 R_{abcd} &\stackrel{(1),(2)}{=} R_{badc} \\
 &\stackrel{(3)}{=} -R_{bcad} - R_{bdca} \\
 &\stackrel{(1)}{=} R_{cbad} - R_{bdca}.
 \end{aligned}$$

Adding these expressions gives

$$\begin{aligned}
 2R_{abcd} &= -R_{adbc} - R_{bdca} + R_{cdab} \\
 &\stackrel{(1)}{=} R_{dabc} + R_{dbca} + R_{cdab} \\
 &\stackrel{(3)}{=} -R_{dcab} + R_{cdab} \\
 &\stackrel{(2)}{=} 2R_{cdab},
 \end{aligned}$$

which shows that $R_{abcd} = R_{cdab}$. One way to deduce property (5) is to write out all 24 terms in $R_{[abcd]}$, and then regroup these to obtain four terms like $R_{a[bcd]}$, which vanish by (3). There's probably a way to do this that doesn't involve writing out all the permutations like a caveperson.

Explicitly, one finds that

$$\begin{aligned}
R_{[abcd]} &= \frac{1}{4!} (R_{abcd} - R_{abdc} + R_{adbc} - R_{adcb} + R_{acdb} - R_{acbd} \\
&\quad - R_{bcda} + R_{bcad} - R_{bacd} + R_{badc} - R_{bdac} + R_{bdca} \\
&\quad + R_{cdab} - R_{cdba} + R_{cbda} - R_{cbad} + R_{cabd} - R_{cadb} \\
&\quad - R_{dabc} + R_{dacb} - R_{dcab} + R_{dcba} - R_{dbca} + R_{dbac}) \\
&= \frac{1}{4} (R_{a[bcd]} - R_{b[cda]} + R_{c[dab]} - R_{d[abc]}) \\
&= 0.
\end{aligned}$$

Conversely, if we assume that $R_{[abcd]} = 0$, then it should be possible (you haven't done this!) to collapse the above expression into a single term proportional to $R_{a[bcd]}$ using symmetries (1), (2), and (4), which will then imply symmetry (3). $\square$

Exercise 6.26

Show that $G_{ab} = 0$ if and only if $R_{ab} = 0$.

Solution. The reverse direction is clear, so consider the forward direction. In n dimensions,

$$\begin{aligned}
G_{ab} = 0 &\implies R_{ab} - \frac{1}{2}g_{ab}R = 0 \\
&\implies R - \frac{1}{2}(n)R = 0 \\
&\implies (2 - n)R = 0.
\end{aligned}$$

Provided that $n \neq 2$, this implies $R = 0$, and substituting back into the equation $G_{ab} = 0$ yields the desired result. $\square$

Exercise 6.27

Establish the identity (6.90). Deduce that the Weyl tensor is trace-free on *all* pairs of indices.

Solution. In $n = 4$ dimensions, the Weyl tensor is defined to be

$$C_{abcd} = R_{abcd} + \frac{1}{2}(g_{ad}R_{cb} + g_{bc}R_{da} - g_{ac}R_{db} - g_{bd}R_{ca}) + \frac{1}{6}(g_{ac}g_{db} - g_{ad}g_{cb})R.$$

Then (recalling that $R_{ab} = R_{ba}$)

$$\begin{aligned} C^a{}_{bad} &= g^{ca}C_{abcd} \\ &= R_{bd} + \frac{1}{2}(\delta_d^c R_{cb} + \delta_b^a R_{da} - 4R_{db} - g_{bd}R) + \frac{1}{6}(4g_{db} - \delta_d^c g_{cb})R \\ &= R_{bd} + \frac{1}{2}R_{db} + \frac{1}{2}R_{db} - 2R_{db} - \frac{1}{2}g_{bd}R + \frac{2}{3}g_{db}R - \frac{1}{6}g_{db}R \\ &= 0, \end{aligned}$$

which is (6.90). [This doesn't use anything special about \$n = 4\$; the trace-free property holds for general \$n\$.](#) To show that the Weyl tensor is trace-free on all pairs of indices, we recall that C_{abcd} possesses the same algebraic symmetries as R_{abcd} . Therefore,

$$\begin{aligned} C^a{}_{bad} &\stackrel{(3)}{=} -C^a{}_{dba} - C^a{}_{adb} \\ &\stackrel{(2)}{=} C_d{}^a{}_{ba} - C^a{}_{adb} \\ &\stackrel{(4)}{=} C_{bad}{}^a - C^a{}_{adb} \\ &\stackrel{(1),(2)}{=} C_{ab}{}^a{}_d - C^a{}_{adb} \\ &= C^a{}_{bad} - C^a{}_{adb}. \end{aligned}$$

Thus $C^a{}_{adb} = 0$, and by working backward, we conclude that $C_d{}^a{}_{ba} = C^a{}_{dba} = 0$ as well. Then

$$C_{db}{}^a{}_a \stackrel{(4)}{=} C^a{}_{adb} = 0 \quad \text{and} \quad C_d{}^a{}_{ab} \stackrel{(1)}{=} -C_d{}^a{}_{ba} = 0.$$

To summarize, we've shown that

$$C^a{}_{abc} = C^a{}_{bac} = C^a{}_{bca} = C_b{}^a{}_{ac} = C_b{}^a{}_{ca} = C_{bc}{}^a{}_a = 0,$$

so all six traces of the Weyl tensor are zero. □

Exercise 6.29

Prove that the null geodesics of two conformally related metrics coincide.

Proof. Let $\bar{g}_{ab} = \Omega^2 g_{ab}$, and suppose $x^a = x^a(u)$ is a null geodesic associated with $\bar{g}_{ab}$. Suppose further that u is an affine parameter. Then

$$\frac{d^2 x^a}{du^2} + \bar{\Gamma}^a_{bc} \frac{dx^b}{du} \frac{dx^c}{du} = 0.$$

By part (ii) of Exercise 6.30, this is equivalent to

$$\frac{d^2 x^a}{du^2} + (\Gamma^a_{bc} + \delta^a_b W_c + \delta^a_c W_b - g_{bc} W^a) \frac{dx^b}{du} \frac{dx^c}{du} = 0,$$

and by simplifying a bit, this becomes

$$\frac{d^2 x^a}{du^2} + \Gamma^a_{bc} \frac{dx^b}{du} \frac{dx^c}{du} = -2W_c \frac{dx^c}{du} \frac{dx^a}{du} + g_{bc} \frac{dx^b}{du} \frac{dx^c}{du} W^a. \quad (1)$$

Since we have a null geodesic, we get

$$\bar{g}_{bc} \frac{dx^b}{du} \frac{dx^c}{du} = 0 \implies g_{bc} \frac{dx^b}{du} \frac{dx^c}{du} = 0, \quad (2)$$

because Ω^2 is nonzero by definition. Then by letting $\lambda(u) = -2\nabla_c(\ln \Omega) dx^c/du$, we see that Eq. (1) reduces to

$$\frac{d^2 x^a}{du^2} + \Gamma^a_{bc} \frac{dx^b}{du} \frac{dx^c}{du} = \lambda(u) \frac{dx^a}{du},$$

which is the equation for a geodesic (with non-affine parameter, unless $\lambda(u) \equiv 0$). By Eq. (2), this is also a null geodesic. Thus, the null geodesics of $\bar{g}_{ab}$ are also null geodesics of g_{ab} , and the same argument shows that the converse is true. [You can see why this reasoning doesn't work for any other type of geodesic – the second term on the right-hand side of \(1\) is nonzero.](#) $\square$

Exercise 6.30

Given two metrics which are conformally related, i.e.

$$\bar{g}_{ab} = \Omega^2 g_{ab},$$

then defining

$$W = \ln \Omega, \quad W_c = \nabla_c(\ln \Omega), \quad W_{cd} = \nabla_c(\nabla_d(\ln \Omega)),$$

show that

(i)

$$W_{cd} = W_{dc}$$

(ii)

$$\bar{\Gamma}^a_{bc} = \Gamma^a_{bc} + \delta^a_b W_c + \delta^a_c W_b - g_{bc} W^a$$

(iii) **WARNING: The book has typos here!**

$$\begin{aligned} \bar{R}^a_{bcd} = R^a_{bcd} + \delta^a_d W_{cb} - \delta^a_c W_{bd} - g_{bd} W^a_c - g_{bc} W^a_d + \delta^a_d W_b W_c - \delta^a_c W_b W_d - \delta^a_c g_{bd} W_e W^e \\ - g_{bc} W_d W^a - \delta^a_d W_b W^c - \delta^a_d g_{bc} W_e W^e - g_{bd} W_c W^a \end{aligned}$$

(iv) Use the definition (6.88) to deduce (6.92).

Solution. (i) This is true because ∇ is torsion-free, so $\nabla_c \nabla_d f = \nabla_d \nabla_c f$ for scalar functions f .

(ii) A direct calculation gives

$$\begin{aligned} \bar{\Gamma}^a_{bc} &= \frac{1}{2} \bar{g}^{ad} (\bar{g}_{bd,c} + \bar{g}_{cd,b} - \bar{g}_{bc,d}) \\ &= \frac{1}{2} \frac{g^{ad}}{\Omega^2} ((\Omega^2 g_{bd})_{,c} + (\Omega^2 g_{cd})_{,b} - (\Omega^2 g_{bc})_{,d}) \\ &= \frac{1}{2} g^{ad} (g_{bd,c} + g_{cd,b} - g_{bc,d}) + \frac{1}{\Omega} g^{ad} (\Omega_{,c} g_{bd} + \Omega_{,b} g_{cd} - \Omega_{,d} g_{bc}) \\ &= \Gamma^a_{bc} + \frac{\Omega_{,c}}{\Omega} \delta^a_b + \frac{\Omega_{,b}}{\Omega} \delta^a_c - \frac{\Omega_{,d}}{\Omega} g^{ad} g_{bc} \\ &= \Gamma^a_{bc} + (\nabla_c \ln \Omega) \delta^a_b + (\nabla_b \ln \Omega) \delta^a_c - (\nabla^a \ln \Omega) g_{bc} \\ &\equiv \Gamma^a_{bc} + \delta^a_b W_c + \delta^a_c W_b - g_{bc} W^a. \end{aligned}$$

(iii) The calculation is quite long, so here are the main ideas. You start from the definition

$$\bar{R}^a_{bcd} = \bar{\Gamma}^a_{bd,c} - \bar{\Gamma}^a_{bc,d} + \bar{\Gamma}^e_{bd} \bar{\Gamma}^a_{ec} - \bar{\Gamma}^e_{bc} \bar{\Gamma}^a_{ed},$$

and then expand the right-hand side using the result of part (ii). You end up with a lot of terms (prepare a big sheet of paper), but many of these cancel out, and it's not too difficult to group

the remaining terms according to (iii). There's only one trick involved – to get rid of the terms like $g_{bd,c}$, you'll need to add and subtract by a certain quantity to make two terms like $\nabla_c g_{bd}$ appear, which then vanish due to metric compatibility. Eventually, you obtain

$$\begin{aligned}\bar{R}^a_{bcd} = & R^a_{bcd} + \delta^a_d W_{cb} - \delta^a_c W_{bd} - g_{bd} W^a_c + g_{bc} W^a_d + \delta^a_c W_b W_d - \delta^a_c g_{bd} W_e W^e \\ & - g_{bc} W_d W^a - \delta^a_d W_b W_c + \delta^a_d g_{bc} W_e W^e + g_{bd} W^a W_c.\end{aligned}$$

The terms in red differ from the ones in the problem statement, but you can double-check this against the results from this Mathematica notebook. The relevant output cell is shown below. Alternatively, you can look at Appendix G of Carroll, which also has an expression for $\bar{R}^a_{bcd}$. His derivation is more sophisticated than what d'Inverno asks of the student in this exercise.

In[78]:= **Conformal[g, gS2][RiemannCD[-μ, -ν, α, -β]] // ToCanonical**

$$\begin{aligned}\text{Out[78]}= & R[\nabla]^\alpha_{\beta\mu\nu} + \frac{\delta^\alpha_\nu g_{\beta\mu} g_{\gamma\lambda} \nabla^\gamma S \nabla^\lambda S}{S^2} - \frac{\delta^\alpha_\mu g_{\beta\nu} g_{\gamma\lambda} \nabla^\gamma S \nabla^\lambda S}{S^2} + \frac{2 g_{\beta\nu} \nabla^\alpha S \nabla_\mu S}{S^2} - \frac{2 \delta^\alpha_\nu \nabla_\beta S \nabla_\mu S}{S^2} + \\ & \frac{\delta^\alpha_\nu \nabla_\mu \nabla_\beta S}{S} - \frac{g^{\alpha\gamma} g_{\beta\gamma} \nabla_\mu \nabla_\gamma S}{S} - \frac{2 g_{\beta\mu} \nabla^\alpha S \nabla_\nu S}{S^2} + \frac{2 \delta^\alpha_\mu \nabla_\beta S \nabla_\nu S}{S^2} - \frac{\delta^\alpha_\mu \nabla_\nu \nabla_\beta S}{S} + \frac{g^{\alpha\gamma} g_{\beta\mu} \nabla_\nu \nabla_\gamma S}{S}\end{aligned}$$

To translate into this problem's notation, note that the code uses S in place of Ω , and that

$$W_{cd} = \frac{\nabla_c \nabla_d \Omega}{\Omega} - \frac{\nabla_c \Omega \nabla_d \Omega}{\Omega^2},$$

which follows from the definition.

(iv) Another long calculation; below are the main results.¹

$$\begin{aligned}\bar{R}_{abcd} = & \Omega^2 (R_{abcd} + g_{ad} W_{cb} - g_{ac} W_{bd} - g_{bd} W_{ac} + g_{bc} W_{ad} + g_{ac} W_b W_d - g_{ac} g_{bd} W_f W^f \\ & - g_{bc} W_d W_a - g_{ad} W_b W_c + g_{ad} g_{bc} W_f W^f + g_{bd} W_a W_c),\end{aligned}$$

$$\bar{R}_{bd} = R_{bd} - 2W_{bd} + 2W_b W_d - 2g_{bd} W_f W^f - g_{bd} W^f_f,$$

$$\bar{R} = \frac{1}{\Omega^2} R - \frac{6}{\Omega^2} (W^f_f + W_f W^f).$$

From here you can substitute into the definition (6.88) to find $\bar{C}_{abcd} = \Omega^2 C_{abcd}$, and raising an index with $\bar{g}^{fa}$ kills the factor on the right-hand side, thus proving (6.92). It would be nicer to do this for general n ; this only proves the assertion for $n = 4$. $\square$

¹This was much more frustrating than it had the right to be, because at some point mid-calculation I mistakenly wrote $\delta^a_a = 1$. Of course, then, none of the terms canceled out properly.

Exercise 7.3

Denoting the transformation matrices by

$$J_{ab} = \left(\frac{\partial x^a}{\partial x'^b} \right), \quad J^{ab} = \left(\frac{\partial x'^a}{\partial x^b} \right),$$

use the argument of §7.3 to show that

$$\partial_c J = J J^{ba} \partial_c J_{ab},$$

where $J = \det(J_{ab})$ is the Jacobian. Hence show from first principles that, if $\mathfrak{T}^a$ is a vector density of weight +1, then $\partial_a \mathfrak{T}^a$ is a scalar density of weight +1.

Solution. By a result from linear algebra, we have

$$J^{ij} = \frac{1}{J} C^{ji}, \tag{1}$$

where C^{ij} denotes the cofactor of J_{ij} . If we fix i , then we can also write the cofactor expansion

$$J = \sum_{j=1}^n J_{ij} C^{ij} \implies \frac{\partial J}{\partial J_{ij}} = C^{ij}. \tag{2}$$

Treating the matrix elements J_{ij} as functions of the unprimed coordinates x^k , we get

$$J = J(J_{ij}(x^k))$$

which implies

$$\begin{aligned} \frac{\partial J}{\partial x^k} &= \frac{\partial J}{\partial J_{ij}} \frac{\partial J_{ij}}{\partial x^k} \\ &= C^{ij} \frac{\partial J_{ij}}{\partial x^k} \\ &= J J^{ji} \frac{\partial J_{ij}}{\partial x^k}, \end{aligned}$$

where we've used Eqs. (1) and (2). Then by relabeling indices, we obtain

$$\partial_c J = J J^{ba} \partial_c J_{ab},$$

as we wanted. Now suppose $\mathfrak{T}^a$ is a vector density of weight +1, so that

$$\mathfrak{T}'^a = J \frac{\partial x'^a}{\partial x^b} \mathfrak{T}^b.$$

Then

$$\begin{aligned}
\partial'_a \mathfrak{T}^a &= \frac{\partial x^c}{\partial x'^a} \frac{\partial}{\partial x^c} \left(J \frac{\partial x'^a}{\partial x^b} \mathfrak{T}^b \right) \\
&= \frac{\partial x^c}{\partial x'^a} \left((\partial_c J) \frac{\partial x'^a}{\partial x^b} \mathfrak{T}^b + J \frac{\partial^2 x'^a}{\partial x^c \partial x^b} \mathfrak{T}^b + J \frac{\partial x'^a}{\partial x^b} \partial_c \mathfrak{T}^b \right) \\
&= (J J^{fe} \partial_c J_{ef}) \underbrace{\frac{\partial x^c}{\partial x'^a} \frac{\partial x'^a}{\partial x^b}}_{\delta_b^c} \mathfrak{T}^b + J \frac{\partial x^c}{\partial x'^a} \frac{\partial^2 x'^a}{\partial x^c \partial x^b} \mathfrak{T}^b + J \underbrace{\frac{\partial x^c}{\partial x'^a} \frac{\partial x'^a}{\partial x^b}}_{\delta_b^c} \partial_c \mathfrak{T}^b \\
&= J \left(\frac{\partial x'^f}{\partial x^e} \frac{\partial x'^g}{\partial x^b} \frac{\partial^2 x^e}{\partial x'^g \partial x'^f} + \frac{\partial x^c}{\partial x'^a} \frac{\partial^2 x'^a}{\partial x^c \partial x^b} \right) \mathfrak{T}^b + J \partial_a \mathfrak{T}^a \\
&= J \partial_a \mathfrak{T}^a.
\end{aligned}$$

In the fourth line, we use the definitions of J^{fe} and J_{ef} , taking care to differentiate with respect to the correct functions, and in the last line, we use a result from Exercise 6.3 to conclude that the expression in parentheses is zero. This shows that $\partial_a \mathfrak{T}^a$ is a scalar density of weight +1. $\square$

Exercise 7.10

Deduce (7.55) from (7.54).

Solution. A straightforward computation. By metric compatibility, we have

$$\partial_e g_{ab} = \Gamma_{ae}^c g_{cb} + \Gamma_{be}^c g_{ac},$$

which means (using torsion-freeness)

$$\begin{aligned}
L_X g_{ab} &= X^e \partial_e g_{ab} + g_{ad} \partial_b X^d + g_{bd} \partial_a X^d \\
&= X^e \Gamma_{ae}^c g_{cb} + X^e \Gamma_{be}^c g_{ac} + g_{ad} \partial_b X^d + g_{bd} \partial_a X^d \\
&= (\partial_a X^c + \Gamma_{ea}^c X^e) g_{bc} + (\partial_b X^c + \Gamma_{eb}^c X^e) g_{ac} \\
&= (\nabla_a X^c) g_{bc} + (\nabla_b X^c) g_{ac} \\
&= \nabla_a X_b + \nabla_b X_a.
\end{aligned}$$

This shows that Killing's equation is equivalent to $\nabla_a X_b + \nabla_b X_a = 0$. More generally, one can replace ordinary derivatives by covariant derivatives in the expression for the Lie derivative of a tensor, provided that the connection is torsion-free. So an alternative calculation is

$$L_X g_{ab} = X^e \nabla_e g_{ab} + g_{ad} \nabla_b X^d + g_{bd} \nabla_a X^d = \nabla_b X_a + \nabla_a X_b.$$

$\square$

Exercise 7.11

Find all the Killing vectors X^a of the three-dimensional Euclidean line element

$$ds^2 = dx^2 + dy^2 + dz^2.$$

[The solution will have six independent constants of integration.] Denoting [these] by $\lambda_1, \dots, \lambda_6$, write the general solution for X^a in the form

$$\lambda_1 \overset{1}{X^a} + \lambda_2 \overset{2}{X^a} + \lambda_3 \overset{3}{X^a} + \lambda_4 \overset{4}{X^a} + \lambda_5 \overset{5}{X^a} + \lambda_6 \overset{6}{X^a}.$$

Find expressions for the vector fields $\overset{i}{X}$, ($i = 1, 2, \dots, 6$) and hence find all values of $[\overset{i}{X}, \overset{j}{X}]$. Interpret the six Killing vector fields in terms of geometrical transformations.

Solution. In flat space all of the Christoffel symbols vanish, so covariant differentiation is equivalent to ordinary differentiation (and in this case we can ignore the distinction between co- and contravariant indices), and Killing's equations read

$$\partial_b X_a + \partial_a X_b = 0.$$

We can differentiate this with respect to x^c to get

$$\partial_c \partial_b X_a + \partial_c \partial_a X_b = 0, \tag{1}$$

and permuting indices gives

$$\partial_a \partial_c X_b + \partial_a \partial_b X_c = 0, \tag{2}$$

$$\partial_b \partial_a X_c + \partial_b \partial_c X_a = 0. \tag{3}$$

Adding (1) and (3), subtracting by (2), then using the commutativity of partial derivatives gives

$$\partial_b \partial_c X_a = 0.$$

This has solution

$$X_a = \omega_{ab} x^b + t_a,$$

where ω_{ab} and t_a are constants of integration. Then since

$$\begin{aligned} \partial_b X_a + \partial_a X_b &= \partial_b (\omega_{ac} x^c + t_a) + \partial_a (\omega_{bc} x^c + t_b) \\ &= \omega_{ac} \delta_b^c + \omega_{bc} \delta_a^c \\ &= \omega_{ab} + \omega_{ba} \\ &= 0, \end{aligned}$$

we see that the constants ω_{ab} can be interpreted as components of an antisymmetric matrix. Thus the general solution for X^a takes the form

$$X^a = \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{pmatrix} + \begin{pmatrix} 0 & \lambda_4 & \lambda_5 \\ -\lambda_4 & 0 & \lambda_6 \\ -\lambda_5 & -\lambda_6 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

From here, we can read off

$$\begin{aligned} X^1_a &= (1, 0, 0), & X^4_a &= (y, -x, 0), \\ X^2_a &= (0, 1, 0), & X^5_a &= (z, 0, -x), \\ X^3_a &= (0, 0, 1), & X^6_a &= (0, z, -y). \end{aligned}$$

The corresponding Killing vector fields are

$$\begin{aligned} \overset{1}{X} &= \frac{\partial}{\partial x}, & \overset{4}{X} &= y \frac{\partial}{\partial x} - x \frac{\partial}{\partial y}, \\ \overset{2}{X} &= \frac{\partial}{\partial y}, & \overset{5}{X} &= z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z}, \\ \overset{3}{X} &= \frac{\partial}{\partial z}, & \overset{6}{X} &= z \frac{\partial}{\partial y} - y \frac{\partial}{\partial z}. \end{aligned}$$

Geometrically, the first three vector fields generate translations in the x , y , and z directions, respectively, while the last three generate rotations around the z , y , and x axes, respectively. From here, computing the Lie brackets is straightforward – we have

	$\overset{1}{X}$	$\overset{2}{X}$	$\overset{3}{X}$	$\overset{4}{X}$	$\overset{5}{X}$	$\overset{6}{X}$
$\overset{1}{X}$	0	0	0	$-\overset{2}{X}$	$-\overset{3}{X}$	0
$\overset{2}{X}$		0	0	$\overset{1}{X}$	0	$-\overset{3}{X}$
$\overset{3}{X}$			0	0	$\overset{1}{X}$	$\overset{2}{X}$
$\overset{4}{X}$				0	$\overset{6}{X}$	$-\overset{5}{X}$
$\overset{5}{X}$					0	$\overset{4}{X}$
$\overset{6}{X}$						0

There is some arbitrariness when it comes to the signs of these expressions, because we're free to multiply these Killing vectors by constants. $\square$

Exercise 7.13

Consider the following operator identity:

$$L_X L_Y - L_Y L_X = L_{[X, Y]}.$$

- (i) Check it holds when applied to an arbitrary scalar function f .
- (ii) Check it holds when applied to an arbitrary contravariant vector field m^a .
- (iii) Deduce that the identity holds when applied to a covariant vector field V_a .

Use the identity to prove that, if X and Y are Killing vector fields, then so is their commutator $[X, Y]$. Given that $\partial/\partial x$ and $-y\partial/\partial x + x\partial/\partial y$ are Killing vector fields, find another.

Solution. (i) We have

$$\begin{aligned} (L_X L_Y - L_Y L_X)f &= L_X(Y^c \partial_c f) - L_Y(X^c \partial_c f) \\ &= X^d \partial_d (Y^c \partial_c f) - Y^d \partial_d (X^c \partial_c f) \\ &= X^d (\partial_d Y^c) (\partial_c f) + \cancel{X^d Y^c \partial_d \partial_c f} - Y^d (\partial_d X^c) (\partial_c f) - \cancel{Y^d X^c \partial_d \partial_c f} \\ &= (X^d \partial_d Y^c - Y^d \partial_d X^c) \partial_c f \\ &= [X, Y]^c \partial_c f \\ &= L_{[X, Y]} f, \end{aligned}$$

where in the third line, we use the equality of mixed partial derivatives.

- (ii) This can be proven by another direct calculation, or you can use the Jacobi identity, together with the fact that $L_X Y^a = [X, Y]^a$:

$$\begin{aligned} (L_X L_Y - L_Y L_X)m^a &= L_X[Y, m]^a - L_Y[X, m]^a \\ &= [X, [Y, m]]^a - [Y, [X, m]]^a \\ &= [X, [Y, m]]^a + [Y, [m, X]]^a \\ &\stackrel{\text{Jacobi}}{=} -[m, [X, Y]]^a \\ &= [[X, Y], m]^a \\ &= L_{[X, Y]} m^a. \end{aligned}$$

- (iii) This uses the product rule for Lie derivatives, as well as the fact that Lie differentiation

commutes with contraction. Let W^a be an arbitrary vector field, so that

$$\begin{aligned}(L_X L_Y - L_Y L_X)(W^a V_a) &= W^a (L_X L_Y - L_Y L_X) V_a + V_a (L_X L_Y - L_Y L_X) W^a \\ &= W^a (L_X L_Y - L_Y L_X) V_a + V_a L_{[X,Y]} W^a\end{aligned}$$

where in the second line we use the result of (ii). We also have

$$\begin{aligned}(L_X L_Y - L_Y L_X)(W^a V_a) &= L_{[X,Y]}(W^a V_a) \\ &= W^a L_{[X,Y]} V_a + V_a L_{[X,Y]} W^a\end{aligned}$$

where we've used the result of part (i). Comparing these two expressions and using the fact that W^a is arbitrary, we get

$$(L_X L_Y - L_Y L_X) V_a = L_{[X,Y]} V_a.$$

(iv) It isn't immediately obvious to me how (ii) and (iii) allow us to generalize the result to arbitrary (r, s) tensors, although I'm happy to just accept the identity. Maybe this would be easier to see in index-free notation. Suppose X and Y are Killing vector fields, so that $L_X g_{ab} = 0$ and $L_Y g_{ab} = 0$. Then

$$\begin{aligned}L_{[X,Y]} g_{ab} &= (L_X L_Y - L_Y L_X) g_{ab} \\ &= 0,\end{aligned}$$

which shows that $[X, Y]$ is also a Killing vector field. So if $\partial/\partial x$ and $-y\partial/\partial x + x\partial/\partial y$ are Killing vector fields, then another one is

$$\left[\frac{\partial}{\partial x}, -y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y} \right] = \frac{\partial}{\partial y}.$$

□

Exercise 7.14

Express $(\nabla_c \nabla_b - \nabla_b \nabla_c)X_a$ in terms of the Riemann tensor. Use this result to prove that any Killing vector satisfies

$$g^{bc} \nabla_b \nabla_a X_c - R_{ab} X^b = 0.$$

Solution. Using the Ricci identity and metric compatibility,

$$\begin{aligned} (\nabla_c \nabla_b - \nabla_b \nabla_c)X_a &= (\nabla_c \nabla_b - \nabla_b \nabla_c)(g_{ad} X^d) \\ &= g_{ad}(\nabla_c \nabla_b - \nabla_b \nabla_c)X^d \\ &= g_{ad} R^d_{ecb} X^e \\ &= R_{aecb} X^e, \end{aligned}$$

hence

$$g^{ac}(\nabla_c \nabla_b - \nabla_b \nabla_c)X_a - R_{eb} X^e = 0.$$

If X^a is a Killing vector, then $\nabla_a X_c = -\nabla_c X_a$, and this equation becomes (since $g^{ab} = g^{ba}$)

$$g^{ac}(\nabla_c \nabla_b + \nabla_b \nabla_c)X_a - R_{eb} X^e = 0.$$

Adding these two equations, using our symmetries, then relabeling indices, we conclude that

$$g^{bc} \nabla_b \nabla_a X_c - R_{ab} X^b = 0.$$

□

Exercise 7.15

By making use of the identity

$$R^a{}_{bcd} + R^a{}_{dbc} + R^a{}_{cdb} \equiv 0,$$

or otherwise, prove that a Killing vector satisfies

$$\nabla_c \nabla_b X_a = R_{abcd} X^d.$$

Solution. We start from the identity derived in Exercise 7.14,

$$\nabla_c \nabla_b X_a - \nabla_b \nabla_c X_a = R_{aecb} X^e. \quad (1)$$

Cycling through the indices gives

$$\nabla_a \nabla_c X_b - \nabla_c \nabla_a X_b = R_{beac} X^e,$$

and if X^a is a Killing vector, then $\nabla_a X_b = -\nabla_b X_a$, so we have

$$\nabla_a \nabla_c X_b + \nabla_c \nabla_b X_a = R_{beac} X^e. \quad (2)$$

We can cycle through the indices one more time and use the Killing vector property to get

$$-\nabla_b \nabla_c X_a + \nabla_a \nabla_c X_b = R_{ceba} X^e. \quad (3)$$

Adding (1) and (2) gives

$$2\nabla_c \nabla_b X_a - \nabla_b \nabla_c X_a + \nabla_a \nabla_c X_b = (R_{aecb} + R_{beac}) X^e,$$

and then we can subtract by (3) and use the symmetries of the Riemann tensor to get

$$\begin{aligned} 2\nabla_c \nabla_b X_a &= (R_{aecb} + R_{beac} - R_{ceba}) X^e \\ &= (R_{cbae} + R_{acbe} - R_{bace}) X^e && \text{interchange of first and second pairs} \\ &= (-R_{bcae} - R_{cabe} + R_{abce}) X^e && \text{antisymmetry in first pair} \\ &= 2R_{abce} X^e. && \text{first Bianchi identity} \end{aligned}$$

Therefore, if X^a is a Killing vector, then

$$\nabla_c \nabla_b X_a = R_{abce} X^e.$$

□